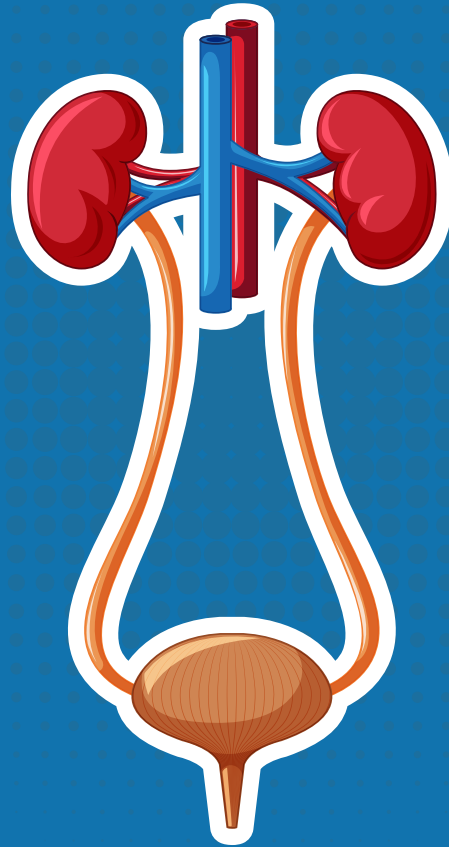




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MINISTRY OF HEALTH-ETHIOPIA

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HEALTHIER CITIZENS FOR PROSPEROUS NATION!

NATIONAL TRAINING ON SCREENING AND MANAGEMENT OF CHRONIC KIDNEY DISEASES



PARTICIPANT MANUAL

June 2025

Addis Ababa, Ethiopia

**NATIONAL TRAINING ON
SCREENING AND MANAGEMENT OF
CHRONIC KIDNEY DISEASES**

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FOREWORD



The National Training Manual on Chronic Kidney Disease (CKD) for healthcare workers is one of several key efforts aimed at implementing interventions to address the rising burden of non-communicable diseases (NCDs) in Ethiopia, as outlined in the Second NCD Strategic Action Plan (2020–2025) launched in July 2020. In developing this manual, a wide range of global guidelines on CKD, as well as best practices in the development and implementation of high-quality clinical practice guidelines, were reviewed and adapted.

The manual is largely based on international and regional clinical practice recommendations. Its approach emphasizes providing practical guidance to healthcare workers—focusing not only on what to do but also on how to do it—rather than offering general recommendations alone.

This manual is designed to support the prevention, screening, diagnosis, treatment, and care of CKD at the primary care and general hospital levels. It also provides clear guidance on when referral to secondary and tertiary levels of care is necessary. Moreover, it is intended to assist policymakers and health managers at all levels in planning human resource requirements and managing CKD programs effectively.

For program managers, the manual outlines key “what to do” and “how to do it” elements, while recognizing that detailed implementation guidance will need to emerge from pilot projects and the best practices developed through them. Nonetheless, this document offers a concise and practical overview of feasible and desirable strategies for CKD prevention and control that will be valuable to program implementers.

Additionally, the manual will be of interest to academic institutions, medical students, and a broad range of multidisciplinary health professionals seeking to familiarize themselves with Ethiopia’s national CKD prevention and control efforts.

It is with great enthusiasm that I encourage all stakeholders to use this first edition as a reference and guiding document for the training of healthcare workers in the clinical management of CKD in Ethiopia.

A handwritten signature in black ink, appearing to read 'H. Solomon'.

Dr Hiwot Solomon (BSC, MPH, PHD)

Diseases Prevention and Control Lead Executive Officer
Ministry of Health, Ethiopia

APPROVAL STATEMENT OF THE MINISTRY

The Federal Ministry of health of Ethiopia has been working towards standardization and institutionalization of In-Service Trainings (IST) at a national level. As part of this initiative, the ministry developed a national in-service training directive and implementation guide for the health sector. The directive requires all in-service training materials fulfill the standards set in the implementation guide to ensure the quality of in-service training materials. Accordingly, the ministry reviews and approves existing training materials based on the IST standardization checklist annexed on the IST implementation guide.

As part of the national IST quality control process, screening and management of chronic kidney disease Training has been reviewed based on the standardization checklist and approved by the ministry in July,2025.



Assegid Samuel Cheru
Human Resource Development and Improvement
Lead executive officer
Ministry of Health- Ethiopia

ACKNOWLEDGMENTS

The Ministry of Health would like to extend its sincere appreciation to all those who contributed to the development of this training package. It is not possible to mention all who participated in the preparation of this document; however, we would like to acknowledge the following professionals:

The following professionals were involved in developing the training manual:

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The MoH would like to pass its gratitude to AstraZeneca, and PSI-Ethiopia, for supporting the development of this national training material which will help us address this deadly but silent disease that afflicts significant proportion of our population. We would also appreciate the invaluable technical support provided by the Ethiopian Kidney Association.

It is hoped that this training material and provider tools will help fill the information and skill gap of our healthcare workers so that they can deliver high quality uninterrupted care to patients with chronic non communicable diseases and their families.

ACRONYMS

ACR	Albumin-to-Creatinine Ratio
ADDIE	Analyze Design Development Implementation
AKD	Acute Kidney Disease
AKI	Acute Kidney Injury
AoCKD	Acute-on-Chronic Kidney Disease
BPH	Benign Prostatic Hyperplasia
CEU	Continuous Educational Unit
CKD	Chronic Kidney Disease
CVD	Cardiovascular Disease
CPD	Continuous Professional Development
DALY	Disability Adjusted Life Years
eGFR	Estimated Glomerular Filtration Rate
ESRD	End Stage Renal Disease
GUTB	Genitourinary Tuberculosis
GFR	Glomerular Filtration Rate
KRT	Kidney Replacement Therapy
LCD	Liquid Crystal Display
LMIC	Low- and Middle-Income Countries
MET	Medical Expulsive Therapy
OPD	Outpatient Department
OUP	Obstructive Uropathy
POD	Post Operative Diuresis
RRT	Renal Replacement Therapy
TURP	Transurethral Resection of The Prostate
UACR	Urine Albumin Creatinine Ratio
UTI	Urinary Tract Infections
UTO	Urinary Tract Obstruction

INTRODUCTION TO THE MANUAL

The increasing burden of non-communicable diseases (NCDs) has been observed in recent years due to urbanization and lifestyle changes. In response, the Ministry of Health (MoH) has placed greater emphasis on the prevention and control of NCDs, as demonstrated by the development of Clinical and Programmatic Management Guidelines on major NCDs, a national NCD treatment protocol, and training materials for health workers in Ethiopia. Moreover, the MoH has developed a strategic plan to prevent and control these diseases. The Ministry also undertook a STEPS survey in 2024 to estimate their burden.

Chronic kidney disease (CKD) is one of the fastest-growing chronic non-communicable diseases globally. It is the 10th leading cause of morbidity and mortality worldwide, with an estimated prevalence of 13.4% among adults. Over 850 million people live with kidney disease—roughly double the number affected by diabetes (422 million) and over 20 times more than those living with cancer (42 million) or HIV/AIDS (36.7 million). By 2040, CKD is projected to become the 5th leading cause of years of life lost (YLL).

Up to 78% of patients with kidney disease live in low- and middle-income countries (LMICs), which are the least equipped to address its consequences. In sub-Saharan Africa, approximately 13% of adults have kidney disease, yet 90% of those with CKD remain unaware of their condition. This figure is even higher in low-income countries like Ethiopia, where routine screening is uncommon. Epidemiological studies in Ethiopia have shown a CKD prevalence ranging from 3.5% to 12.2% in the community. CKD is also among the top 10 NCD-related causes of death in Ethiopia, responsible for approximately 26 deaths per 100,000 population.

There is a need for early detection of CKD in LMICs to reduce its associated morbidity, mortality, and economic burden. Studies show that early detection and treatment of CKD is feasible even in resource-limited settings. In sub-Saharan Africa, CKD typically affects individuals aged 20 to 50, with end-stage renal disease (ESRD) developing approximately 20 years earlier among populations of African descent compared to other ethnic groups in Western countries (45 years vs. 63 years).

CKD has multiple risk factors that are categorized as modifiable and non-modifiable. Identifying these is crucial for effective prevention, early detection, and management. Modifiable risk factors can be addressed through lifestyle changes, medical treatment, or early intervention. These include:

- **Medical conditions:** diabetes mellitus (DM), hypertension, cardiovascular disease, obesity, dyslipidemia, autoimmune diseases, infections, and exposure to nephrotoxins
- **Lifestyle factors:** including smoking, unhealthy diet, sedentary lifestyle, and dehydration

While CKD is more manageable in its early stages, most people are unaware of their condition until it progresses. The catastrophic costs associated with treating advanced CKD—including expensive medications, dialysis, and transplant services in many countries in African including Ethiopia struggle to provide care for their people.

In Ethiopia, the best way to reduce the CKD burden lies in amplifying efforts to identify the unique risk factors associated with kidney disease, making tools available for early diagnosis and treatment and enhance the capacity of health workers in managing CKD at all levels of care, especially at primary health care and general hospital level. Providing clear guidance to health workers will support improved decision-making for CKD screening and management.

Therefore, the Ethiopian Ministry of Health in collaboration with its partners has developed this training manual on CKD. The CKD training manual is premised on three levels of prevention: primary prevention that aims at appropriate prevention and management of CKD risk factors, secondary prevention that embraces early diagnosis, timely referral, and delay of progression; and tertiary prevention aimed at managing complications.

This manual aims to strengthen the national NCD program and support CKD prevention and control. It is intended for healthcare providers at health centers and hospitals at both primary and general levels. The purpose of this training manual is to provide guidance on early screening, staging, and management of CKD among at-risk populations. We hope it will support healthcare workers in providing a comprehensive care package to individuals affected by CKD. Program managers and health service leaders may also use this manual to determine the diagnostics, supplies, devices, and equipment needed for CKD prevention and management.

RATIONALE FOR THIS TRAINING

Insight gathering assessment conducted before the course design showed that CKD services at hospital level lack both depth and quality. There is no systematic approach to screening, diagnosis, treatment, or follow-up for patients with CKD. At health center level, this gaps are more significant. This training manual has been developed to improve the quality of CKD care at general hospital level, strengthen referral systems between hospitals and health centers, and decentralize CKD services to the primary health care level.

Core competencies:

Upon successful completion of the training, participants should be able to:

- Diagnose CKD in adults and children, including women
- Manage CKD and its complications in adults, children and women
- Manage major infections in patients with CKD
- Practice proper recording and reporting of CKD-related data and indicators

COURSE SYLLABUS



Course description: This three-day course is designed to equip participants with the basic knowledge, skills and attitudes, required to effectively diagnose and manage early-stage chronic kidney disease patients. It also describes appropriate management of CKD complications and criteria for timely referral of late-stage CKD patients to nephrologists.



Course goal: To effectively diagnose and manage chronic kidney disease and its complications at the primary health care and general hospital levels.



COURSE OBJECTIVES

By the end of this course the participants will be able to:

- Discuss the definition, epidemiology and risk factors of CKD.
- Explain the diagnosis of CKD
- Describe a comprehensive approach to management of CKD.
- Identify the diagnostic criteria and methods for obstructive uropathy.
- Identify appropriate management for complications of CKD.
- Discuss the nutritional management of CKD patients.
- Describe the management of major infections and routine vaccinations in CKD
- Discuss the peculiarities of CKD diagnosis and management in children and women
- Describe prevention of CKD, screening and management of CKD at different levels and the monitoring and evaluation of the CKD program.



TRAINING METHODS

- Interactive presentations
- Case scenarios
- Small group discussion
- Individual reflection
- Demonstration
- Guided practice



TRAINING MATERIALS/AIDS

- CKD participant manual
- CKD facilitator guide
- PowerPoint slides
- Laptop
- LCD Projector
- Marker
- Flip chart
- Plaster
- DM and HTN/CKD cohort register

- DM and HTN /CKD screening tally sheet
- DM and HTN/CKD patient intake form
- DM and HTN/CKD patient follow-up form



PARTICIPANT SELECTION CRITERIA

The target audience is health professionals working at primary health care (health centers and primary hospitals) and general hospitals, including:

- BSc nurses,
- Health officers,
- General Practitioners, and
- Internists



TRAINER SELECTION CRITERIA

Qualified trainer/s for CKD training:

- TWG members who participated in the development of the CKD training manual
- Internists who are certified ToTs in CKD course.
- Individuals with basic training on CKD and basic facilitation skills
- Proficient clinician/s (General practitioners, Health Officers, BSc nurses) that are competent service provider/s for CKD and received Training of Trainers (TOTs).



METHODS OF EVALUATION

PARTICIPANT

Formative: Pretest, Exercise, progressive assessment

Summative: Post test

COURSE

- Daily Evaluation.
- End of course evaluation

CERTIFICATION CRITERIA

- A score of at least 70 % on the post course Knowledge Assessment for basic training and 80% on ToT.

Continuing Educational Unit (CEUs)= 15 CEUs

COURSE DURATION

- Three days



SUGGESTED CLASS SIZE

- Each Session/Room should accommodate no more than 25 participants
- Three trainers are required to facilitate this training.



Training Venue: Accredited CPD Center

COURSE SCHEDULE

Time	Sessions	Facilitator	Co facilitator
Day 1			
8:30-9:00AM	Registration of participants		
9:00-9:10 AM	Welcoming and keynote address		
9:10-9:40 AM	Pre-test on CKD		
9:40-10:00 AM	Course Goals and objectives		
10:00-10:15 AM	Tea/Coffee Break		
10:15AM-11:45PM	Chapter 1: Definition, spectrum of kidney disease, epidemiology, risk factors and prevention of CKD		
11:45AM-12:30PM	Chapter 2: Screening and diagnosis of CKD		
12:15-1:30 PM	Lunch		
1:30 – 2:45PM	Chapter 2: Screening and diagnosis of CKD...continued		
2:45-3:30 PM	Chapter 3: Comprehensive Management of CKD		
3:30 – 3:45 PM	Tea Break		
3:45-5:30PM	Chapter 3: Comprehensive Management of CKD...continued		
Day Two			
8:30-8:45AM	Recap of day one		
8:45 – 10:15AM	Chapter 3: Comprehensive Management of CKD...continued		
10:15-10:30 AM	Tea Break		
10:30AM-12:30PM	Chapter 4: Obstructive Uropathy		
12:30 – 1:30 PM	Lunch		
1:30-3:30PM	Chapter 5: Complications of CKD		
3:30-3:45 PM	Tea Break		
3:45-5:30PM	Chapter 5: Complications of CKD... continued		

Day Three			
8:30-8:45AM	Recap of day two		
8:45-10:00AM	Chapter 6: Nutrition in CKD		
10:00-10:15AM	Tea Break		
10:30AM-12:00 PM	Chapter 7: Management of major infections and routine vaccination in CKD		
12:00-12:30 PM	Chapter 8: CKD in special population		
12:30-1:30 PM	Lunch		
1:30-3:00 PM	Chapter 8: CKD in special population ... continued		
3:00-3:30 PM	Chapter 9: CKD Program Management		
3:30-3:45 PM	Tea Break		
3:45-5:00 PM	Chapter 9: CKD Program Management... continued		
5:00-5:30 PM	Post Test		

CHAPTER 1: DEFINITION, SPECTRUM OF KIDNEY DISEASE, EPIDEMIOLOGY, RISK FACTORS AND PREVENTION OF CKD



TIME ALLOCATED: 1 hour and 30 minutes.



Chapter description: This chapter describes the definition of CKD based on established criteria, explores the spectrum of kidney disease, reviews the standardized nomenclature, and discusses the epidemiology of CKD, with a focus on global trends and the burden of disease in Ethiopia. It also explains key risk factors for CKD with prevention strategies.

Primary objective

- Discuss the definition, epidemiology, risk factors, and prevention of CKD.

Enabling objectives

By the end of this chapter, participants will be able to:

- Define Chronic Kidney Disease (CKD)
- Explain the current nomenclature used in CKD
- Describe the spectrum of kidney disease.
- Summarize the global CKD prevalence, trends, and its socioeconomic impact.
- Describe the national burden of CKD
- Identify and explain modifiable and non-modifiable risk factors for CKD
- Discuss prevention strategies of CKD

Chapter Outline

- 1.1. Definition of CKD
- 1.2. Nomenclature of CKD
- 1.3. Spectrum of kidney disease
- 1.4. Epidemiology of kidney disease
- 1.5. Risk factors of CKD
- 1.6. Prevention of CKD
- 1.7. Chapter summary

1.1. DEFINITION OF CHRONIC KIDNEY DISEASES

CKD is defined as the presence of functional (estimated GFR < 60 ml/min/1.73m²) and/or structural abnormalities of the kidney persisting beyond 3 months, with implications for health.

Markers of Kidney damage (1 or more)	Albuminuria (ACR > 30 mg/g [> 3 mg/mmol]) Urine sediment abnormalities Persistent hematuria Electrolyte and other abnormalities due to tubular disorders Abnormalities detected by histology Structural abnormalities detected by imaging History of Kidney transplantation
Decreased GFR	GFR < 60 ml/min per 1.73 m ² (GFR categories G3a-G5)

Table 1: Criteria for CKD (either of the following present for a minimum of 3 months)

1.2. CURRENT CKD NOMENCLATURE USED BY KDIGO

CKD is classified based on Cause, Glomerular filtration rate (GFR) category (G1–G5), and Albuminuria category (A1–A3), abbreviated as CGA.

- CAUSE: Identify underlying causes such as DM, Hypertension, glomerulonephritis, polycystic kidney disease or obstructive uropathy.
- GFR Categories (G1-G5)
 - G1: GFR ≥ 90 mL/min/1.73m² (normal or high)
 - G2: GFR 60-89 mL/min/1.73m² (mildly decreased)
 - G3a: GFR 45-59 mL/min/1.73m² (mildly to moderately decreased)
 - G3b: GFR 30-44 mL/min/1.73m² (moderately to severely decreased)
 - G4: GFR 15-29 mL/min/1.73m² (severely decreased)
 - G5: GFR < 15 mL/min/1.73m² (kidney failure)
- Albuminuria categories (A1-A3)
 - A1: <30 mg/g (normal to mildly increased)
 - A2: 30-300 mg/g (moderately increased)
 - A3: >300 mg/g (severely increased)

CGA staging of CKD: examples of nomenclature			
Cause	GFR Category	Albuminuria category	Criterion for CKD
Diabetic kidney disease	G5	A3	Decreased GFR, Albuminuria
Idiopathic focal sclerosis	G2	A3	Albuminuria
Kidney transplant recipient	G2	A1	History of kidney transplantation
Polycystic kidney disease	G2	A1	Imaging abnormality
Vesicoureteral reflex	G1	A1	Imaging abnormality
Distal renal tubular acidosis	G1	A1	Electrolyte abnormality

Table 2: CGA staging of CKD: Adopted from KDIGO 2012 clinical practice guideline for the evaluation and management of CKD

The CGA classification facilitates:

- Disease severity and progression assessment
- Appropriate management strategies
- Standardized diagnosis, management and communication between providers

1.3. SPECTRUM OF KIDNEY DISEASE

Kidney disease encompasses a wide range of conditions that affect kidney function and structure. It is broadly categorized into acute kidney disease (AKD) and chronic kidney disease (CKD) with varying degrees of severity and progression.

Acute Kidney Disease (AKD): Refers to a spectrum of AKI or other acute kidney dysfunction lasting less than 3 months. It Includes:

- **AKI:** rapid decline in kidney function, often reversible (e.g., prerenal azotemia, acute tubular necrosis).
- **Non-AKI AKD:** Acute but non severe kidney dysfunction not meeting AKI criteria.

Common causes of AKD include Dehydration, sepsis, nephrotoxins, ischemia, or obstruction.

Prognosis: can resolve or progress to CKD

Chronic Kidney Disease (CKD): Long term (≥ 3 months) structural or functional kidney abnormalities with potential health implications. Common causes of CKD include diabetes mellitus, hypertension, chronic glomerulonephritis, Polycystic Kidney disease or obstructive uropathy.

Prognosis: progressive, potentially leading to ESRD

- **Acute-on-chronic kidney disease (AoCKD):** Acute deterioration of kidney function in a patient with pre-existing CKD Often triggered by superimposed AKI events like infections, dehydration, or nephrotoxic drugs.
- **End-Stage-Kidney-Disease (ESRD):** Advanced CKD (GFR<15ml/min/1.73m²) requiring Renal Replacement Therapy (RRT) in the form of Hemodialysis, peritoneal dialysis, or kidney transplantation.
- **Kidney Disease with Normal or high GFR:** Early stages of CKD or conditions like hyperfiltration in Diabetes, where GFR is preserved but kidney damage is present (e.g., proteinuria, imaging abnormalities)

Importance of spectrum awareness: Helps in early detection, intervention, and management to Slow disease progression; Differentiate between reversible (AKD) and irreversible (CKD) conditions and guides treatment from addressing acute causes to managing chronic complications.

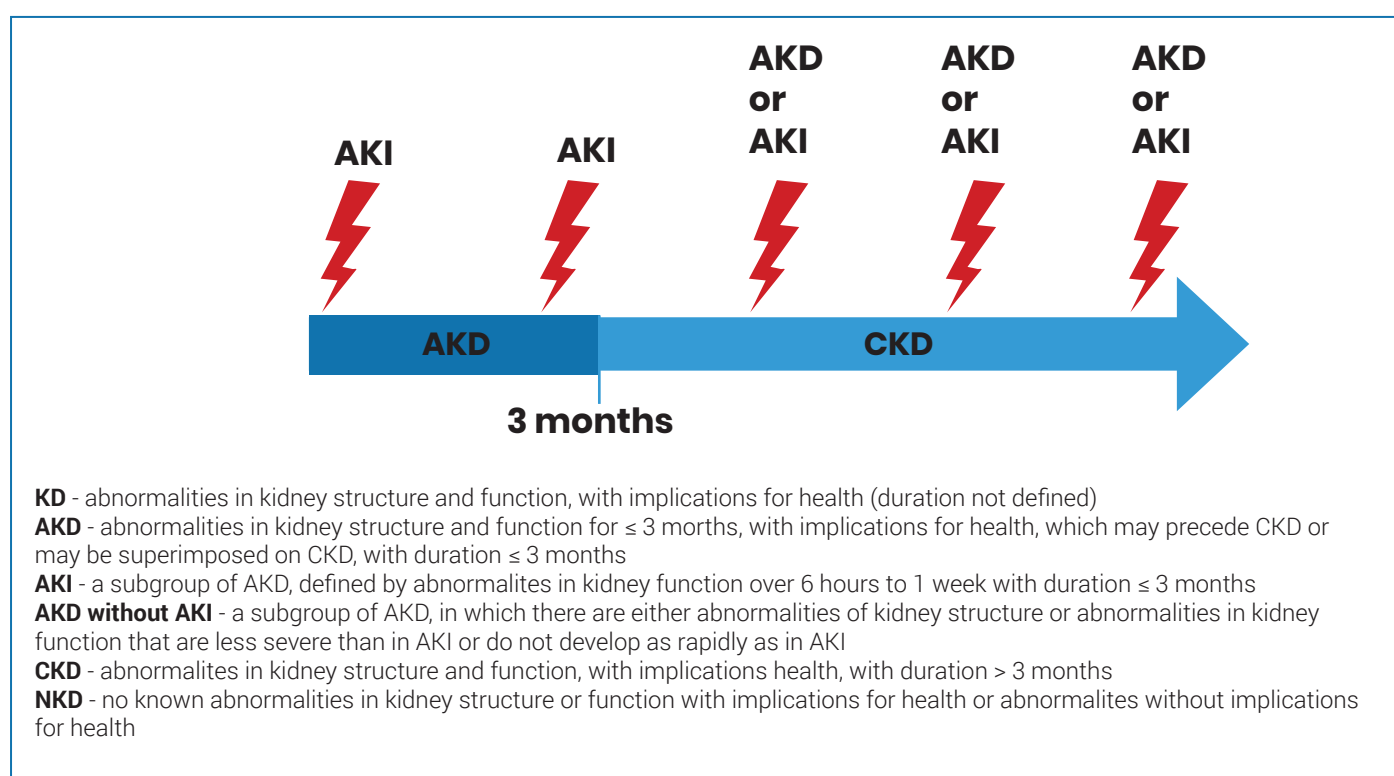


Figure 1: Conceptual model of the time course of AKI, AKD, and CKD. AKI, acute kidney injury; AKD, acute kidney disease and disorders; CKD, chronic kidney disease

Adopted from KDIGO 2021 consensus conference

1.4. EPIDEMIOLOGY OF CKD

Chronic kidney disease (CKD) is an increasingly global public health issue, affecting one in ten adults worldwide. Over 850 million people are living with kidney disease, which is roughly double the number of people who live with diabetes (422 million) and twenty times more than those with cancer (42 million) or HIV/AIDS (36.7 million).

Up to 78% of patients with kidney disease live in low- and middle-income countries which are the least equipped to deal with its consequences. In Sub-Saharan Africa, an estimated 13% of adults have some form of kidney disease (Stanifer, 2014). However, 90% of people with kidney disease do not know that they have the disease. This figure is higher in low-income countries like Ethiopia which do not routinely screen for kidney disease.

There is a need for early detection of CKD in LMICs in order to reduce the related morbidity, mortality and economic costs. Studies suggest that early detection and treatment of CKD is feasible even in low-resource settings. In sub-Saharan Africa, CKD more commonly affects individuals aged between 20 and 50 years, with the onset of end-stage renal disease (ESRD) occurring approximately 20 years earlier in populations of African descent compared with other ethnic groups in Western countries (45 years vs. 63 years).

The global increase in the incidence and prevalence of CKD is mainly driven by the rise in the prevalence of type 2 diabetes mellitus (T2D), hypertension, obesity and aging. Additionally, infectious diseases such as HIV contribute significantly to CKD in Sub-Saharan Africa.

ESRD requires renal replacement therapy (RRT) in the form of either dialysis or kidney transplant, both of which are costly to individuals and society, imposing a major burden on health systems.

A recent study in 12 LMICs reported a CKD prevalence of 14.3% in the general population and 36.1% in high-risk populations. These findings are consistent with those of a previous meta-analysis of 21 studies in LMICs, in which 13.9% of the general population had CKD.

A community-based study conducted among adults in Southwest Ethiopia showed a prevalence of CKD to be 7.4 %.

A 2016 cross-sectional study at the outpatient diabetes clinic of Dessie Referral Hospital, Northeast Ethiopia, reported a CKD prevalence of 26.3%. However, only 10.6% of individuals with CKD were informed of their condition by a healthcare professional.

A hospital-based study at the University of Gondar Hospital, Northwest Ethiopia, found CKD prevalence among diabetes patients to be 17.3% (using the MDRD equation) and 14.3% (using the CKD-EPI equation).

Another hospital-based cross-sectional study conducted at Jimma University among hypertensive and diabetes mellitus patients showed a prevalence of 26%.

A systematic review and meta-analysis estimated CKD prevalence among patients with chronic illnesses in Ethiopia to be 21.71%, with the highest prevalence recorded in Oromia (32.55%).

Chronic kidney disease has emerged as one of the leading causes of mortality worldwide and is among the few non-communicable diseases with an increasing death toll over the past two decades.

In 2017, CKD was the 12th leading cause of years of life lost globally, rising from 36th in 1990 and 19th in 2013. CKD is projected to become the fifth leading global cause of years of life lost by 2040.

The high prevalence and significant adverse impact of CKD necessitates enhanced efforts for prevention, early detection, and treatment. Early identification and management of CKD are critical to reducing cardiovascular disease risks, progression to ESRD, and mortality.

1.5. RISK FACTORS OF CKD

Activity 1.5. Case scenario



Instruction:

Following your facilitator's presentation, form small groups of four to six participants

Read the case scenario given below

Discuss and **answer** the questions among your group

Present to the larger group

Case: A 40-year-old female from a rural area in Ethiopia presents with newly diagnosed hypertension. She has no family history of kidney disease but works as a farmer and consumes a diet high in salt.

Discussion Questions:

1. What risk factors for CKD are present in this patient?
2. How might socioeconomic and environmental factors in Ethiopia contribute to her risk of CKD?
3. What public health measures could reduce CKD prevalence in similar populations?

Time: 10 minutes

CKD has multiple risk factors, categorized into modifiable and non-modifiable types. Identifying these factors is critical for prevention, early detection, and management.

Modifiable risk factors

These factors can be addressed through lifestyle changes, medical intervention, or early treatment.

- **Medical conditions:** DM, hypertension, cardiovascular disease, obesity, dyslipidemia, autoimmune diseases, infections, and exposure to nephrotoxins
- **Lifestyle factors:** Smoking, unhealthy diet, sedentary lifestyle, and dehydration.

Non-modifiable risk factors

- Demographics
 - Age:** CKD risk increases with age, especially after 60 years.
 - Sex:** Males may have a slightly higher risk, though females often progress faster in advanced stages.
- **Genetics and family history:** Polycystic kidney disease (PKD), Family history of CKD, and APOL1 genetic variants
- **Ethnicity and socioeconomic factors:** Higher prevalence in African Americans, Hispanics, and South Asians
- **Low socioeconomic status:** limited access to healthcare, unhealthy diets, and delayed diagnosis.

Emerging risk factors:

- **Environmental factors:** exposure to heavy metals (e.g., lead, cadmium) and air pollution.
- **Gut microbiome dysbiosis:** altered gut bacteria linked to systemic inflammation and CKD progression.
- **AKI:** previous AKI episodes increase the risk of CKD development.

Domains	Examples
Common risk factors	Hypertension Diabetes Cardiovascular disease (including heart failure) Prior AKI/AKD
People who live in geographical areas with high prevalence of CKD	Areas with endemic CKD Areas with the high prevalence of APOL1 genetic variants Environmental exposures
Genitourinary disorders	Structural urinary tract disease Recurrent kidney calculi
Multisystem diseases/chronic inflammatory conditions	Systemic lupus erythematosus Vasculitis HIV HBV HCV
Iatrogenic (related to drug treatments and procedures)	Drug-induced nephrotoxicity and radiation nephritis
Family history or known genetic variant associated with CKD	Kidney failure, regardless of identified cause Kidney disease is recognized to be associated with genetic abnormality (e.g., PKD, APOL1-mediated kidney disease, and Alport syndrome)
Gestational conditions	Preterm birth Small gestational size Pre-eclampsia/eclampsia
Occupational exposures that promote CKD risk	Cadmium, lead, and mercury exposure Polycyclic hydrocarbons Pesticides

Table 3: Risk factor for CKD (Adopted from KDIGO 2024 Clinical practice guideline for the evaluation and management of CKD)

1.6. PREVENTION OF CKD

Activity 1.6. Individual reflection



1. What is CKD?
2. Could you list risk factors for CKD?

Time Allocated: 10 minutes

Prevention of CKD focuses on addressing modifiable risk factors, promoting early detection, and adopting a kidney-friendly lifestyle. Intervention at various stages across the course of the disease can prevent or delay progression to the next stage. The strategy identifies three broad levels of prevention: primary, secondary, and tertiary.

Primary prevention measures aim to eliminate or reduce exposure to factors that cause ill health or disease. For CKD, this involves strategies to reduce the incidence and prevalence of risk factors such as diabetes and high blood pressure, in order to reduce the number of people at risk of developing CKD. Primary prevention strategies for CKD include:

- Screening high - risk groups for CKD
- Public awareness and education:
 - Health campaigns: educate the public about CKD risk factors, symptoms, and prevention through community outreach, media, and schools.
 - Lifestyle awareness: promote healthy dietary practices, regular exercise, and avoidance of harmful practices like smoking and alcohol misuse.
- Focus on High-Risk Groups: Target individuals with diabetes, hypertension, obesity, or a family history of CKD
- Strengthen healthcare infrastructure`
 - Expand access to primary care: establish NCD clinics for early detection and management of CKD risk factors
 - Routine screening programs: test blood pressure, blood glucose, and urine protein at primary health care centers; offer affordable or free screening for at-risk populations.
 - Capacity building: train healthcare providers in early identification and management of CKD
- Control of Non-communicable Diseases (NCDs)
 - Diabetes management: Improve access to insulin and oral antidiabetic medications
 - Educate patients about blood sugar monitoring and glycemic control
 - Hypertension control: Ensure availability of affordable antihypertensive drugs
 - Promote regular BP monitoring at health posts and health centers.
- Address Infectious Diseases
 - Early diagnosis and treatment of recurrent or chronic UTIs to prevent CKD

- Improve access to ART and hepatitis vaccination programs
- Implement prevention and treatment programs for parasitic infections linked to kidney damage.

Secondary prevention: Consists of measures for early detection of disease to allow prompt and effective intervention to prevent the disease from becoming established.

Early detection and management of the disease in its initial stages help to prevent progression and complication:

- This approach focuses on minimizing CKD progression and addressing underlying causes,
- Early diagnosis and monitoring
- Routine screening: nationwide for at-risk populations can help identify CKD in its early stages.
- Education and Awareness: raising public awareness about the importance of kidney health and the risks of CKD through media campaigns, health programs, and community outreach.
- Managing underlying conditions: Hypertension control, Diabetes management, Lipid management, Lifestyle modifications, Dietary changes, Physical activity and Smoking cessation
- Pharmacologic interventions:
 - RAAS inhibitors: can help protect the kidneys by reducing proteinuria and lowering BP
 - SGLT2 Inhibitors: have shown promise in preventing CKD progression and improving heart and kidney outcomes.
- Preventing and Managing complications
 - Anemia Management: Managed with ESAs and iron supplementation
 - Electrolyte Imbalance: Monitoring and correcting electrolyte disturbances such as hyperkalemia, hyperphosphatemia, and metabolic acidosis is critical
 - Cardiovascular risk reduction: Statin therapy, antiplatelet drugs, and proper management of hypertension are vital in reducing heart disease and stroke risks
- Access to care and specialist support:
 - Primary care provider involvement: Strengthening the role of primary healthcare providers in managing CKD patients through training and resource allocation.
 - Referral to nephrologists: Early referral of advanced CKD cases for specialized care and providing interventions like dialysis or kidney transplantation if needed

Tertiary prevention strategies: Focuses on managing advanced stages of CKD to prevent complications, improve quality of life, and delay the progression to ESRD. At this stage, the goal is to minimize kidney damage, manage symptoms, and prepare for renal replacement therapies (such as dialysis or kidney transplantation) and provide Palliative care for end-stage renal disease.

Primary prevention	Secondary prevention	Tertiary prevention
AIM: Prevent and minimize risk of occurrence of CKD Strategies • Health educates about CKD and its risk factors • Promote healthy lifestyle such as increased physical activity, cessation of smoking, reduction of alcohol intake and dietary salt restriction • Screen and manage risk factors for CKD like DM/ HTN, SCD, infections • Restrict use of nephrotoxic medicines such as NSAIDs, aminoglycosides like gentamycin and herbal medicines • Encourage mothers to adequately attend antenatal care to prevent low birth weight, prematurity, neonatal infections, childhood malnutrition	AIM: Detect early & treat CKD Strategies: • Screen those at risk of CKD • Offer appropriate treatment for CKD • Screen and treat for risk factors for CKD	AIM: Prevent progression of & treat complications of CKD Strategies: • Treat and control hypertension and proteinuria using appropriate medication • Screen and treat N complications

Table 4: prevention strategies in low- and middle-income countries (LMICs)
Adopted from kidney international reports 2020

1.7. CHAPTER SUMMARY

- Chronic Kidney Disease (CKD) is a progressive condition characterized by a gradual decline in kidney function over time.
- CKD is defined based on a persistent reduction in the glomerular filtration rate (GFR) and/or evidence of kidney damage lasting for at least three months.
- Understanding the nomenclature and classification of CKD, as outlined by standard guidelines like KDIGO, is crucial for uniform diagnosis and management.
- The spectrum of kidney disease ranges from normal kidney function to end-stage kidney disease (ESKD), with acute-on-chronic kidney disease representing an intermediate but distinct condition.
- Epidemiologically, CKD poses a significant global health burden, with increasing prevalence and substantial socioeconomic impacts.
- In Ethiopia, CKD is an emerging public health challenge, fueled by factors like limited awareness, late diagnosis, and resource constraints.
- Preventing CKD requires a multifaceted approach, emphasizing early detection, lifestyle modification, and effective management of comorbid conditions like Ethiopia.

CHAPTER 2: SCREENING AND DIAGNOSIS OF CHRONIC KIDNEY DISEASES



TIME ALLOCATED: 2 hours and 10 minutes



Chapter description: In this chapter, risk factors for chronic kidney disease, objectives of screening and screening tools will be covered. CKD grading will be discussed using glomerular filtration rate (GFR) and albuminuria categories. The teaching methods involve interactive lectures, seminar presentations and case scenarios with small group discussions.



Primary objective

By the end of this chapter the participants will be able to:

- Diagnose Chronic kidney disease

Enabling objectives

- Define chronic kidney disease
- Describe objectives of screening for CKD
- Interpret UACR, urine dipstick and serum creatinine test results
- Describe the 5-step plan for CKD evaluation and referral
- Describe grading of CKD

Chapter Outline:

- 2.1. Screening of CKD
- 2.2. Diagnostic evaluation of CKD
- 2.3. Grading of CKD
- 2.4. Chapter summary

2.1 SCREENING OF CKD



Activity 2.1

Case Scenario

Instruction:

- Make small groups of 4 to 6 participants
- Read the case study in your manual
- Write your answers on a flip chart
- Present your answers to the larger group

W/ro Tenagne is a 55-year-old woman who is on follow up for Type II diabetes. She is taking Metformin 1gm PO BID. She claims she has good glucose control. She wants to know if she needs to have a screening for CKD.

- a) Who needs to be screened for CKD?
- b) What are the screening tools?
- c) How frequently should someone be screened for CKD?

Time: - 15 minutes

Early kidney disease is usually asymptomatic, and clinicians do not always check for it. Decreased kidney function is closely connected with cardiovascular disease, diabetes, high blood pressure, and other adverse health conditions.

The morbidity and mortality from CKD and its associated consequences, such as elevated rates of cardiovascular disease, hyperlipidemia, anemia, and bone and mineral disease, could be significantly decreased with targeted early detection.

Screening results can guide therapy adjustments, including better lifestyle choices and medication dosages.

A strategy to manage this burden must include early kidney disease detection programs, evidence-based treatment of CKD and its associated illnesses, such as diabetes and hypertension, and education of health professionals and populations at risk.

2.1.1 Objectives of screening

Early detection of any chronic disease, including CKD, provides greater opportunities to reduce morbidity as treatments can be initiated earlier in the disease course. Given the asymptomatic

progression of CKD, systematic testing of people with risk factors for CKD is the only method that would detect CKD at early stages and allow the initiation of appropriate treatments.

Early detection and management reduce the burden of CKD by improving the quality of care and quality of life for all patients. Given the high healthcare costs of kidney failure, especially in LMICs, early screening offers economic benefits.

2.1.2. Criteria for CKD screening

Test people at risk for and with chronic kidney disease (CKD) using both urine albumin measurement and assessment of glomerular filtration rate (GFR).

The highest priority for screening is those with the following conditions:

- Hypertension,
- Diabetes
- CVD, including heart failure.
- Prior acute kidney injury/recurrent AKI

In addition to the above group, patients with the following conditions should also be tested:

- Family history of kidney disease
- Multisystem diseases/chronic inflammatory conditions
- Age above 60 years.
- Obesity

2.1.3. Screening tools

Screening tools for CKD

- I. Urine albumin-to-creatinine ratio
- II. Urine dipstick
- III. Serum creatinine
- IV. Serum cystatin c (if available)
- V. Ultrasound

I. Urine albumin to creatinine ratio

Albumin-to-creatinine ratio (ACR) is the first method of preference to detect elevated protein.

The recommended method to evaluate albuminuria is to measure urinary ACR in a spot urine sample.

ACR is calculated by dividing the albumin concentration in milligrams by the creatinine concentration in grams.

Albuminuria categories in CKD		
Category	ACR (mg/g)	Terms
A1	<30	Normal to mildly increased
A2	30-300	Moderately increased
A3	>300	Severely increased

Table 5: Albuminuria categories

II. Urine dipstick

It is the most inexpensive method to detect albuminuria. A routine dipstick is not sensitive enough to detect small amounts of urine protein. (Refer table 3)

Activity 2.1.3

Case scenario A

Instruction:

- Make small groups of 4 to 6 participants
- Read the case study on your manual

Ato Abera is a 65-year-old man. He has been diabetic for the past 3 years. He is on metformin 500mg PO BID and glibenclamide 5m PO daily. He is a merchant and is currently able to carry out his daily activities. He claims he is adherent to his medication and follow-up. On his last visit urine dipstick and UACR were done as part of his follow-up and the results were urine protein +1 and UACR 250mg/g respectively. Six months back, he had the same result.

Discussion Questions:

- Discuss among your group which category his result of albuminuria falls under.
- Does this patient meet the definition for CKD? Why or why not?

Time: 15 minutes



III. Serum creatinine

- In adults at risk for CKD, it is recommended to use creatinine-based estimated glomerular filtration rate (eGFR_{Cr}).
- GFR is the product of the average filtration rate of each nephron, the filtering unit of the kidneys, multiplied by the number of nephrons in both kidneys.
- The normal level for GFR is approximately:
 - 130 mL/min/1.73 m² for men
 - 120 mL/min/1.73 m² for women,

With considerable variation among individuals according to age, sex, body size, physical activity, diet, blood pressure, glycemia, pharmacotherapy, and physiologic states such as pregnancy

- Creatinine is derived from:
 - The metabolism of phosphocreatine in muscle,
 - From dietary meat intake or
 - Creatine supplements.
- Creatinine generation is proportional to muscle mass, which can be estimated based on:
 - Age,
 - Sex, and
 - Body size,
 - But many other factors can affect the creatinine generation.

How can we calculate an estimated glomerular filtration rate (eGFR_{Cr})?

- **Recommended equation is**
 - 2021 CKD-EPI creatinine equation
 - <https://www.mdcalc.com/calc/3939/ckd-epi-equations-glomerular-filtration-rate-gfr>

Activity 2.1.3



Case scenario B continued from activity 2.1.3 case scenario A

Instruction:

- Make small groups of 4 to 6 participants
- Read the case study on your manual
- Practice with the equation
- Refer to the Heat Map

Following the UACR result, you sent a serum creatinine test for Ato Abera. The result is 1.4 mg/dL.

- a) Using the 2021 CKD-EPI creatinine equation calculate his eGFR.
- b) Considering his eGFR and UACR, stage his CKD (refer to the Heat map)
- c) Discuss the importance of identifying the grade in guiding management and predicting progression.

Time: 15 minutes

IV. Ultrasound

This imaging test is recommended to evaluate disorders of the kidneys, ureters, and the urinary bladder.

2.1.4 Frequency of screening

Albuminuria or proteinuria measurement and serum creatinine measurement and eGFR calculation should be done for all:

Diabetes

- CKD screening should begin at the time of diagnosis of type 2 diabetes (T2D) and be repeated **annually**, as evidence of CKD is often already present at diagnosis.
- For type 1 diabetes (T1D), CKD screening is recommended starting 5 years after diagnosis.

Hypertension - CKD screening should begin at the time of diagnosis and be repeated **annually**

Cardiovascular disease - CKD screening should begin at the time of diagnosis and be repeated **annually**

CKD screening should be performed **every two years** for the following individuals:

- Those who are obese
- Individuals with a family history of kidney disease
- Adults over the age of 60 years

For the initial evaluation of UACR >300-700 mg/g or pcr > 500-1000mg/g or 24-hour proteinuria >0.5-1 gm, the result should be confirmed by early morning urine sample of UACR or UPCR

IF initial UACR is >700mg/g or UPCR >1000mg/g, a repeat sample is not needed.

2.2 DIAGNOSTIC EVALUATION OF CKD

- In individuals without risk factors for CKD, the pretest probability of CKD is low. Therefore, any unexpected results should be verified before making a diagnosis of CKD.
- In people with risk factors for CKD, there is a higher probability that the person does have CKD even with an unexpected finding. Subsequent testing should be performed to confirm the diagnosis and to complete the evaluation, as required.

Activity 2.2



Role play

Instruction:

- Make two groups
- Two volunteers to act in the two parts from each group
- Read the case study on your manual

Ato Dagne is a 55-year-old man who is a security guard in a private company. He has been recently diagnosed with type 2 DM and started metformin. He comes to your clinic for a refill and asks you about complications of diabetes. What do you recommend in terms of screening for CKD?

Points to be raised

- Indication for screening for CKD
- Frequency of screening for CKD
- Objective of screening for CKD

Time -15 minutes

Evaluation of chronicity

Proof of chronicity (duration of a minimum of 3 months) can be established by:

- I. Review of past measurements/estimations of GFR;
- li. Review of past measurements of albuminuria or proteinuria and urine microscopic examinations.
- lii. Imaging findings such as reduced kidney size and reduction in cortical thickness;
- Iv. Kidney pathological findings such as fibrosis and atrophy;
- V. Medical history, especially conditions known to cause or contribute to CKD;
- Vi. Repeated measurements within and beyond the 3-month period.

Do not assume chronicity based upon a single abnormal level for eGFR and ACR, as the finding could be the result of a recent acute kidney injury.

Activity 2.2.1



Case Scenario

Instructions: You will be divided into 4 small groups and discuss and answer the questions based on case scenario below.

W/ro Hiwot is a 60-year-old female who has been hypertensive for 7 years now. She has had follow-up in different facilities in the last few years. She is taking hydrochlorothiazide and Nifedipine for 2 years.

Questions

- a. Is she a candidate for screening for CKD?
- b. If yes, which test would you like to order?

Time 10 minutes

Activity 2.2.2



Case Scenario

Instruction: Based on your case study on activity 2.2.1, the following additional information is provided. Return to your previous small group discuss and answer the following questions.

Her results are as follows:

- Urine dipstick for protein =+ 2
- UACR = 350 mg/g
- Serum creatinine = 2 mg/dL
- Abdominal ultrasound showed left kidney measuring 8cm and right kidney 8.5 cm

- a. What is your diagnosis?
- b. Is this CKD?
- c. What evidence of chronicity is there in this case?

Time 15 minutes

A 5-step plan for CKD evaluation and referral

1. Know the criteria for CKD

- Abnormalities of kidney structure or function, present for > 3 months, with implications for health
- Either of the following must be present for > 3 months:
 - Markers of kidney damage (one or more)
 - $\text{GFR} < 60 \text{ mL/min/1.73m}^2$

2. Recognize risk factors. CKD risk factors include, but are not limited to the following:

- Diabetes
- Hypertension
- Family history of kidney disease
- Age 60 or older (GFR declines normally with age)
- Race/U.S. ethnic minority status - African Americans, Hispanics, Asians/Pacific Islanders, and American Indians
- Frequent NSAID use
- History of acute kidney injury

3. Screen for CKD with simple tests

- Urine Protein
- Spot urine for albumin-to-creatinine ratio (ACR) to detect albuminuria
- Serum creatinine to estimate glomerular filtration rate (GFR)

4. Classify CKD to guide testing and treatment

- Assign GFR category
- Assign albuminuria category
- Determine prognosis of CKD based on GFR and albuminuria category

5. Implement a clinical action plan based on patient's CKD classification

- Consider co management with a nephrologist if the clinical action plan cannot be carried out
- Refer to a nephrologist when $\text{GFR} < 30 \text{ mL/min/1.73 m}^2$ or $\text{ACR} > 300 \text{ mg/g}$

2.3. GRADING OF KIDNEY DISEASE



Activity 2.3: Case scenario

Instruction

- Form small groups of 4 to 6 participants
- Read the case scenario in your manual
- Write your answers on a flip chart
- Present your group work

Case

A 52-year-old male presents to the clinic with fatigue and mild ankle swelling for the last three months.

His laboratory tests reveal:

- GFR: 50 mL/min/1.73 m²
- Urine ACR: 200 mg/gm
- Blood pressure: 150/90 mmHg

Discussion Questions:

1. Does this patient meet the definition of CKD? Why or why not?
2. What stage of CKD does he fall into, based on GFR and albuminuria?
3. Why is staging important in guiding management and predicting disease progression?

Time: 10 minutes

Grading of kidney disease helps to assess disease severity, progression risk, and inform management Strategies. Staging of chronic kidney disease requires an estimation of the glomerular filtration rate (derived from serum creatinine or cystatin C) and the measurement of albuminuria.

According to the Kidney Disease Improving Global Outcomes (KDIGO) CKD working group, kidney disease is staged according to eGFR (G1 to G5) and severity of albuminuria (A1, A2, A3)

KDIGO: Prognosis of CKD by GFR, albuminuria categories and urine dipstick test				Persistent albuminuria categories Description and range		
				A1	A2	A3
				Normal to mildly increased	Moderately increased	Severely increased
				<30 mg/g <3 mg/mmol	30-300 mg/g 3-30 mg/mmol	>300 mg/g >30 mg/mmol
				Negative	Trace/+	++ or greater
GFR categories (ml/min/1.73 m ²) Description and range	G1	Normal or high	≥90			
	G2	Mildly decreased	60-89			
	G3a	Mildly to moderately decreased	45-59			
	G3b	Moderately to severely decreased	30-44			
	G4	Severely decreased	15-29			
	G5	Kidney failure	<15			

Green: low risk (if no other markers of kidney disease, no CKD); Yellow: moderately increased risk; Orange: high risk; Red: very high risk. GFR, glomerular filtration rate.

Figure 2: Adopted from KDIGO 2024 clinical practice guideline for the evaluation and management of CKD

NB: For purposes of this training manual, proteinuria measured by urine dipstick can be used in cases where the capacity to determine albuminuria is not available

Categories	UACR (mg/gm)	Protein reagent strip	PCR (mg/gm)
Normal to mildly increased	< 30	Negative	< 150
Moderately increased	30- 300	Trace/+	150- 500
Severely increased	<300	+ + or greater	>500

Table 6: Urine dipstick equivalence of UACR

Clinical implications of staging:

- G1-G2 and A1/Negative dipstick protein: often asymptomatic; focus on monitoring and addressing risk factors (e.g., hypertension, diabetes)
- G3a-G3b and A2/+ Trace dipstick protein: Increased risk of complications such as cardiovascular disease; early intervention is crucial.
- G4-G5 and A3/++ or greater dipstick protein: Advanced CKD; preparation for renal replacement therapy (dialysis or transplantation) may be necessary

2.4 CHAPTER SUMMARY

- Early kidney disease is usually asymptomatic
- Because CKD progresses asymptotically, the only way to identify CKD early and start the right treatments is to systematically test those who have risk factors for the disease.
- Those with Diabetes and hypertension are given top priority for screening:
- Recommended equation for estimated glomerular filtration rate (eGFR_{cr}) 2021 CKD-EPI creation
- Annual screening for patients with diabetes, Hypertension, and cardiovascular disease
- Screening tools for CKD are Urine albumin to creatinine ratio, Urine dipstick, Serum creatinine, Ultrasound and Serum cystatin c (if available)

CHAPTER 3: COMPREHENSIVE MANAGEMENT OF CKD



Time Allocated: 4 hours



Chapter description: This chapter describes the comprehensive approach to management of CKD



Primary objective:

By the end of this chapter, participants will be able to describe a comprehensive approach to management of CKD.

Enabling Objectives

By the end of this chapter, participants will be able to:

- List major components of comprehensive CKD management;
- Describe the strategies of delaying CKD progression;
- Discuss strategies of CVD Risk reduction and RFs management;
- Explain optimal management of hypertension and DM in CKD patients;
- Describe causes of further injury to the kidneys in patients with CKD;
- Describe the criteria for timely referral of CKD patients to nephrologists;
- Discuss components of palliative care in advanced CKD patients.

Chapter outline:

- 3.1. Major components of CKD management.
- 3.2. Strategies of delaying CKD progression.
- 3.3. Strategies of CVD Risk reduction and RFs management.
- 3.4. Optimal management of hypertension and DM in CKD patients.
- 3.5. Causes of further injury to the kidneys in patients with CKD.
- 3.6. Nutritional and psychosocial support
- 3.7. Criteria for timely referral of CKD patients to nephrologists.
- 3.8. Palliative care in advanced CKD patients.
- 3.9. Chapter summary

3.1. MAJOR COMPONENTS OF CKD MANAGEMENT

Introduction

CKD is a complex disease with deleterious impacts on patients and their families. A wide range of preventive and therapeutic interventions ranging from lifestyle modification to renal transplantation are required for optimal outcomes. It needs a multi-disciplinary team approach ranging from primary health care workers to nephrologists. A key goal for healthcare providers should be to identify people at risk and to start preventive strategies and treatments early in the course of CKD to maximize potential benefits.

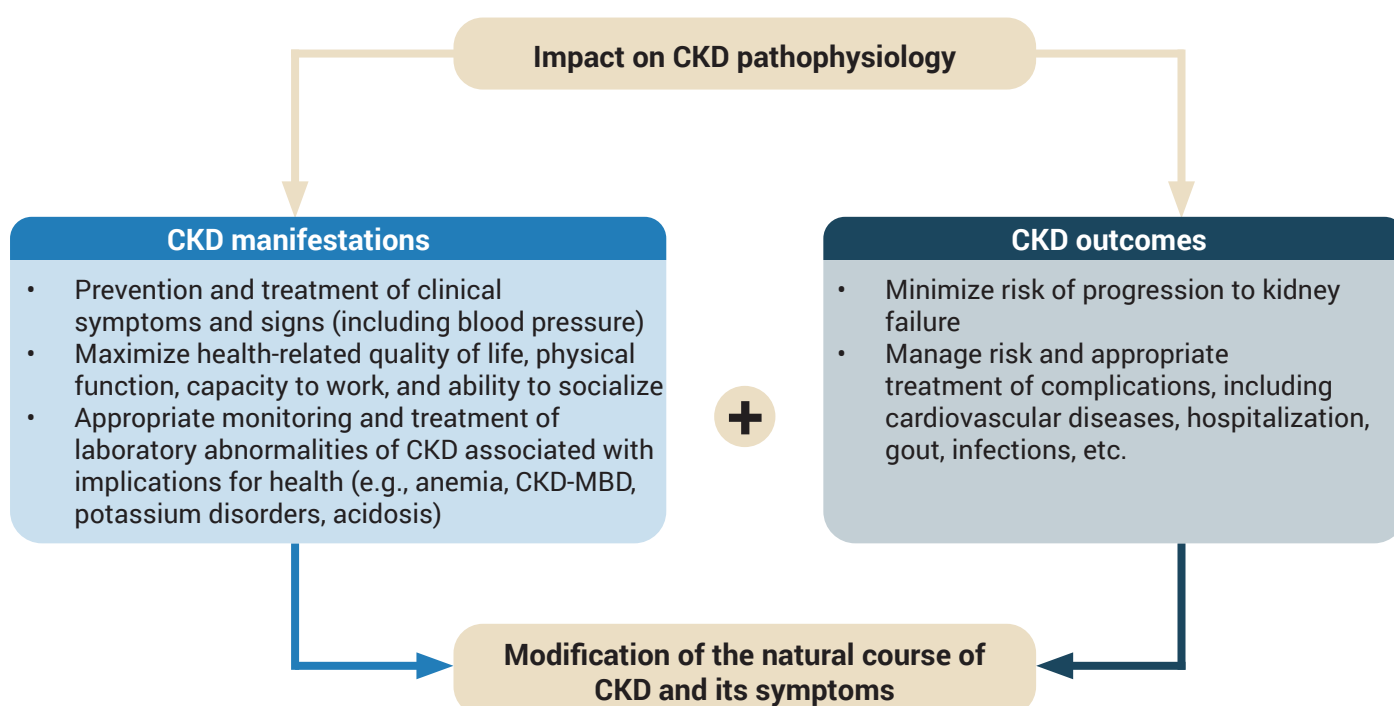


Figure 3: chronic kidney disease (CKD) treatment and risk modification. CKD-MBD, chronic kidney mineral disease, mineral and bone disorders.

Comprehensive CKD Management Includes:

- Screening and preventive strategies (see chapter one and chapter two).
- Etiologic diagnosis and staging (see chapter two).
- Delaying of disease progression. Main interventions that delay the progression of CKD include:
 - Optimal management of RFs and CVD Risk reduction.
 - Optimal management of hypertension.
 - Optimal management of DM.
 - Lipid management
 - Avoidance of further injury to the kidneys.

- Prevention and optimal management of CKD complications (see chapter 5).
- Psychosocial and nutritional support (see chapter 6 for further reading).
- Early referral to nephrologist.
- Preparation for renal replacement therapy (RRT) (see chapter 5).
- Palliative care

Different stages of CKD have different risks of progression; and clinical management action plan depends on the stage of CKD (See table 7).

Stage	Description	GFR (mL/min/1.73 m ²)	Action
	At increased risk	≥ 90	Screening CKD risk reduction
1	Kidney damage with normal or ↑↑GFR	≥ 90	Diagnosis and treatment. Treatment of comorbid conditions. Slowing of progression. Cardiovascular disease risk reduction.
2	Kidney damage with mildly ↓GFR	60-89	Estimating progression.
3	Moderately ↓GFR	30-59	Evaluating and treating complications.
4	Severely ↓GFR	15-29	Preparation for kidney replacement therapy.
5	End-stage renal disease (ESRD)	< 15 (or dialysis)	Replacement (if uremia is present).

Table 7: Stages of CKD: A clinical action plan



Activity 3.1

Group Discussion:

Instruction:

- Your facilitator will divide you into two small groups; select group presenter.
- Discuss and write down major components of comprehensive CKD management to delay CKD progression.
- Make presentation of your answers for the larger group.

Time allowed: 10 minutes

3.2 DELAYING PROGRESSION OF CKD TO ESRD

Early diagnosis of CKD and delaying progression to ESRD are vital in the management of CKD.

These measures reduce costs and improve patients' quality of life and survival. Risk factors associated with CKD progression, CVD, and other CKD complications are highly interrelated, and so is their management.

Main interventions that **delay the progression of CKD** include:

- Optimal management of RFs and CVD Risk reduction;
- Optimal management of hypertension;
- Optimal management of diabetes mellitus (DM);
- Lipid management
- Avoidance of further injury to the kidneys;
- Prevention and optimal management of CKD complications (see chapter 5);
- Psychosocial and nutritional support (see chapter 6 for further reading).
- Early referral to nephrologist

3.3 OPTIMAL MANAGEMENT OF RFS AND CVD RISK REDUCTION

CKD remains largely asymptomatic; the symptoms develop slowly. Screening for the RFs and for renal dysfunction is very important. Early identification of at-risk patients and preventive interventions improve outcomes. People with CKD are at increased risk of CVD, a key feature of which is structural heart disease, heart failure, and sudden death. Up to 50% of patients with stage 5 CKD have cardiovascular disease. In light of the high prevalence of cardiovascular disease and increased perioperative morbidity in patients with CKD, a thorough cardiovascular risk assessment is indicated.

Increased risk of atherosclerotic disease also accompanies CKD. These risks increase progressively as eGFR declines. Risk of death from CVD exceeds the risk of progression to kidney failure for the majority of people with CKD.

The diagnosis of cardiac disease can be more complex and challenging in CKD, with many standard tests needing careful consideration in people with CKD. In people with CKD, the same principles should be used to manage atherosclerotic risk as in people without CKD.

Prevention of ASCVD should consider non-pharmacologic interventions (dietary, physical activity, lifestyle etc) and pharmaceutical interventions which target traditional cardiovascular risk factors (e.g., BP and dyslipidemias), as well as CKD-MBD, which accelerates vascular calcification.

We recommend oral low dose aspirin for prevention of recurrent ischemic cardiovascular disease events (i.e., secondary prevention) in people with CKD and established ischemic cardiovascular disease (1C). We recommend the use of non-vitamin K antagonist oral anticoagulants (NOACs) in preference to vitamin K antagonists (e.g. warfarin) for thromboprophylaxis in atrial fibrillation in people with CKD G1–G4 (1C).

Non-pharmacologic Management of CKD (Lifestyle Modification)

Lifestyle modification is the first line modality in management of all people diagnosed with CKD. Health care should emphasize lifestyle changes among patients with CKD to delay the progression of the disease. These lifestyle changes include stopping tobacco use, ensuring proper nutrition, limiting alcohol use, and increasing physical activity.

The healthcare worker should effectively communicate to patients and their care givers about health improvement goals, diet, and exercise to slow the progression of CKD. Encourage people with CKD to undertake physical activity compatible with cardiovascular health, tolerance, and level of frailty; achieve an optimal body mass index (BMI); and not to use tobacco products. Referral to providers and programs (e.g., psychologists, renal dietitians or accredited nutrition providers, pharmacists, physical and occupational therapy, and smoking cessation programs) should be offered were indicated and available.

Physical Activity:

We recommend that people with CKD be advised to undertake moderate intensity physical activity for a cumulative duration of at least 150 minutes per week, or to a level compatible with their cardiovascular and physical tolerance (1D). Recommendations for physical activity should consider age, presence of other comorbidities, and access to resources. People with CKD should be advised to avoid sedentary behavior. For people at higher risk of falls, healthcare providers should provide advice on the intensity of physical activity (low, moderate, or vigorous) and the type of exercise (aerobic vs. resistance, or both). Physicians should consider advising/ encouraging people with obesity and CKD to lose weight.

Physical activity is beneficial in patients with CKD through slowing down the development and progression of disease. Patients should have an individualized exercise program based on risk factors, comorbidities, and stage of disease. Before any exercise program, patient should be assessed for safety of the program:

- We recommend that people with CKD be advised to undertake moderate- intensity physical activity for a cumulative duration of at least 150 minutes per week, i.e. at least 30 minutes of mild to moderate exercise per day for 5 days per week.

-
- Recommendations for physical activity should consider age, ethnic background, presence of other comorbidities, and access to resources.
 - People with CKD should be advised to avoid sedentary behavior. For people at higher risk of falls, healthcare providers should provide advice on the intensity of physical activity (low, moderate, or vigorous) and the type of exercise (aerobic vs. resistance, or both).
 - Physicians should consider advising/encouraging people with obesity and CKD to lose weight.

Dietary Care/Nutrition:

CKD patients are at risk of malnutrition for a number of reasons. Some diets are also associated with worsening renal dysfunction. The optimal renal diet should be composed of low Na⁺, low K⁺, low phosphorus, and limited protein.

Advise people with CKD to adopt healthy and diverse diets with a higher consumption of plant-based foods compared to animal-based foods and a lower consumption of ultra processed foods. Plant-based diets use proportionately more plant-based food. Diets such as Dietary Approaches to Stop Hypertension (DASH) are rich in fruits, vegetables, whole grains, and low-fat dairy foods. A Mediterranean diet pattern is built around vegetables, fruits, herbs, nuts, beans, whole grains, and seafood but also includes moderate amounts of dairy, meat, and eggs.

Use renal dietitians or accredited nutrition providers to educate people with CKD about dietary adaptations regarding sodium, phosphorus, potassium, and protein intake, tailored to their individual needs, and severity of CKD and other comorbid conditions. (For detailed nutrition recommendations refer Chapter 6)

Smoking and alcohol cessation should be part of the management process.

Health Education (healthy life style promotion)

All patients and their caretakers at all levels of care should be empowered with information concerning the CKD including causes, risk factors, symptoms, treatment and possible complications and prognosis. The ministry of health recommends linkage of patients to CKD support care networks and provision of information about kidney disease using available media platforms. The healthcare worker should effectively communicate to patients and their care givers about health improvement goals, diet, and exercise to slow the progression of CKD.

3.4. OPTIMAL CONTROL OF HYPERTENSION AND DM IN CKD PATIENTS

3.4.1. Optimal control of hypertension



Activity 3.4.A. Case Scenarios

Instruction: Read the following case scenarios and discuss the points mentioned in teams. Your instructor will create groups of three each. Select one individual to present the group's opinion after discussion.

Case: A 60-year-old female patient with known hypertension, who has been on irregular follow-up and was not taking medications, presented with persistent high blood pressure readings ranging between 150/90 mmHg and 170/100 mmHg. The patient has bilateral leg swelling. Serum creatinine = 1.8 mg/dl, Urea = 80 mg/dl; 24 hours urine protein = 2,300 mg/24 hours in 2 liters of urine.

1. What additional investigations do you want to order?
2. What is the Blood pressure treatment target?
3. With what medications do you start your treatment for blood pressure control?
4. What additional measures should be taken for optimal outcome?

Time allocated – 20 minutes

Hypertension is both a cause and a complication of CKD. The condition is associated with an increased risk of stroke and heart attack- the main complications of CKD; and accelerates the progression of CKD. Hypertension should be managed to prevent cardiovascular disease and reduce risks of kidney disease progression.

Hypertension causes and promotes progression of CKD and CKD causes hypertension. When patients present with both CKD and HPN, it is often difficult to know which one precedes: the “egg and Chicken dilemma”. In CKD, HTN occurs as a result of disturbed fluid and electrolyte regulation.

KDIGO recommends that adults with high BP and CKD be treated with a target systolic blood pressure (SBP) of <120 mm Hg, when tolerated, using standardized office BP measurement (2B).

However, applying the recommended SBP target of <120 mm Hg to BP measurements obtained in a non-standardized manner is potentially hazardous. Standardization emphasizes adequate preparations for BP measurement, not the type of equipment. Clinicians can reasonably offer less intensive BP-lowering therapy in patients with limited life expectancy or symptomatic postural hypotension. Hypertension should be treated to a < 130/80mmHg target using non-standardized conventional office blood pressure measurement.

Lifestyle changes:

Lifestyle changes are integral to the management of hypertension in CKD and should be encouraged. They are cost effective and important preventive and controlling strategies for Hypertension. The usual impact of each lifestyle change is 4-5 mmHg decrease in SBP and 2-4 mm Hg decrease in DBP. With the biggest impacts being changes to diet and exercise: The DASH diet (low sodium, high potassium, fruits and vegetables...) may decrease SBP by approximately 11 mm Hg. Recommendations for physical activity include 90 to 150 minutes of aerobic and/or dynamic resistance exercise per week and/or 3 sessions per week of isometric resistance exercises.

Pharmacotherapy:

In both diabetic and non-diabetic adults with CKD, start antihypertensive when office BP is consistently > 140/90 mmHg to maintain a BP of < 130/80 mm Hg. Individualize BP targets and antihypertensive agents according to age, coexistent cardiovascular disease and other comorbidities, and risk of progression of CKD.

Early intervention with ACE-Is or ARBs can reduce progression and cardiovascular risk by up to 50%. ACE-Is or ARBs are first line antihypertensives. Better proteinuria suppression can be attained with low Na diet and diuretics. When using ACE –Is or ARBs avoid volume depletion; avoid ACEi and ARB in combination and with supplemental K⁺ or K⁺-sparing drugs. Risk/benefit should be carefully assessed in medically fragile patients. ACE inhibitors and ARBs should be discontinued during pregnancy.

We recommend starting renin-angiotensin-system inhibitors (RASi) (angiotensin- converting enzyme inhibitor [ACEi] or angiotensin II receptor blocker [ARB]) for people with CKD and moderately to severely increased albuminuria (G1 – G4, A2- A3) with or without diabetes.

We recommend avoiding any combination of ACEi, ARB, and direct renin inhibitor (DRI) therapy in people with CKD, with or without diabetes.

RASi (ACEi or ARB) should be administered using the highest approved dose that is tolerated to achieve the benefits described because the proven benefits were achieved in trials using these doses. Hyperkalemia associated with the use of RASi can often be managed by measures to reduce the serum potassium levels rather than decreasing the dose or stopping RASi. Continue ACEi or ARB therapy unless serum creatinine rises by more than 30% within 4 weeks following initiation of treatment or an increase in dose.

Consider reducing the dose or discontinuing ACEi or ARB in the setting of either symptomatic hypotension or uncontrolled hyperkalemia despite medical treatment, or to reduce uremic symptoms

while treating kidney failure (estimated glomerular filtration rate [eGFR] <15 ml/min per 1.73 m²). Changes in BP, serum creatinine, and serum potassium should be checked within 2–4 weeks of initiation or increase in the dose of a RASi.

Check labs within 2 to 4 weeks after initiation:

- If serum creatinine increase is less than 30 %, continue and monitor.
- If the increase is more than 30%, stop ACEIs.

If HTN is not controlled with RAAS inhibition or if it is contraindicated, diuretics and/or CCBs are preferred. Amlodipine or sustained release of nifedipine can be used. Calcium channel blockers are the first line in post-transplant patients. Beta-blockers are first line if there is CAD or HF.

Multiple medications (2 or more drugs) are often needed to control hypertension adequately in most people with CKD. All groups of drugs should be considered with the aim being to reduce blood pressure to target. We recommend ACE inhibitors or ARB as the first-line class of drugs for BP lowering in hypertensive patients with proteinuria.

Additional antihypertensive agents can be chosen based on cardiovascular indications and comorbidities as shown below (Table 8 and Figure 4).

DRUG CLASS	CONSIDERATION
ACEi / ARB	Preferred for BP lowering especially in CKD patients with proteinuria ACEi and ARBs should not be combined Can be used in all stages of CKD. CAUTION ⚠ : STOP if Serum Potassium > 5.5 mmol/L DO NOT INITIATE If eGFR < 15 mL/min/1.73m²
Thiazide diuretics	Effective in all stages of CKD as adjunct antihypertensive therapy especially in presence of fluid overload. They can be used if the eGFR is >45 mL/min/1.73m ² but can be effective at low levels of eGFR, when sequentially combined with loop diuretics. Chlorthalidone and indapamide, the thiazide like diuretics are preferred over hydrochlorothiazide due to their longer half-lives and higher potency.
Loop diuretics	Should not be the preferred drug for the treatment of hypertension. Can be used as additional medicine in patients with fluid overload in all stages of CKD even when eGFR is severely reduced to <30 mL/min/1.73m ² Doses of furosemide are 20-120 mg/day, but higher doses (up to 500 mg/day) may be required, especially at lower levels of eGFR.

β-blockers	Alternative agents after ACEi/ARBs for controlling hypertension in patients with CKD and systolic heart failure. Vasodilatory β-blockers (carvedilol, nebivolol) preferred in CKD
Calcium channel blockers	DHP CCBs such as amlodipine are used with ACEi or ARB (synergistic action in BP reduction) Non-DHP CCBs (e.g., verapamil and diltiazem) should be considered in CKD with proteinuria. Nifedipine should be avoided
Mineralocorticoid receptor antagonists	Preferred in resistant hypertension (failed BP control with >3 classes of drugs) Finerenone preferred in CKD with diabetes CAUTION ⚠: Use with ACEi/ARB may be associated with hyperkalemia



Table 8: Considerations in the use of antihypertensive agents for BP lowering in CKD

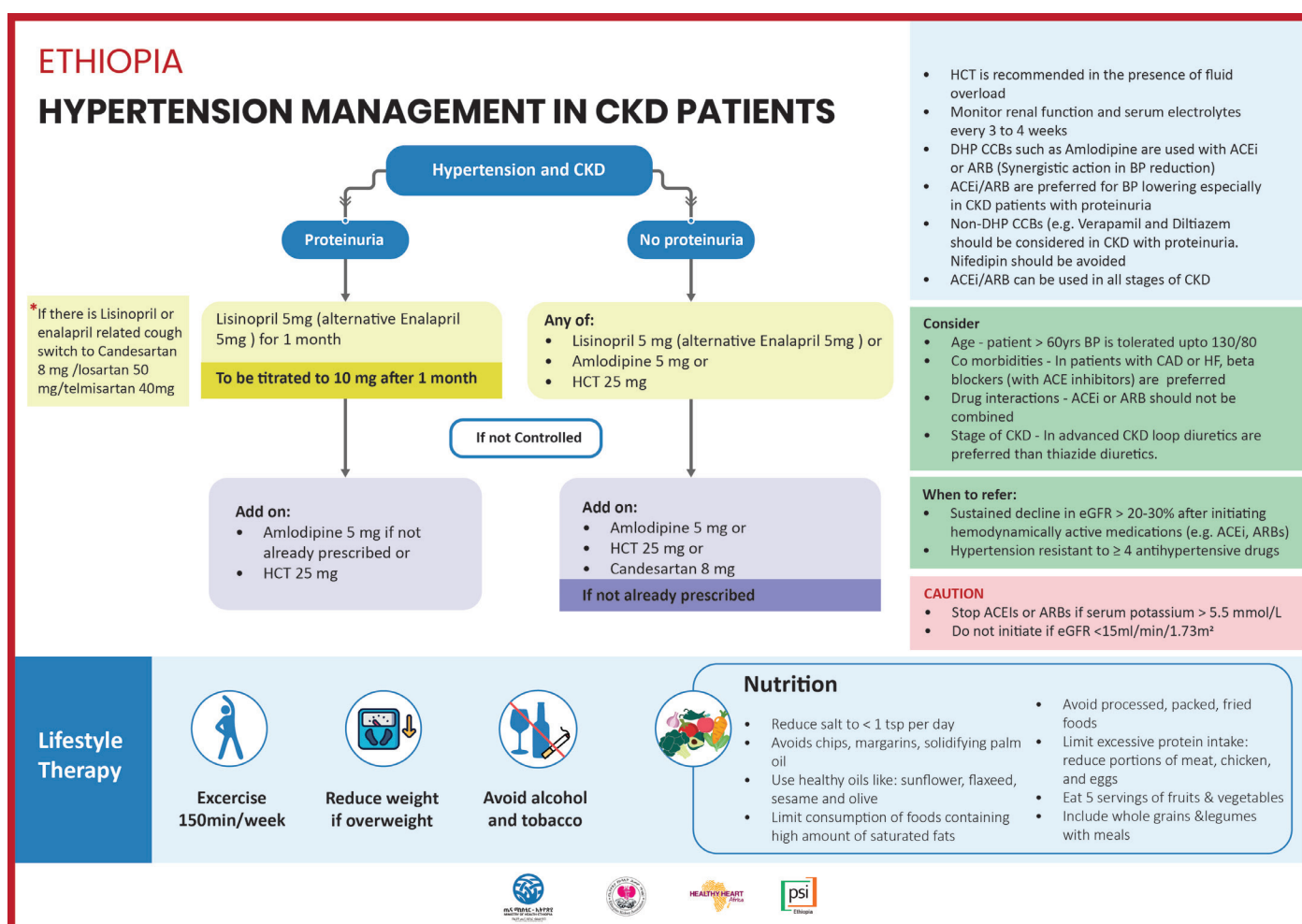


Figure 4: Algorithm for management of Hypertension in CKD

ECF Volume expansion:

ECF volume expansion (edema) significantly contributes to hypertension in CKD. Optimal diuresis is important to achieve optimal outcomes. Thiazide diuretics are preferred if eGFR is >30 ml/min. Loop diuretics are preferred in heart failure (HF) and when GFR is <30 ml/min. In advanced CKD, higher doses of diuretics are required to achieve optimal effect. Torsemide is more readily absorbed and has a longer half-life than furosemide. A combination of loop and thiazide diuretics may be used in resistant edematous states, with careful monitoring of electrolytes and volume status.

Activity 3.4.B. Case Scenarios for group discussion



Instruction: Read the following case scenarios and discuss the points mentioned in teams.

Your instructor will create groups of three each. Select one individual to present the group's opinion after discussion.

Case: A 45 years old man was admitted for heart failure secondary to ischemic heart disease. He is diabetic and hypertensive for the past 3 years. He uses 3-liter oxygen and has signs of fluid overload on physical examination. Blood pressure upon admission was 160/100 mmHg.

Serum Creatinine is 5.9 mg/dl, Serum Na⁺ = 125 meq/l, Serum K⁺ = 6 meq/l

1. What additional investigations do you want to order?
2. What is the Blood pressure treatment target?
3. What Blood pressure lowering medications do you start your treatment with?
4. Discuss options of blood sugar control, medications, and targets.
5. What other management issues should be discussed with the patient and family?

Time allocated - 20 minutes

3.4.2. Optimal Control of Diabetes Mellitus

Diabetes should be detected early and managed optimally with a comprehensive strategy to reduce the risks of kidney disease progression and cardiovascular disease. This is a multidisciplinary approach that includes a foundation of lifestyle modifications and diet with first-line drug therapies according to clinical characteristics.

CKD in patients with DM is often diagnosed through routine screening. CKD screening should start at diagnosis of Type II DM because evidence of CKD is often already apparent at this time. For Type I DM, screening is recommended commencing 5 years after diagnosis, before which CKD is uncommon.

Non- Pharmacological measures to control DM (Diet and Lifestyle modifications):

Patients with diabetes and CKD should consume an individualized diet high in vegetables, fruits, whole grains, fiber, legumes, plant-based proteins, unsaturated fats, and nuts; and lower in processed meats, refined carbohydrates, and sweetened beverages. It is recommended to maintain a protein intake of 0.8 g protein/kg (weight)/d for those with diabetes and CKD not treated with dialysis. Patients treated with hemodialysis, and particularly peritoneal dialysis, should consume between 1.0 and 1.2 g protein/kg (weight)/day.

Sodium intake should be <2 g of sodium per day (<5 g of sodium chloride per day) in patients with diabetes and CKD.

Patients with diabetes and CKD are advised to undertake moderate-intensity physical activity for a cumulative duration of at least 150 minutes per week, or to a level compatible with their cardiovascular and physical tolerance.

Pharmacologic Management for Glycemic Control:

Optimal glycemic control is one of the important measures to delay progression of CKD.

Hemoglobin A1c (HbA1c) is used to monitor glycemic control in patients with diabetes and CKD. Its reliability decreases as CKD progresses. Self-monitoring of blood glucose (SMBG) may help prevent hypoglycemia and improve glycemic control when glucose-lowering therapies associated with the risk of hypoglycemia are used. Individualized HbA1c target ranging from <6.5% to <8.0% in patients with diabetes and CKD not treated with dialysis is recommended. The target choice depends on:

- Severity of CKD
- Presence and severity of macrovascular complications
- Comorbidities
- Life expectancy
- Presence of Hypoglycemic unawareness
- Resources for hypoglycemia management

Patients who are in advanced forms of CKD with shorter life expectancy and comorbidities should be put in higher HgbA1C target groups. Declining kidney function may necessitate changes to diabetes medications and renal cleared drugs (See figures below).

Glycemic monitoring and targets in patients with diabetes and CKD	
Recommendation 1	Using hemoglobin A1c (HbA1c) to monitor glycemic control in patients with diabetes and CKD (1C)
Recommendation 2	An individualized HbA1c target ranging from <6.5 to 8.0 % in patients with diabetes and CKD not treated with dialysis (Figure 5) (1C)

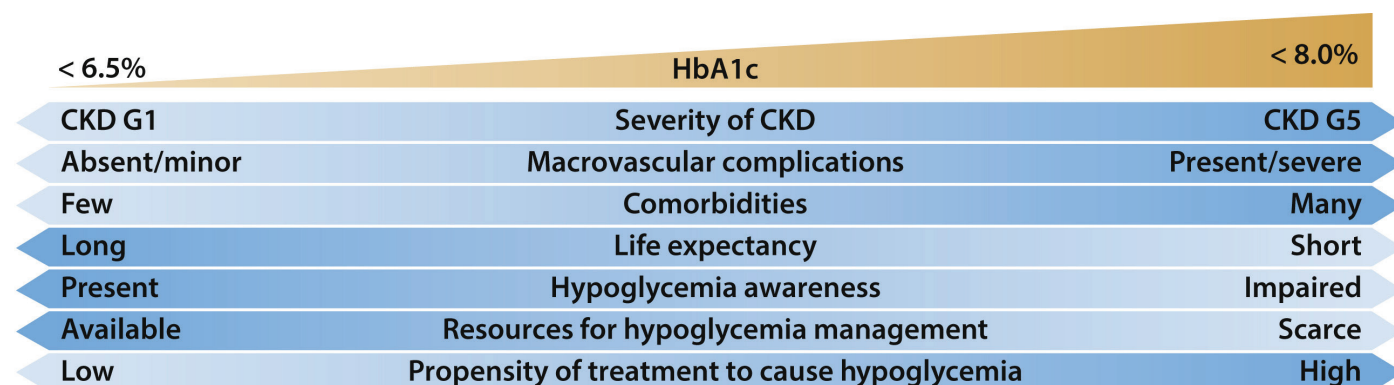


Figure 5: Factors guiding decisions on individual glycated hemoglobin (HbA1c) targets, CKD, G1, estimated glomerular filtration rate (eGFR) > 90ml/min per 1.73 m2, G5, eGFR <15ml/min per 1.73m2

Glycemic control is based on insulin for type 1 diabetes (T1D) and a combination of metformin and sodium-glucose cotransporter-2 inhibitors (SGLT2i) for type 2 diabetes (T2D).

- **Metformin** may be given when the estimated glomerular filtration rate (eGFR) is >30 ml/min per 1.73 m2, and
- **SGLT2i** should be initiated when eGFR is >20 ml/min per 1.73 m2 and continued as tolerated, until dialysis or transplantation is initiated.
- **Glucagon-like peptide-1 receptor agonists (GLP-1 RA)** are preferred glucose-lowering drugs for people T2D if SGLT2i and metformin are insufficient to meet glycemic targets or if they are unable to use SGLT2i or metformin.

A reversible decrease in the eGFR with commencement of SGLT2i treatment may occur and is generally not an indication to discontinue therapy. Once SGLT2i is initiated, it is reasonable to continue SGLT2i even if the eGFR falls below 20 ml/min per 1.73 m2, unless it is not tolerated or kidney replacement therapy is initiated. It is reasonable to withhold SGLT2i during times of prolonged fasting, surgery, or critical medical illness (when people may be at greater risk for ketosis).

Recommendations for SGLT2i use in adults with CKD following (1A):

- eGFR > 20 ml/min per 1.73 m2 with urine ACR >200 mg/g, or
- Heart failure, irrespective of level of albuminuria.

SGLT2i initiation or use **does not necessitate alteration of frequency of CKD monitoring**, and the reversible decrease in eGFR on initiation is generally not an indication to discontinue therapy. We suggest treating adults with eGFR 20 to 45 ml/min per 1.73 m² with urine ACR <200 mg/g with SGLT2i (2B).

Drugs like Insulin, sulphonyl urea (preferably Glipizide, Gliclazide, or Glimepiride), and DPP4 Inhibitors can be added to the above regimen as needed. However, close monitoring is required as patients with CKD are also at high risk of developing hypoglycemia in the presence of sulphonyl Reas.

Insulin therapy is often the treatment of choice in:

- Critical illness
- Need for urgent blood sugar lowering
- Stage 4/5 CKD

Dose adjustments for insulin may be required depending on eGFR:

- When eGFR is **15-50 ml/min/1.73 m²**, reduce dose by **25%**.
- When eGFR is **<15 ml/min/1.73 m²**, reduce dose by **50%** to decrease hypoglycemia risk.

Nonsteroidal MRA is most appropriate for adults with T2D who are at high risk of CKD progression and cardiovascular events, as demonstrated by persistent albuminuria despite other standard-of-care therapies.

We **suggest** a nonsteroidal MRA with proven kidney or cardiovascular benefit for adults with T2D who have:

- eGFR **>25 ml/min per 1.73 m²**
- Normal serum potassium concentration
- Albuminuria **>30 mg/g** despite maximum tolerated dose of renin-angiotensin system inhibitors (**RASi**) (2A).

A nonsteroidal MRA may be added to a RASi and SGLT2i for treatment of T2D and CKD in adults. To mitigate risk of hyperkalemia, select people with consistently normal serum potassium concentration and monitor serum potassium regularly after initiation of a nonsteroidal MRA.

Steroid MRA may be used for treatment of:

- Heart failure
- Hyperaldosteronism
- Refractory hypertension

However, they may cause **hyperkalemia** or a reversible decline in glomerular filtration, especially among people with low GFR.

In adults with T2D and CKD who have not achieved individualized glycemic targets despite metformin and SGLT2 inhibitor treatment, or who are unable to use those medications, we recommend a long-acting GLP-1 RA (1B). The choice of GLP-1 RA should prioritize agents with documented cardiovascular benefits.

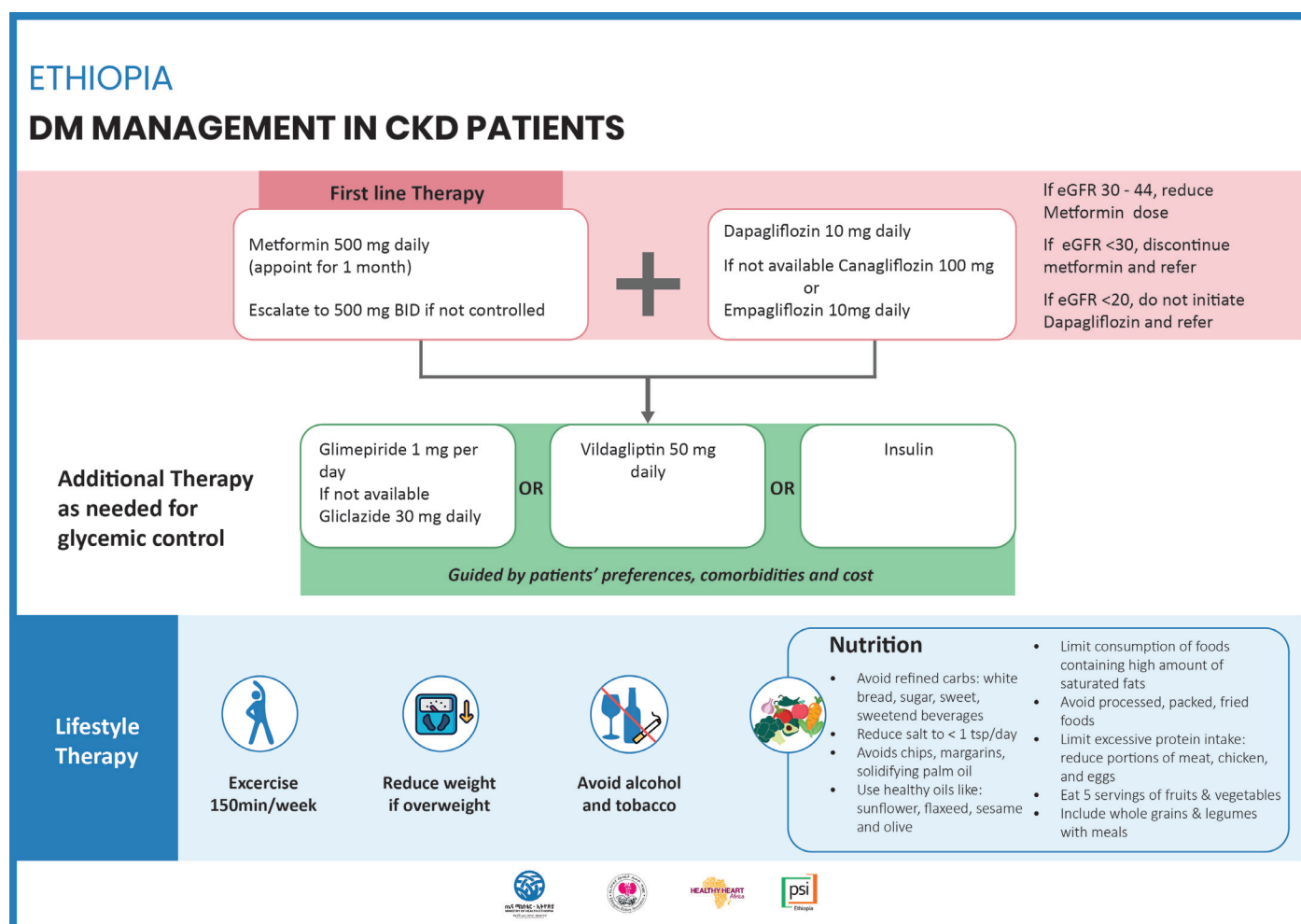


Figure 6: Algorithm of Diabetes Management in CKD

A statin is recommended for all patients with T1D or T2D and CKD. Aspirin generally should be used lifelong for secondary prevention among those with established cardiovascular disease and may be considered for primary prevention among patients with a high risk of atherosclerotic cardiovascular disease (ASCVD).

3.4.3. Lipid management

Dyslipidemia in CKD is frequently characterized by high triglycerides, low HDL cholesterol, and an increased proportion of low-density lipoprotein (LDL) particles, which are small and oxidized. In adults with newly identified CKD, it has been recommended to evaluate their lipid profiles (total cholesterol, LDL cholesterol, HDL cholesterol, and triglycerides), but follow-up lipid measurements are not required for the majority of people. This is because treatment initiation is based on risk, and the benefits of

statin-based therapy have been shown to be independent of the level of cholesterol levels. For those with a total cholesterol > 290 mg/dl and a personal or family history of premature ischemic heart disease (e.g., an event before the age of 60 years in an individual or first-degree relative), it is important to consider familial disease and refer to a specialist. The benefits of lowering LDL cholesterol using statin-based therapies on the risk of ASCVD are well established in people with and without CKD. There are clear recommendations on when to initiate such therapies:

- **In adults aged >50 years with eGFR <60 ml/min/1.73 m²** but not treated with chronic dialysis or kidney transplantation (GFR categories G3a–G5), we recommend treatment with a statin or statin/ezetimibe combination (1A).
- **In adults aged >50 years with CKD and eGFR >60 ml/min/1.73 m²** (GFR categories G1–G2), we recommend treatment with a statin (1B).
- **In adults aged 18–49 years with CKD** with CKD but not treated with chronic dialysis or kidney transplantation, we suggest statin treatment in people with one or more of the following (2A):
 - Known coronary disease (myocardial infarction or coronary revascularization),
 - Diabetes mellitus,
 - Prior ischemic stroke, or
 - Estimated 10-year incidence of coronary death or nonfatal myocardial infarction >10%.

3.5. AVOIDANCE OF FURTHER INJURY TO THE KIDNEYS

Activity 3.5. Case Scenario



Instruction: Read the following case scenarios and discuss the points mentioned in teams.

Your instructor will create groups of two participants each. Select one individual to present the group's opinion after discussion.

Case: A 45-year-old male known hypertensive and CKD patient presented to your hospital with fever, headache, diarrhoea and vomiting. He was started with oral antibiotics, analgesics and ORS at a local clinic; and referred for better management. He used to take amlodipine 10 mg po daily and enalapril 5 mg po daily for his hypertension. He looks dehydrated; he is conscious but lethargic. B/P = 90/50 mmHg; PR = 120/min, febrile; maintains saturation.

1. Investigations revealed: Leucocytosis with left shift; mild anemia, Cr = 5.5 mg/dL (it was 2 last month); Urea = 112 mg/dL; Serum K⁺ = 5.8; Na⁺ = 145. What is your assessment of current condition?
2. What additional investigations do you want to order?
3. Outline the management plan.

Time allocated - 15 minutes

Even minor further injuries to the already compromised kidneys accelerate the progression of CKD. Inter-current AKIs and nephrotoxin exposures (mainly drugs) are major offenders. Therefore, common causes of intercurrent AKI (infections, volume depletion, nephrotoxin ...) should be prevented and/or promptly treated. Optimal management of comorbidities is also crucial.

3.5.1. Intercurrent illnesses (Superimposed AKI on CKD)

In addition to exacerbation of their original kidney disease, patients with CKD with a recent decrease in eGFR may be suffering from an underlying reversible process, which, if identified and corrected, may result in the recovery of function. This is sometimes termed **“acute on chronic kidney disease.”** These could occur in the following situations.

Hypovolemia and decreased kidney perfusion (such as vomiting, diarrhea, diuretic use, bleeding, myocardial dysfunction, sepsis) and the administration of drugs that lower the eGFR (such as nonsteroidal anti-inflammatory drugs [NSAIDs] and ACE inhibitors or ARBs) are common causes of potentially reversible declines in kidney function.

The heart and kidneys are closely related. The dysfunction of either of these organs leads to a functional deterioration of the other. Heart failure (HF) and CKD share risk factors, such as diabetes, hypertension, and coronary artery disease. Additionally, acute and chronic heart failure leads to worsening of the kidney function.

Most drugs used in the treatment of heart failure also have benefits in the management of CKD through blood pressure reduction and control of albuminuria. However, these same drugs can lead to deterioration in kidney function and expose the patient to complications such as hyperkalaemia, leading to dialysis. Health workers should anticipate worsening kidney function and electrolyte abnormalities that may occur during heart failure treatment.

The choice of HF therapies should be individualized based on the CKD stage. **Goal-Directed Medical Therapy (GDMT)**—the cornerstone of pharmacological treatment in heart failure with reduced ejection fraction (HFrEF)—should be used in CKD patients. GDMT includes:

- Angiotensin receptor neprilysin inhibitors (ARNI)
- Evidence-based β -blockers
- Mineralocorticoid receptor antagonists
- SGLT2 inhibitors: Dapagliflozin/empagliflozin or canagliflozin are preferred

3.5.2. CKD and Medications

Medication management and drug stewardship in CKD is an important component of the care of people with CKD. Medications can be highly beneficial, but some may be toxic, are excreted by the kidney, may have narrow therapeutic windows, or may have no proven benefit or indication in people with CKD. Drug stewardship refers to the effective, safe, and sustainable use of medications by all staff and physicians, encompassing the whole cycle of medication use. Medications need to be prescribed responsibly, monitored for efficacy and safety, and when they no longer serve their intended purpose, discontinued.

Abnormal kidney function results in altered pharmacokinetics and pharmacodynamics, and for people with CKD, as the GFR worsens, so does the prevalence of polypharmacy and comorbidities. People with CKD are at increased risk of medication errors and inappropriate prescribing (reported to be up to 37% in ambulatory outpatient studies and up to 43% in long-term care studies). Thus, improved understanding and collaboration with pharmacists in developing care plans and performing medication review is strongly recommended.

People with CKD have reduced ability to excrete medications and/or their metabolites (which may increase adverse event risk or exaggerate/diminish efficacy) and increased sensitivity to medications (e.g. those bound to albumin in states of hypoalbuminemia such as nephrotic syndrome).

Additional issues include nephrotoxicity, diminished tolerance of side effects in the context of coexisting comorbidities or older age, and lack of adequate evidence for either benefit or harm of specific compounds, due to historical exclusion of people with advanced CKD from most clinical trials.

Review and limit the use of over-the-counter medicines and dietary or herbal remedies that may be harmful for people with CKD. Consider GFR when dosing medications that are cleared by the kidneys.

Thus, healthcare providers should consider the indication, benefit-risk profile, and potential nephrotoxicity while balancing accessibility, availability, local health policies, cultural practices, affordability, and patient preferences. Also perform thorough medication review periodically and at transitions of care to assess adherence, continued indication, and potential drug interactions because people with CKD often have complex medication regimens and are seen by multiple specialists. Where available, consultation with pharmacists as part of the multidisciplinary team is encouraged to ensure optimized comprehensive medication management and to improve pharmaco equity.

Several classes of drugs are renally eliminated. There are a number of safety issues that need to be given due attention to prevent further deterioration of kidney function:

- Any medication with >30% renal clearance needs dose adjustment for CKD;

-
- Consider kidney function and current eGFR (not just SCr) when prescribing medications;
 - Minimize pill burden as much as possible;
 - Remind CKD patients to avoid NSAIDs;
 - Avoid Dual RAAS blockade;
 - Avoid bisphosphonates for eGFR <30 ml (about 1.01 oz)/min; and
 - Radio contrast agents should be used very cautiously.

Certain drugs also interfere with either creatinine secretion or the assay used to measure the serum creatinine. These include cimetidine, trimethoprim, cefoxitin, and flucytosine. In these settings, there will be no change in the true GFR; the clinical clue that this may have occurred is the absence of a concurrent elevation in the blood urea nitrogen (BUN).

Perioperative drug management:

We recommend that medications such as metformin, renin-angiotensin-system inhibitors, diuretics, aminoglycosides, and nonsteroidal anti-inflammatory drugs (NSAIDs) be discontinued 48-72 hours before elective surgery to prevent complications (eg, hypotension, metabolic acidosis, or hyperkalemia). Sodium glucose cotransporter-2 inhibitors should be withheld at least 3-4 days prior to elective surgery.

3.6. NUTRITIONAL AND PSYCHOSOCIAL SUPPORT

CKD patients are at risk of malnutrition for a number of reasons. Some diets are also associated with worsening renal dysfunction. The optimal renal diet should be composed of low Na⁺, low K⁺, low phosphorus, and limited protein.

Lowering salt intake to <2 g per day of sodium (corresponding to 5 g of sodium chloride) is also recommended. Diets rich in phosphorus and potassium need to be avoided as well. If possible, the involvement of a dietician or nutritionist is important (See chapter 6 for details).

Advanced CKD is significantly associated with socioeconomic burden and psychosocial problems that affect both patients and families. Therefore, addressing these problems in consultation with the patient, family, and social support groups is part of the comprehensive care for CKD patients.

3.7. EARLY REFERRAL TO NEPHROLOGIST



Activity 3.7. Group Discussion

Instruction: Discuss and present the possible benefits of referring the following CKD patient for Nephrologist evaluation.

Your instructor will create groups of two. Select one individual to present the group's opinion after discussion.

Group 1. Thirty-two years old man with one year history of hypertension and presented with generalized body swelling decreased urine output, proteinuria and many RBCs on urinalysis.

Group 2. A 65 years old diabetic, hypertensive and cardiac patient whose creatinine has been > 3 mg/dl since the past 4 months.

Time allocated – 20 minutes

Early identification and referral to specialist kidney care services for people with CKD has the potential to reverse, delay, or prevent progression of disease and is a key focus of international initiatives in the context of the global “epidemic” of kidney disease. The goals of early identification and referral to specialist kidney care services are several-fold and include:

- Ensuring a specific diagnosis for CKD is sought, where appropriate;
- Acute kidney injury or abrupt fall in GFR;
- Active urinary sediments: Persistent albuminuria (ACR > 300 mg/g) and hematuria;
- Atypical Progression of CKD;
- Resistant Hypertension;
- Persistent abnormalities of serum potassium;
- Recurrent or extensive nephrolithiasis;
- Hereditary kidney diseases; and/or
- Cause of CKD is uncertain;
- Provision of specific therapy based on diagnosis;
- Slowing/arresting CKD progression;
- Evaluation and management of comorbid conditions;
- Prevention and management of CVD;
- Identification, prevention, and management of CKD-specific complications (e.g., malnutrition, anemia, bone disease, and acidosis);
- Planning and preparation for RRT (e.g., choice of modality, vascular access-placement and care, and preemptive transplantation);

- GFR <30 ml/min/1.73m (G4-G5); or A >3%–5% 5-year risk of requiring RRT measured using a validated risk equation.
- Psychosocial support; and
- Provision of conservative care and palliative care options where required.

When to Refer a CKD Patient for a Nephrologist Evaluation	
I. Based on the Causes	• Uncertain etiology of CKD • Hereditary kidney diseases • Recurrent or extensive nephrolithiasis (kidney stones)
II. eGFR or Risk of Kidney Replacement Therapy (RRT)	• >3-5% 5-year risk of requiring KRT, measured with a validated risk equation • eGFR <30 mL/min/1.73 m² • Sustained decline in eGFR >20-30% after initiating hemodynamically active medications (e.g., ACE inhibitors, ARBs)
III. Proteinuria With or Without Hematuria	• Persistent significant albuminuria (ACR ≥300 mg/g or AER ≥300 mg/24 h or PCR ≥500 mg/g or PER ≥500 mg/24 h), especially with hematuria • ≥2-fold increase in albuminuria in follow-up • ACR consistently >700 mg/g • Presence of urinary red cell casts or RBC >20/HPF, sustained and unexplained
IV. Other Indications	• Hypertension resistant to ≥4 antihypertensive drugs • Persistent electrolyte abnormalities (e.g., hyperkalemia, hypokalemia) • Metabolic acidosis • Unexplained anemia • Mineral and bone disorders • Malnutrition related to CKD

Table 9: When to Refer a CKD Patient for a Nephrologist Evaluation

Studies have shown late referral is significantly associated with poor outcomes like increased morbidity, hospitalization, and mortality. Early referral and improved diagnosis create opportunities for strategic preservation of kidney function. Therefore, early diagnosis and referral to a nephrologist is one of the major strategies to delay and reduce these deleterious impacts of CKD (see table 10 and figure 7).

Consequences of late referral	Benefits of early referral
Severe hypertension and fluid overload, and increased cardiovascular comorbidity	Delay requirement for KRT, better management of CVD, and comorbid conditions
Low prevalence of permanent access	Reduced need for urgent dialysis using temporary access
Delayed referral for transplant	Greater choice of treatment options and pre-emptive transplantation
Higher initial hospitalization rate	Reduced hospital length of stay and costs
Higher 1-year mortality rate	Lower 1-year mortality rate
Less choice of KRT modality	Increased informed freedom of choice of KRT modality
Worse psychosocial adjustment and increase in DALYs	Early access to psychosocial counseling and support for care partners
Malnutrition	Improved nutritional status

Table 10: Benefits and consequences of early versus late referral

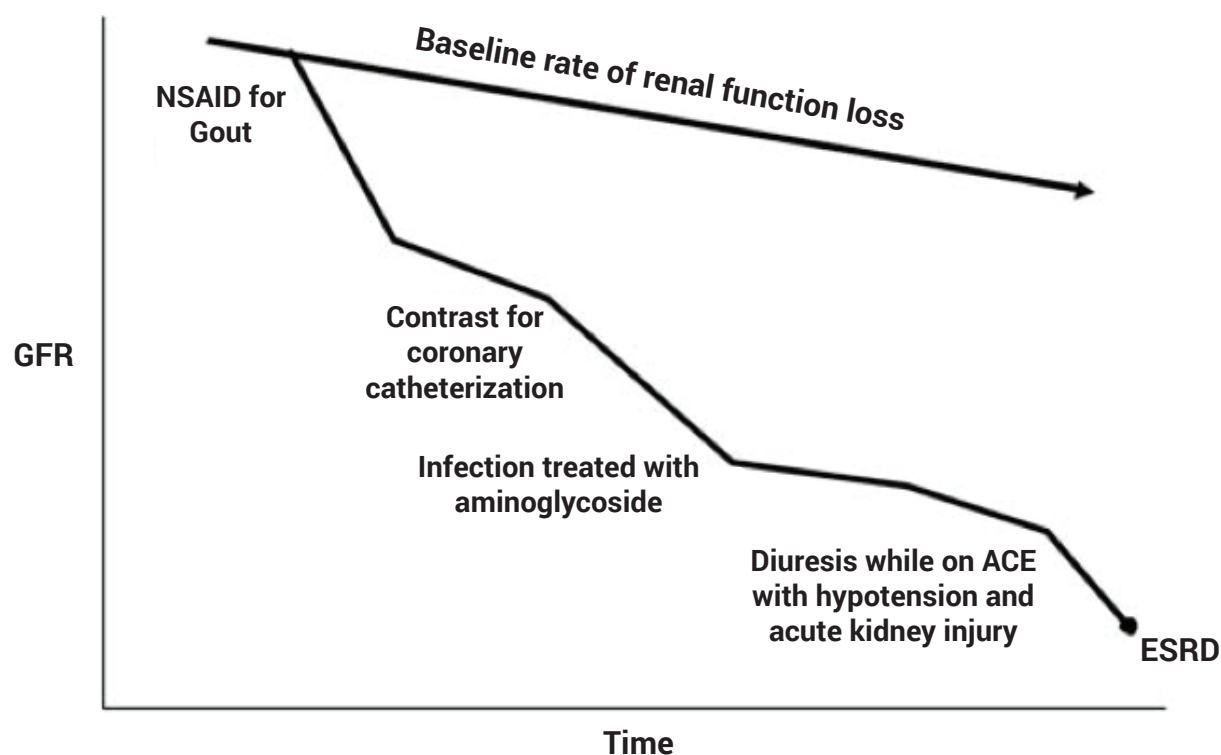


Fig 7: Impact of Early CKD detection with a patient safety approach

3.8. PALLIATIVE CARE IN CHRONIC KIDNEY DISEASE

In Ethiopia, there is a high burden of CKD due to the burgeoning epidemics of communicable and non-communicable diseases. However, access to life-saving renal replacement therapy, including dialysis and kidney transplantation, is severely limited. One of the most difficult situations encountered by healthcare professionals dealing with end stage kidney disease (ESKD) patients is the management of a conservative or palliative care pathway for many patients who have no other option.

For many older patients with progressive CKD, including those with ischemic heart disease, dialysis provides little or no benefit and managing these patients conservatively (i.e., without dialysis) may prolong survival with better preservation of function and quality of life and fewer acute care admissions.

Conservative Management of Kidney Failure

Patients may elect to withhold dialysis. We agree with the 2024 kidney disease: Improving Global Outcomes (KDIGO) guideline that conservative management of end-stage kidney disease (ESKD) should be an option for all patients who decide not to pursue kidney replacement therapy. Conservative care includes the management of symptoms, advance-care planning, and provision of appropriate palliative care.

3.9 CHAPTER SUMMARY

- CKD is a complex disease with deleterious impacts on patients and their families.
- A wide range of preventive and therapeutic interventions—from lifestyle modification to renal transplantation—is required for optimal outcomes.
- Effective management of CKD requires a **multidisciplinary team approach**, involving everyone from primary health care workers to nephrologists.
- The risk of progression varies across the different stages of CKD, and clinical management plans should be tailored accordingly.
- The main goals of CKD management are early **diagnosis, risk stratification**, and **staging**, to enable timely prevention and initiation of treatment.
- Key strategies for delaying CKD progression include managing risk factors (e.g., cardiovascular disease), optimizing hypertension and diabetes control, avoiding further kidney injury, and ensuring timely referral to a nephrologist.
- The number of patients with advanced CKD—especially those who are older, frail, have multiple comorbidities, and poor functional status—is increasing. These patients require comprehensive palliative care to reduce their complex burden and suffering.

CHAPTER 4: OBSTRUCTIVE UROPATHY



Time Allocation: 2 hours



Chapter Description: This chapter focuses on the diagnostic criteria and methods used to identify obstructive uropathy, including history taking, physical examination, laboratory investigations, imaging studies management and complications



Chapter Objective:

By the end of this chapter, learners will be able to identify the diagnostic criteria and methods for obstructive uropathy.

Enabling Objectives:

- Discuss anatomy pathophysiology of obstructive Uropathy
- Describe a comprehensive history and physical examination relevant to obstructive uropathy
- Describe proper laboratory investigations and imaging modalities relevant to obstructive uropathy.
- Describe the principles of emergency management for obstructive uropathy.
- Describe the appropriate medications for symptom relief and management.
- Describe further management based on the underlying cause of the obstruction

Outline

- 4.1. Definition of Obstructive Uropathy
- 4.2. Pathophysiology and Cause of Obstructive Uropathy
- 4.3. Clinical manifestation of obstructive uropathy
- 4.4. Diagnosis of Obstructive Uropathy
- 4.5. Management of obstructive uropathy
- 4.6. Chapter Summary

4.1 DEFINITION OF OBSTRUCTIVE UROPATHY



Activity 4.1

Group Discussion:

Instruction: Read and discuss the definition and differences of the following clinical conditions in groups and present to the plenary.

Acute kidney injury, **Acute kidney diseases**, chronic kidney disease, Obstructive uropathy and Obstructive Nephropathy.

Time: 10 Minutes

Obstructive uropathy (OUP) refers to an impediment in the regular flow of urine downstream due to either benign or malignant obstructions in the urinary tract, resulting in hydronephrosis and gradual renal impairment. As per the European Association of Urology (EAU) treatment guidelines, OUP is a critical condition that can escalate to sepsis and/or acute kidney injury (AKI), necessitating urgent intervention. Often “obstructive uropathy” and “obstructive nephropathy” are used interchangeably.

However, the term obstructive nephropathy should be reserved for those individuals who have renal impairment because of blockage or impairment of urinary flow. There are different renal clinical conditions that should be clearly differentiated.

4.2 PATHOPHYSIOLOGY AND CAUSE OF OBSTRUCTIVE UROPATHY

Obstructive uropathy can be classified based on three primary criteria: degree, duration, and site of obstruction.

Degree of Obstruction:

High Grade (Complete): Refers to a total blockage of urinary flow.

Low Grade (Partial): Indicates a partial blockage of urinary flow.

Duration of Obstruction:

Acute: Develops suddenly, often caused by conditions like nephrolithiasis.

Chronic: Develops gradually over weeks to months. Chronic obstruction, often associated with congenital genitourinary anomalies and prostatic disease, is more likely to lead to obstructive nephropathy and CKD, while acute obstruction is usually identified more quickly.

Site of Obstruction:

Upper Tract Obstruction: Blockage occurring above the ureterovesical junction. Typically, unilateral.

Lower Tract Obstruction: Blockage occurring below the ureterovesical junction. More commonly bilateral.

Understanding these classifications is crucial for accurate diagnosis and appropriate management of obstructive uropathy.

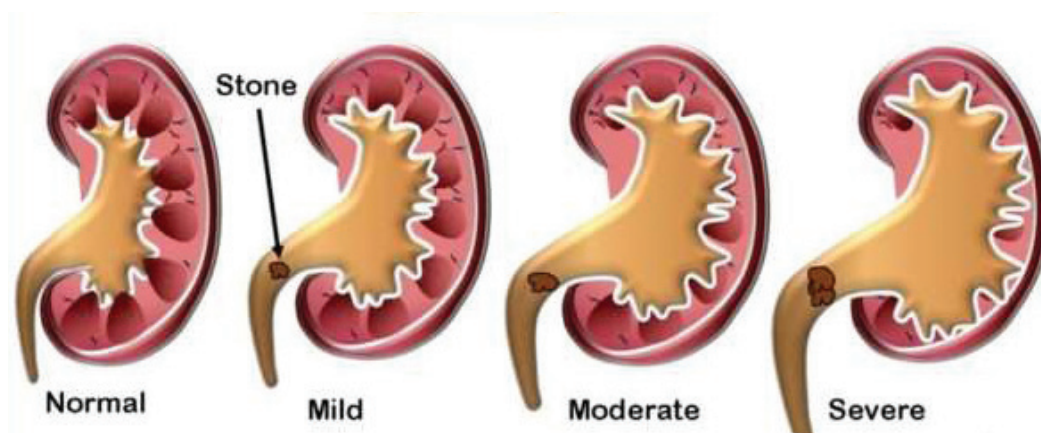


Figure 8: Hydronephrosis

4.2.1 Pathophysiology of Obstructive Uropathy

Hydronephrosis develops due to obstruction in the urinary flow. This obstruction leads to increased hydrostatic pressure within the renal collecting system. Elevated intraglomerular pressure subsequently affects the glomerular filtration rate (GFR). The severity and duration of the obstruction dictate the extent of renal function impairment.

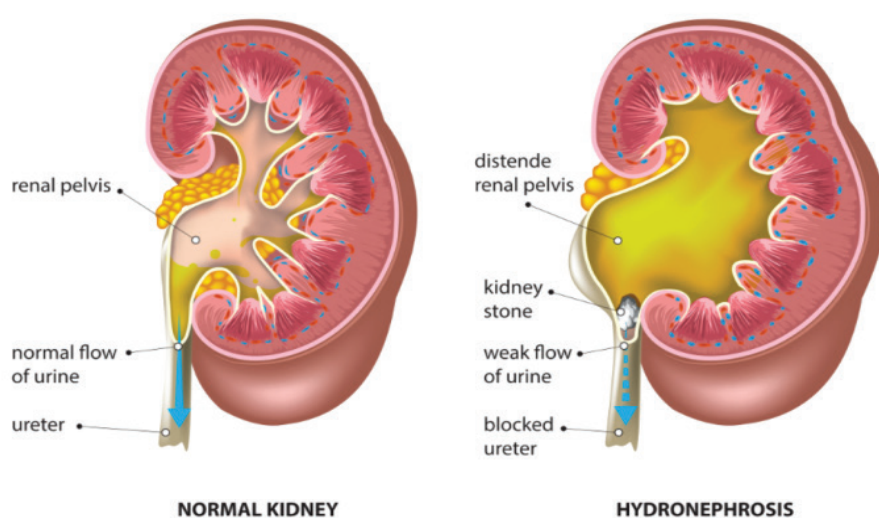


Figure 9: Normal Kidney vs Hydronephrosis

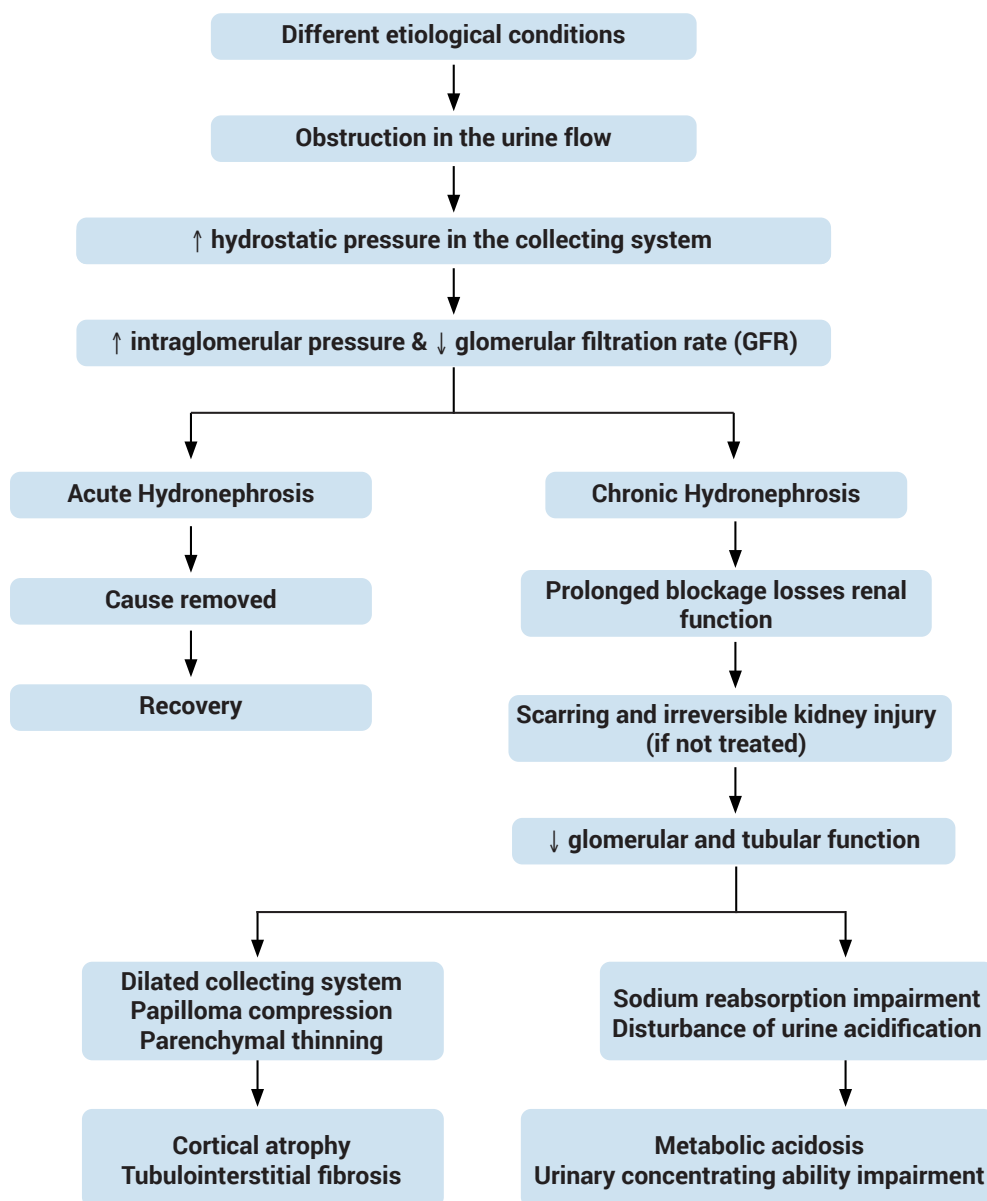


Figure 10: Pathophysiology of Hydronephrosis

4.2.2 Causes of Obstructive Uropathy

Activity 4.2

Think-Pair- Share

Instruction: Be in pairs, think over the following question individually, discuss in pairs and finally share your opinions with the plenary.

List the common causes of OUP?

Time: 5 minutes



Many conditions can cause obstructive uropathy, which may be acute or chronic, partial or complete, and unilateral or bilateral. The most common causes differ by age:

Children: Anatomic abnormalities (including posterior urethral valves or stricture and stenosis at the ureterovesical or ureteropelvic junction)

Young adults: Urolithiasis, Urethral Stricture Disease

Older adults: Benign prostatic hyperplasia (BPH) or prostate cancer, Bladder Tumor, Urolithiasis, Urethral stricture and retroperitoneal or pelvic tumors (including metastatic cancer)

I. Prostatic Disorders

Benign Prostatic Hyperplasia (BPH) and Prostatic Cancer

In males, obstruction is much more common in the reproductive tract, predominately related to disorders of the prostate, both benign hypertrophy and malignancy. Prostatic disorders contribute to approximately 5% of ESRD in men older than 65 years of age. Relief of the obstruction with medications and/or surgery is recommended. However, if obstruction is chronic, its relief may not improve GFR.

II. Urolithiasis (Urinary Stones)

Urolithiasis refers to the formation of urinary stones in the kidneys, ureters, bladder, or urethra. Large urinary stones or multiple stones can cause obstruction at various points in the urinary tract, leading to obstructive uropathy. This obstruction can impair kidney function and result in symptoms such as severe flank pain, hematuria, recurrent infections, and urinary retention.

III. Prostatic cancer is a malignant tumor that arises from the cells of the prostate gland. It is one of the most common cancers in men. Similar to BPH, prostatic cancer can cause obstruction of the urethra due to the enlargement of the prostate gland or due to the presence of a tumor mass. This obstruction can result in obstructive uropathy and associated complications.

IV. Bladder Tumor

Bladder tumors are growths of abnormal cells that develop in the lining of the bladder. A bladder tumor can cause obstructive uropathy by extending into the bladder neck or invading the ureteric orifices, leading to urinary retention (acute or chronic) and upper urinary tract damage, respectively. This obstruction can result in obstructive uropathy and associated symptoms.

V. Urethral Stricture

A urethral stricture is a narrowing of the urethra due to scar tissue formation, inflammation, or injury. Urethral strictures can obstruct the flow of urine through the urethra, leading to urinary retention, difficulty in urination, and potential damage to the bladder and kidneys.

VI. Pelvic Malignancies

Pelvic malignancies refer to cancers that originate from various organs within the pelvic region, such as the bladder, prostate, uterus, cervix, and rectum.

Pelvic malignancies can directly invade or compress the urinary tract, leading to obstruction of urine flow. This obstruction can result in obstructive uropathy and associated urinary symptoms.

VII. Genitourinary Tuberculosis (GUTB)

Genitourinary tuberculosis is a form of tuberculosis that affects the urinary tract, including the kidneys, ureters, bladder, and urethra. GUTB can cause inflammation, scarring, and strictures within the urinary tract, leading to obstruction of urine flow.

4.3 CLINICAL MANIFESTATION OF OBSTRUCTIVE UROPATHY

Activity 4.3



Individual Reflection:

Instruction: Read the following question and reflect individually.

What are the common presenting symptoms of Obstructive uropathy?

Time: 5 minutes

The common presenting symptoms of obstructive uropathy can vary depending on the cause, location, and severity of the obstruction. Here are some of the typical symptoms associated with obstructive uropathy:

- **Decreased Urine Output (Oliguria):** Reduced amount of urine passed, which can sometimes progress to anuria (no urine output).
- **Difficulty Urinating (Dysuria):** Pain or discomfort during urination.
- **Urinary difficulty (LUTS):** Urinary Frequency, Urgency, Hesitancy, Incomplete Emptying of the Bladder,
- **Flank Pain:** Pain in the side (flank) of the body, which may be severe and can radiate to the groin.
- **Lower Abdominal Pain:** Discomfort or pain in the lower abdominal region.
- **Hematuria:** Blood in the urine, which can be visible (gross hematuria) or microscopic.
- **Urinary Retention:** Inability to completely empty the bladder, leading to a feeling of fullness or pressure in the bladder.
- **Overflow Incontinence:** Involuntary leakage of urine due to an overfull bladder.
- **Hydronephrosis:** Swelling or enlargement of the kidneys due to the build-up of urine caused by the obstruction.
- **Urinary Tract Infections (UTIs):** Recurrent UTIs may occur due to stagnant urine providing a conducive environment for bacterial growth.
- **Nausea and Vomiting:** These symptoms can occur as a result of severe pain or due to the associated kidney dysfunction.

Some individuals may experience mild symptoms initially, while others may present with severe symptoms requiring urgent medical attention. Early diagnosis and intervention are crucial to prevent complications and preserve kidney function.

4.4 DIAGNOSIS OF OBSTRUCTIVE UROPATHY

4.4.1 The clinical presentation of obstructive uropathy varies depending on whether the underlying process is acute or chronic.

Acute obstructive uropathy tends to be more symptomatic,

- May present with abdominal pain or
- Flank pain,
- Referred pain to the labia or testicle
- Nausea, vomiting, and changes in the passage of urine ranging from anuria to nocturia and polyuria.

In chronic obstruction, individuals may be

- Completely asymptomatic.
- Changes in urine flow such as decrease in the force of the urinary stream, dribbling of urine, inability to initiate micturition, urinary frequency, and nocturia.

4.4.2 Physical examination is variable depending on the acuteness and cause of the obstructive process.

- Infants and children may have a palpable abdominal mass in association with hydronephrosis.
- In some instances, a distended bladder is palpable in the suprapubic region of patients with lower tract obstruction.
- In women, a pelvic examination may reveal a prolapsed uterus, or pelvic masses associated with malignancy. In men, a rectal examination should be performed to evaluate for prostatic hypertrophy or nodules.

4.4.3 Diagnostic Investigations

A. Laboratory investigations

Laboratory Findings in Obstructive Uropathy:

Serum Creatinine:

- An increase in serum creatinine is generally uncommon in patients with obstruction of one kidney unless there is underlying chronic kidney disease.
- However, bilateral obstruction or unilateral obstruction in patients with a single kidney or preexisting kidney disease can lead to elevated serum creatinine levels.

Urinalysis Findings:

- Normal or may show minimal white or red blood cells.
- Significant hematuria might be evident if the urinary tract obstruction (UTO) is caused by a ureteral stone, sloughed papilla, or an intraluminal malignancy (commonly bladder cancer).
- The presence of crystalluria can indicate the presence of kidney stones and might hint at their composition.
- **Pyuria** indicates the presence of inflammation in the urine.
- Infection is indicated by **leukocyte esterase or nitrites**.
- **Hematuria** can be caused by stones, cancer, or infection.

Electrolytes:

- Patients with chronic obstructive uropathy often exhibit Serum electrolyte disturbances. The common one is Hyperkalemia.

Additional Investigations: Total Prostate specific antigen (PSA)

B. Diagnostic imaging modalities:

- The diagnosis of obstructive uropathy relies on distinguishing between lower and upper tract obstructions.
- Renal ultrasonography
- Lower tract obstruction is typically identified by catheterizing the bladder, revealing a high postvoid residual of ≥ 200 ml of urine.
- The upper tract obstruction remains unresolved even after bladder drainage.
- CT urography
- Cystoscopy
- MRI etc...

4.5 MANAGEMENT OF OBSTRUCTIVE UROPATHY

4.5.1 Principles of emergency management of obstructive uropathy.

General Principles of Urinary Tract Obstruction

A. Identifying the Site of obstruction

Lower Tract Obstruction:

Cause:

- Either bladder dysfunction or bladder outlet obstruction.
- Initial Treatment: Bladder decompression is the initial treatment. This involves **the placement of a bladder catheter**.
- Upper Tract Obstruction:
- Placement of a percutaneous nephrostomy tube (PCN) or ureteral stent

B. Determining the Urgency of Treatment in Urinary Tract Obstruction (UTO):

- Emergencies (Relief within several hours):
- Urosepsis: Presence of urosepsis or signs indicating impending sepsis due to a urinary tract infection.
- Severe Metabolic Abnormalities: Such as hyperkalemia.
- Urgencies (Relief within 24 hours):
- Acute Kidney Injury (AKI):
- Bilateral UTO or Obstruction of a Single Functioning Kidney:
- Severe Chronic Kidney Disease or Unclear Duration of Kidney Impairment:

Activity 4.5 Case scenario:



Instruction: Read and analyse the following case and respond to the questions that follow.

Case:

A 60 year old known hypertensive patient presented to emergency with failure to pass of urine and lower abdominal pain of one day duration. He has associated bilateral flank pain of two weeks duration. He was complaining severe lower urinary symptoms like frequency, intermittency, feeling of incomplete emptying and post-micturition dribbling, urgency, poor stream and nocturia of one year duration. Objectively he is in acute pain, and his blood pressure is 145/90, Pulse rate of 92 per minute. His bladder is palpable on the supra pubic palpation. On Ultrasound scanning, he has moderate bilateral hydroureteronephrosis and distended bladder. His prostate volume is 70 ml.

1. How do you approach this patient?
2. What are the core principles in management of this patient?

Time: 15 Minutes

Indications for Catheterization:

- Management of Urinary Retention
- In critically ill patients or for fluid management/diagnostic tests.
- During Surgery: To assess fluid status and prevent bladder overdistention during prolonged or large-volume procedures.
- Post-surgery: For surgeries of the genitourinary tract or adjacent structures (urologic, gynecologic, colorectal).
- Management of Specific Conditions: Hematuria associated with clots, Immobilized patients (e.g., stroke, pelvic fracture), Neurogenic bladder.
- Intravesical pharmacologic therapy (e.g., bladder cancer).

Contraindications:

- Absolute Contraindication
- Urethral injury, typically associated with pelvic trauma. Evaluate blood at meatus or gross hematuria; urologic consultation and urethroscopy may be needed.
- Relative Contraindications
- Urethral stricture, Recent urinary tract surgery (urethra, prostate, bladder), Presence of an artificial sphincter. In such cases, consult a urologist or urogynecologist.

4.5.2 Pharmacological management

Activity 4.5.2



Group Discussion:

Instruction: Form 5 groups and discuss on the following question for 10 minutes and present to the plenary.

How do you optimize patients with OUP besides relieving the obstruction?

Time: 15 minutes

A. Symptomatic Relief:

Pain Management: Opioids may be prescribed to alleviate pain associated with obstructive uropathy.
Urinary Retention: Catheterization may be required to relieve acute urinary retention.

B. Medical Therapies:

- Fluid and Electrolyte Management
- Alpha-blockers
- Antibiotics: Infections, particularly urinary tract infections (UTIs),

4.5.3 Planning further management

Step 1: Diagnosis of Obstructive Uropathy (UTO)

Confirm the level of obstruction: Kidney, Ureter, Bladder, or Urethra.

Step 2: Treatment Based on Etiology

(Refer for further surgical management if at primary health care)

1. Urolithiasis (Refer for confirmation of diagnosis and/or management)

2. Prostatic Disorders (Bladder Outlet Obstruction)

- Medical Management like Alpha-blockers (e.g., Tamsulosin), 5-alpha reductase inhibitors (e.g., Finasteride)

- Refer for Surgical Management

3. Bladder Dysfunction

- Spinal cord injury: Neurogenic bladder management
- Multiple sclerosis: Symptomatic treatment
- Diabetic autonomic neuropathy
- Glycemic control, symptomatic treatment
- Chronic urinary retention
- Clean intermittent catheterization (CIC), medications

4. Ureteral Strictures

- Temporary Management: Indwelling ureteral stent or Percutaneous nephrostomy
- Refer for Definitive Repair: Ureteral reimplantation, Ureteroureterostomy, Ileal ureteral transposition, and other endoscopic procedures.

5. Urethral Stricture: refer for surgical management

- Optical urethrotomy, dilatation,
- Anastomotic urethroplasty
- Graft and flap reconstructive urethroplasty.

6. Extrinsic Ureteral Compression (retroperitoneal or pelvic malignancies):

- Temporary Management: Indwelling ureteral stents, Nephrostomy tubes
- Refer for Definitive Therapy: Treat underlying cause

4.5.3 Monitoring schedule for Obstructive uropathy patients

- RFT serum creatinine and blood urea/BUN/ levels: serial measurements over time help us track changes in renal functions & effectiveness of our management.
- Estimated Glomerular filtration(eGFR): this should be done based on serum creatinine level, age and other factors.
- Urinalysis: to further know if there is kidney damage
- Fluid balance: input and output should be monitored. Post obstructive over diuresis leads to dehydration, and electrolyte imbalance. Decrease urine output after management may indicate progression of acute kidney injury or failure.
- Monitoring and Serial Measurements of serum Electrolyte; In summary, monitoring a patient with obstructive uropathy allows healthcare providers;

4.5.4 Complications associated with obstructive uropathy

- During catheterization: sudden deflation causes reactive bleeding of bladder.
- After catheterization: lack of monitoring of urine output leads to dehydration, electrolyte and fluid imbalance. Patients may have large amount of urine output which needs close follow up like pathologic post obstructive diuresis
- Bladder spasms: Irritation caused by the catheter can trigger involuntary muscle contractions in

the bladder, resulting in bladder

- Hematuria; due to trauma during catheter insertion (urethra and bladder injury),.
- Post-obstructive diuresis (POD) is characterized by prolonged urine production, typically exceeding 200 ml over two consecutive hours or 3000 ml in 24 hours, following the relief of urinary retention or obstructive uropathy.

4.6. CHAPTER SUMMARY

- Obstructive uropathy is a blockage in the urinary tract that leads to urine buildup, causing hydronephrosis and potential kidney damage.
- Causes in Children: - Congenital issues such as posterior urethral valves or renal pelvic dilatation.
- In Adults:- Urolithiasis (kidney stones), Benign Prostatic Hyperplasia (BPH) or prostatic cancer, Pelvic malignancies affecting the ureters, Strictures due trauma or surgery
- Symptoms include Pain (especially in the flank), Dysuria, Hematuria, and frequent urination, nocturia, Oliguria or anuria in severe cases. Nausea and vomiting may occur in acute cases
- Physical Exam: Palpation of the abdomen or flank may reveal tenderness or swelling.
- Imaging: Ultrasound (US): Initial test for hydronephrosis.
- CT scan: For better visualization of stones, tumors, or strictures
- Laboratory Tests: Elevated serum creatinine suggests renal impairment
- Urinalysis: Can show hematuria, pyuria (pus in the urine), or signs of infection
- Emergency management: Catheterization (urethral or suprapubic) to relieve obstruction.
- Definitive management: Surgery to treat the underlying cause (e.g., stone removal, tumor excision, BPH treatment). Antibiotics if infection is present.

CHAPTER 5: COMPLICATIONS OF CHRONIC KIDNEY DISEASES



Time allocated – 3hours and 45 minutes



Chapter description: This chapter describes the major complications of CKD and proposed management for each.



Primary Objective: By the end of this chapter, participants will be able to:

- Identify appropriate management for complications of CKD.

Enabling Objectives: By the end of this chapter, participants will be able to:

- Describe diagnosis and treatment of anemia in CKD appropriately
- Identify CKD-MBD diagnostic modalities and treatments
- Discuss neuropsychiatric, dermatologic, endocrine, and other uremic complications of CKD
- Discuss diagnosis and management of fluid overload in CKD
- Describe the basics of hyperkalemia diagnosis and management
- Explain preparation of patients with advanced CKD for renal replacement therapy

Chapter Outline

- 5.1 Anemia in CKD
- 5.2 CKD-MBD
- 5.3 Neuropsychiatric manifestations of CKD
- 5.4 Dermatologic Manifestation of CKD
- 5.5 Endocrine manifestation of CKD
- 5.6 Other Uremic Complications of CKD
- 5.7 Hyperkalemia
- 5.8 Volume Overload in CKD
- 5.9 Preparation for RRT
- 5.10 Chapter Summary

5.1 ANEMIA IN CKD



Activity 5.1 Case Scenarios

Instruction - Read the following case scenarios and discuss the points mentioned in teams.

Your instructor will create groups.

1. Sara, a 58-year-old female with CKD Stage 5, Type 2 Diabetes, and Hypertension, presents with increased fatigue, pallor, and decreased exercise tolerance over the past month.

Outline anemia work up for Sara

2. Hailu, a 22-year-old male with Stage 4 CKD, presents with symptoms of fatigue and pallor over the past few weeks. Her medical history includes Type 2 Diabetes and Hypertension. Recent lab tests reveal a hemoglobin (Hb) concentration of 10.2 g/dl, a transferrin saturation (TSAT) of 25 % and serum ferritin levels of 80 ng/ml.

Outline the anemia management plan for Hailu

3. Alem, a 32-year-old male with Stage 4 CKD and hypertension, presents with worsening fatigue, shortness of breath, and decreased exercise tolerance over the past two months. His recent lab tests reveal a hemoglobin concentration of 9.5 g/dl, transferrin saturation (TSAT) of 28%, and serum ferritin of 150 ng/ml.

Outline the anemia management plan for Alem

Time Allocated – 30 minutes

DEFINITION: Diagnose anemia in adults and children >15 years with CKD when the Hb concentration is <13.0 g/dl (<130 g/l) in males and <12.0 g/dl (<120 g/l) in females. Diagnose anemia in children with CKD if Hb concentration is <11.0 g/dl (<110 g/l) in children 0.5–5 years, <11.5 g/dl (115 g/l) in children 5–12 years, and <12.0 g/dl (120 g/l) in children 12–15 years.

Screening:

- CKD without anemia - annually in patients with CKD stage3, twice per year in patients with CKD 4–5ND, every 3 months in patients with CKD 5HD and CKD 5PD.
- CKD patients with anemia not being treated with an ESA: AT every 3 months in CKD 3–5TH and CKD 5PD, monthly in patients with CKD 5HD.

ferritin $\geq$ 500 ng/mL, unless otherwise clinically indicated.

In people with anemia and CKD not receiving dialysis or treated with peritoneal dialysis (CKD G5PD), we suggest initiating iron if,

- ferritin <100 ng/ml (<100 $\mu\text{g/l}$) and transferrin saturation (TSAT) $<40\%$, or
- ferritin ≥ 100 ng/ml (≥ 100 $\mu\text{g/l}$) and <300 ng/ml (<300 $\mu\text{g/l}$), and TSAT $<25\%$.

In people with CKD treated with iron, it is reasonable to withhold iron if ferritin ≥ 700 ng/ml (≥ 700 $\mu\text{g/l}$) or TSAT $\geq 40\%$.

Oral iron therapy: Generally little benefit, a trial in pre-dialysis patients, with iron stores checked after 4 – 6 weeks of treatment.

Oral iron formulation

Oral Iron formulation	Dose per Tablet	Elemental iron, per tablet	Starting dose	Considerations
Ferric citrate	1 gm	210 mg	CKD not receiving dialysis: 1 tablet, 3 times daily	In CKD not receiving dialysis, help with phosphate-binding
			CKD G5D: 2 tablets, 3 times daily	In CKD G5D, as a phosphate binder
Ferric maltol	30 mg	30 mg	1 tablet, 2 times daily	Taken between meals
Ferrous sulphate	325 mg	65 mg	1 tablet, 3 times daily	Taken between meals
Ferrous fumarate	325 mg	106 mg	1 tablet, 2 times daily	Gastrointestinal side effects, dark green stools
Ferrous gluconate	300 mg	35 mg	4–6 tablets, daily	Less gastrointestinal side effects and better bioavailability
Liposomal iron	30 mg	30 mg	1 tablet, daily	Less gastrointestinal side effects and better bioavailability
Heme iron polypeptide	12 mg	12 mg	1 tablet, 3–4 times daily	Less gastrointestinal side effects and better bioavailability

Table 12: Oral iron formulations, treatment regimen, and factors influencing the choice between different formulations

Intravenous iron formulations and treatment regimen

Intravenous iron formulations	Elemental iron concentration	Maximum single dose	Minimum infusion time for maximum dose	Minimum injection time Considerations	Considerations
Low-molecular weight iron dextran	50 mg/ml	20 mg/kg	15 min for 50 mg, 100 mg/min 4–6 hours	>60 min	Hypersensitivity lower than high molecular weight dextran
Iron sucrose	20 mg/ml	200 mg	15 min	5 min	For people with CKD G1–G5 not receiving HD, it requires multiple patient visits as 1000 mg cannot be given in a single sitting. (5 doses of 200 mg over 5 weeks)
Ferric gluconate	12.5 mg/ml	125 mg	60 min	10 min	Ferric gluconate in sucrose complex (250 mg 4 doses weekly)
Ferric carboxymaltose	50 mg/ml	750 mg (FDA) 1000 mg (EMA)	15 min	7.5 min (FDA) 15 min (EMA)	Full dose can be given in 1 or 2 sittings (750 mg 2 doses 1 week apart) May cause hypophosphatemia, especially in people with early CKD and kidney transplant recipients
Ferric derisomaltose /Iron isomaltoside	100 mg/ml	1000 mg (FDA) 20 mg/kg (EMA)	20 min	250 mg/min (max. 500 mg) (EMA)	Full dose can be given in single sitting
Ferumoxytol	30 mg/ml	510 mg	15 min	15 min	Full dose can be given in single Sitting Hypersensitivity (due to bolus dosing) rarely occurs

Table 13: Intravenous iron formulations and treatment regimen

When to start ESA?

Address all correctable causes of anemia prior to initiation of ESA therapy.

- CKD ND: Hb <10.0 g/dL
 - Hb > 10g/dL - ESA not to be initiated.
- CKD 5D initiate ESA when Hgb 9-10gm/dl
- Target Hb; not to exceed > 11.5g/dL

Initiation of ESA type therapy is individualized.

- Epoetin alfa and beta CKD not receiving dialysis: 4,000 or 10,000 units weekly or every 2 weeks CKD G5D: 50-100 units/kg, 3 times weekly (may round to convenient dose in units)
- Darbepoetin CKD not receiving dialysis: 40-100 µg every 2–4 weeks CKD G5D: 0.45 µg/kg weekly or 0.75 µg/kg every 2 weeks (may round to convenient dose: 25, 40, 60, 100, 150, or 200 µg (300 µg and 500 mcg also available).
- Methyl polyethylene glycol-epoetin beta CKD not receiving dialysis: 50-120µg every two weeks or 120

ESA Therapy Dose Modification: If HB level increases by >1gm/dl/2wks, then reduce the dose of EPO by 25%. If HB level is not increased by >1gm/dl/ month, then increase the dose of EPO by 25%.

New agents like hypoxia-inducible factor-prolyl hydroxylase inhibitors (HIF-PHIs). In people with anemia and CKD, administer HIF-PHIs at the lowest dose needed to improve symptoms attributable to anemia and to avoid RBC transfusions. People with anemia and CKD, including those with ESA hypo responsiveness, do not use ESAs and HIF-PHIs in combination. In people with anemia and CKD, discontinue HIF-PHI after 3–4 months if a desired erythropoietic response has not been achieved.

Drug doses

- Daprodustat 24 mg daily, Desidustat 150 mg 3 times per week, Enarodustat 8 mg daily, Molidustat 200 mg daily, Roxadustat 3.0 mg/kg body weight 3 times per week, Vadadustat 600 mg daily.

In people with anemia and CKD treated with roxadustat, monitor thyroid stimulating hormone and free T3 and T4 after 4 weeks of therapy initiation. In people with anemia and CKD, HIF-PHIs should be avoided in those at increased risk of adverse events like Active cancer. Polycystic kidney, Proliferative retinal disease, Pulmonary arterial, hypertension, Prior cardiovascular events, Prior thromboembolic events, Hepatic impairment Seizures, but kidney transplant anemia and in children not established.

Transfusion in CKD

In people with anemia and CKD, red blood cell (RBC) transfusion is part of a comprehensive treatment strategy, carefully weighing risks and benefits in a shared decision-making process.

In people with anemia and CKD eligible for organ transplantation, avoid, when possible, RBC transfusions to minimize the risk of all sensitizations.

Benefits of RBC transfusions may outweigh its harms in people in whom:

- ESA or HIF-PHI therapy is ineffective (e.g., hemoglobinopathies, bone marrow failure, ESA or HIF-PHI resistance)
- ESA or HIF-PHI therapy is harmful (e.g., previous or current malignancy, previous stroke)

We recommend basing the decision to transfuse a person with CKD and chronic anemia on symptoms and signs caused by anemia rather than an arbitrary Hb threshold. In people with CKD and acute anemia consider RBC transfusion when the benefits outweigh the risks, including:

- When rapid correction of anemia is required to stabilize the patient's condition (e.g., acute hemorrhage, unstable coronary artery disease)
- When rapid preoperative Hb correction is required.

5.2 CHRONIC KIDNEY DISEASE – METABOLIC BONE DISEASE (CKD-MBD)

Activity 5.2. Case Scenario



Instructions - Read the following case scenarios and discuss the points mentioned in teams. Your instructor will create groups.

1.Hana, a 55-year-old female with Stage 4 CKD, hypertension, and Type 2 diabetes, presents with generalized muscle weakness, lower back pain, and difficulty walking over the past few months.

Outline how to screen CKD-MBD for HANA

2.Bekele was diagnosed with CKD Stage 5 three years ago. He has a history of secondary hyperparathyroidism and has been receiving treatment to manage his condition.

Discuss the frequency of biochemical tests using the participant manual for CKD-MBD Bekele

3.Amanuel, a 59-year-old male with Stage 5 CKD, hypertension, and secondary hyperparathyroidism, presents with muscle weakness, bone pain, and difficulty walking over the past few months. Recent lab tests reveal biochemical abnormalities, including low serum calcium, high serum phosphate, elevated PTH levels (> 10X UNL), and low vitamin D. He is diagnosed with CKD-MBD, and a comprehensive management plan is implemented.

Discuss the management plan for Mr Amanuel

Time allocated – 30 minutes

Definition: Chronic kidney disease—mineral and bone disorder (CKD- MBD) is a common complication of chronic kidney disease and is a part of broad-spectrum disorders of mineral metabolism.

Pathophysiology

Reduced renal mass leads to reduced activity of 1-hydroxylase in the renal tubule and thus failure to increase calcitriol production when required. However, circulating calcitriol concentrations begin to fall at stage 3 CKD either as a direct result of phosphate retention or as a secondary effect via FGF-23 stimulation.

Lowered calcitriol leads to reduced calcium absorption from the gut and proximal tubule, thus causing a tendency to hypocalcemia which is counteracted by increased PTH production and secretion. The net effect is secondary hyperparathyroidism (i.e. abnormally high PTH concentrations as an appropriate response to hypocalcemia), which further aggravates hyperphosphatasemia (positive feedback). Reduced glomerular filtration of phosphate leads to phosphate retention, usually becoming clinically evident at stage 4–5 CKD. Therefore, CKD-MBD results from complex and inter-related metabolic derangements of Vitamin D, calcium, phosphorus, and bone development.

Calcium Metabolism:

Calcium plays a crucial role in muscle and cardiac function as well as bone structure. 99% of calcium is bound and associated with bone structures. The majority of free form is found in the intracellular and extracellular fluid compartments. Plasma concentration is tightly regulated by the actions of PTH and calcitriol (1, 25-dihydroxycholecalciferol). Normal serum calcium levels range from 8.5 to 10.5 mg/dl. Maintenance of calcium homeostasis depends on:

- The adaptation of intestinal Ca^{2+} absorption to the needs.
- The balance between bone formation and resorption, and
- Urinary excretion of calcium.

Phosphate Metabolism:

Phosphorus plays a crucial role in cell structure and metabolism. It regulates enzymatic activity, serves as an essential component of nucleic acids and phospholipid membranes, and it is structural component of teeth and bones as hydroxyapatite. Less than 1% circulates in serum as HPO_4^{2-} and H_2PO_4^- ions, in a 4: 1 ratio at physiologic pH of 7.4. Normal serum phosphate levels range from 2.8 to 4.5 mg/dl. It is mainly regulated by intestines, kidneys, and bones through interaction of PTH, calcitriol, and FGF-23. It is widely found in milk products, meat, eggs, cheese, yogurt, cereals, cola, chocolate, soft drinks, nuts, and is used extensively as food additives. 60% to 75% of total phosphate intake is absorbed by the intestine stimulated by calcitriol. Calcium, magnesium, and aluminum bind to phosphate, limiting its absorption. Kidneys play a major role in controlling extracellular phosphate homeostasis. Phosphate is freely filtered and is reabsorbed primarily in the proximal tubule depending on the body's needs. FGF 23 plays a crucial role in phosphate balance. It is phosphaturic. It also

decreases serum phosphate levels by inhibiting calcitriol production. Normally, phosphate excretion equals intestinal absorption.

Clinical manifestations of CKD-MBD:

This includes clinical, biochemical, and imaging abnormalities related to the uremic state resulting in a complex disorder called CKD-MBD. The spectrum of disorders includes:

- Abnormal concentrations of serum calcium, phosphate, and magnesium.
- Disorders of parathyroid hormone (PTH) and vitamin D metabolism, and
- Skeletal abnormalities in imaging.

In the early stages of CKD, CKD-MBD is mostly asymptomatic, with clinical manifestations often preceded by biochemical or imaging abnormalities. It typically begins in stage 3 CKD and progressively worsens, leading to serious complications.

Clinical consequences of CKD-MBD:

In advanced CKD muscle weakness, fibrosis of cardiac muscle, bone deformities and fractures, and extra skeletal calcifications including vascular calcifications may occur.

Secondary hyperparathyroidism: Is characterized by progressive increases in parathyroid hormone (PTH) levels as a response to hypocalcemia. It stimulates osteoclastic activity resulting in cortical bone resorption and marrow fibrosis.

Tertiary hyperparathyroidism: When PTH production may become autonomous with acceleration of bone destruction and vascular calcification.

Renal osteodystrophy: May manifest as aches and pains in the lower back, hips, and legs, which are often aggravated by weight bearing. Pain around joints may occur as a result of acute arthritis and peri-arthritis due to hyperphosphatemia (periarticular deposition of calcium phosphate crystals).

Vascular Calcifications: In CKD-MBD, there is a greater proportion of calcification in the arterial media which causes vascular stiffness and hypertension. Calciphylaxis is a condition where small cutaneous blood vessels become calcified, leading to acute, painful necrosis and ulceration of the skin.

Work Up of CKD-MBD:

To identify and monitor the consequences of CKD-MBD, a number of sequential laboratory and imaging tests need to be done regularly. These include serum levels of Ca²⁺, PO₄, and PTH, alkaline phosphatase and vitamin D; and radiologic imaging for renal osteodystrophy. The frequency of doing these biochemical tests is summarized below:

KDIGO recommendations for frequency of biochemical testing in CKD					
CKD Stage	Calcium	Phosphorus	PTH	Alkaline phosphatase	Calcidiol (25(OH)D) _a
3	Every 6 – 12 months	Every 6 – 12 months	Baseline _x	Baseline	Baseline
4	Every 3 – 6 months	Every 3 – 6 months	Every 6 – 12 months	Every 6 – 12 months	Baseline
5 and 5D	Every 1 – 3 months	Every 1 – 3 months	Every 3 – 6 months	Every 3 – 6 months	Baseline

_x Other guidelines, including those produced by NICE in the UK, do not advocate routine testing of serum PTH in CKD stage 3.

_a Calcitriol (1,25(OH) 2 D) is not routinely measured, as it is present in only tiny amounts in serum, and the assay is technically difficult and expensive.

Table 14: Work up for CKD-MBD

TREATMENT OF CKD-MBD:

Goals: Keep serum Ca²⁺ and PO₄ within the normal range, keep bone turnover and strength as near normal as possible, keep serum PTH appropriate to these objectives, and prevent the development of parathyroid hyperplasia.

Comprehensive management of CKD-MBD comprises a variety of preventive and therapeutic strategies including:

- Calcium supplementation,
- Dietary phosphate restriction,
- Phosphate binders,
- Vitamin D analogues,
- Calcimimetics, and
- Parathyroidectomy.

Treatment of Hypocalcemia:

Hypocalcemia should be corrected because it is a potent stimulus for PTH secretion. Free calcium should be measured in hypoalbuminemic patients, or it should be corrected for Hypoalbuminemia. Target is to normalize serum Ca²⁺ with appropriate calcium intake ± supplementation (including calcium-containing binders).

Mild to Moderate CKD - calcium supplements such as calcium carbonate, taken between meals and increasing doses as required. Assess vitamin D status by measurement of 25-hydroxyvitamin D and correct it if below 30 ng/ml. Active vitamin D sterols should be considered if hyperparathyroidism or hypocalcemia persists.

Management of Hyperphosphatemia:

Hyperphosphatemia increases CKD progression and increases the risk of all-cause mortality. Phosphate lowering therapy is the cornerstone of CKD-BMD management. It comprises dietary phosphate restriction, oral phosphate binders to prevent intestinal absorption, suppression of PTH synthesis and secretion, and removal of phosphate through adequate dialysis. Calcitriol or vitamin D analogues (e.g. alfacalcidol, paricalcitol) are used to suppress PTH production.

The absence of hyperphosphatemia doesn't indicate a normal parathyroid function in mild to moderate CKD. Thus, don't delay initiation of therapy until frank hyperphosphatemia develops. Dietary phosphate restriction should be started in CKD stage 2 or 3. It reduces CKD progression and increases serum calcitriol levels. If possible, involvement of a dietitian or nutritionist is essential. Cautious protein restriction and avoidance of phosphate rich diets, particularly processed foods and food additives are important.

Phosphate Binders: Different phosphate binders are used to limit intestinal phosphate absorption and achieve target phosphate levels. Are required as renal function deteriorates and hyperphosphatemia develops. Calcium based or non-calcium-based phosphate binders can be used depending on serum calcium level.

Calcium based phosphorus binders: Calcium carbonate or calcium acetate taken with meals effectively binds phosphates and limits their absorption. They are effective phosphate binders in 60% to 70% of patients on HD, dose is adjusted based the binding capacity of phosphate but limit the ingestion of elementary calcium to 1500 mg/day particularly in low turnover disease (PTH < 2X of the upper limit of normal). Avoid these agents if there is hypercalcemia since it may lead to vascular and soft tissue calcification.

Sevelamer Hydrochloride: Starting dose of sevelamer carbonate tablets is 800/mg orally three times per day with meals. Titrate 800 mg per meal in two-week intervals for adult patients as needed to obtain serum phosphorus target. In case of metabolic acidosis, it is better to use sevelamer carbonate.

These agents do not increase calcium load, are associated with fewer episodes of hypercalcemia, and help reduce soft tissue calcification. They also attenuate the progression of coronary and aortic calcification in hemodialysis patients. Additionally, they reduce LDL cholesterol while increasing HDL cholesterol.

Approximate potential phosphate-binding capacities of commonly used agents is: Calcium carbonate:

1 g binds 40 mg, Calcium acetate: 1 g binds 45 mg, Sevelamer: 1 g binds 36 mg, Lanthanum carbonate: 1 g binds 93 mg, Aluminum hydroxide (liquid): 1 g binds 60-75 mg.

Treatment of hyperparathyroidism: Optimize management of modifiable factors, including hypocalcemia, hyperphosphatemia and vitamin D deficiency. Use of calcitriol and vitamin D analogs is recommended for patients with CKD G4– G5 with severe and progressive hyperparathyroidism. In patients with CKD G5D, we suggest maintaining intact PTH levels in the range of approximately 3 to 9 times the upper normal limit. Avoid PTH over suppression of Adynamic bone disease. In patients with CKD G3a–G5D with severe hyperparathyroidism who fail to respond to medical or pharmacological therapy, we suggest parathyroidectomy.

Vitamin D analogues: KDIGO guidelines recommend screening in CKD stages 3 – 5, if Serum 25(OH) D <30ng/mL (<75nmol/L) represents insufficiency. Active analogues like alfacalcitriol 0.25 to 1 microgram po daily can be used. It can also be treated with cholecalciferol in formulations with calcium, e.g. calcium 600mg with D3 400 units (one tablet BID). An alternative is cholecalciferol D3 as 20,000-unit capsules: two capsules once daily for 7 days, then maintenance 2 x 20,000 unit capsules 1/month.

Calcimimetics: are indicated for the treatment of secondary hyperparathyroidism in patients with stage 5 CKD or on dialysis if not responding to vitamin D analogues. Starting dose is 30 mg once daily with food or shortly after a meal. Serum calcium and serum phosphorus should be measured within 1 week, and PTH should be measured 1 to 4 weeks after initiation or dose adjustment of Cinacalcet. Cinacalcet should be titrated no more frequently than every 2 to 4 weeks through sequential doses of 60, 90, 120, and 180 mg once daily to target iPTH consistent with the NKF-K/DOQI recommendation for CKD patients on dialysis of 150-300 pg/mL. PTH levels should be assessed no earlier than 12 hours after dosing with Cinacalcet.

Monitoring Response to Therapy: Poor control of serum PO₄, serum Ca²⁺, calcium phosphate product (CaxP), and PTH are all associated with increased CV morbidity and mortality. An increase in CaxP is associated with soft tissue and cardiac calcification.

Regular monitoring of response to therapy should be assessed to achieve recommended targets (summarized in the following table).

CKD Stage	GFR Range (mL/min/1.73 m ²)	Phosphorus (mg/dl)	Calcium (Corrected, mg/dl)	Intact PTH (pg/mL)
3	30 to 59	2.7 to 4.6	8.4 to 10.2	35 to 70
4	15 to 29	2.7 to 4.6	8.4 to 10.2	70 to 110
5 (KDIGO)	Less than 15, dialysis	Normal range	Normal range	150 to 600 (2 to 9 × normal)

Table 15: Recommended target serum values in CKD-MBD treatment

5.3 NEUROPSYCHIATRIC MANIFESTATIONS OF CKD



Activity 5.3 – 5.6 Case Scenarios for group discussion

Instructions - Read the following case scenarios and discuss the points mentioned in teams. Your instructor will create groups.

Discuss diagnosis and which case scenarios require emergency dialysis.

1. A 65-year-old known stage 5 CKD patient presented with a new onset generalized tonic-clonic seizure and depressed level of consciousness.

Serum Creatinine – 12.4 mg/dl, Urea – 302 mg/dl, RBS – 104 mg/dl, Brain MRI – Unremarkable

2. A 35-year-old man on follow-up for CKD presented with fatigue, poor appetite, and frequent vomiting of ingested matter of 01-week duration. The patient came after being lost to follow-up for one year. Blood pressure – 160/90 mmHg, Urine output- 1200 ml/24 hours, Serum Creatinine – 4.2 mg/dl, Urea – 72 mg/dl, Serum K – 3.2

3. A 47-year-old female who is on follow-up for hypertension and CKD, presented complaining of itching over her back and upper extremities for the past two months. Serum Creatinine – 6.2 mg/dl, Urea – 184 mg/dl, Serum K – 4.8, Serum Ca – 0.81, Serum PO4 – 8.4

4. A 38 years old female known stage 5 CKD patient presented with persistent bilateral nasal and gum bleeding two days duration

Serum Creatinine – 12.2 mg/dl, Urea – 394 mg/dl, Hgb – 7.3 g/dl

5. A 75-year-old stage 4 CKD patient was admitted for Acute watery diarrhea with superimposed AKI. He is dehydrated and urine output is < 50 ml/24 hours. Recent Cr – 7.8 mg/dl, Urea – 242 mg/dl, Serum K – 3.2

6. The above patient developed rapid and shallow breathing. ABG PH < 7.1, Serum K – 3.8, Creatinine rose to 8.4 mg/dl, Urea – 302 mg/dl. Urine output is 500 ml/24 hours after hydration.

7. A 46-year-old known RVI, diabetes, and stage 5 CKD patient came with cough, poor appetite, and generalized body swelling of one month duration. Chest X-ray – Bilateral pleural effusion (More on the left side), cardiomegaly. Echocardiography – Massive pericardial effusion. Serum Cr – 8.4 mg/dl. Urea – 252 mg/dl.

8. A 25-year-old stage 5 CKD patient on follow-up presented with one week history of anterior chest pain and shortness of breath. Patient has pericardial friction rub on pericardial auscultation. Serum Creatinine – 8.4 mg/dl. Urea – 252 mg/dl. Echocardiography – Mild pericardial effusion.

Time allocated – 30 minutes

Uremic Neuropathy/Encephalopathy

Dysfunction of the central and peripheral nervous system, including encephalopathy, polyneuropathy, and mononeuropathy, are important complications of kidney failure. Sensory dysfunctions, characterized by restless leg or burning feet syndromes, are frequent presentations of uremic neuropathy.

Early clinical features of Uremic Encephalopathy include lethargy, irritability, disorientation, hallucination, rambling speech, and asterixis. Generalized seizures can also occur. A comma is unusual but it can occur. Other rare signs and symptoms include focal signs such as hemiparesis and focal myoclonus. These complications are usually absolute indications for the initiation of dialysis.

We recommend the initiation of dialysis in patients with uremic neuropathy and/or Encephalopathy.

5.4 DERMATOLOGIC MANIFESTATION OF CKD

Uremic Pruritis

Pruritis is common among stage 5 CKD. Factors identified for pruritis development in CKD are hyperphosphatemia, hypercalcemia, hypermagnesemia, xerosis, and uremic toxins. Uremic pruritis causes serious discomfort and skin damage, often associated with disturbance of day and night rhythm, sleeping disorders, depression, anxiety, and diminished quality of life.

Treatments start with ruling out other common dermatologic problems. Correcting serum Calcium-phosphorus abnormalities, and using emollients, antihistamines, phototherapy, opioid receptor agonists and gabapentin are the next steps of treatment.

5.5 ENDOCRINE MANIFESTATION OF CKD

- Impaired renal clearance: prolactin; insulin; glucagon; calcitonin; leptin.
- Increased secretion: parathyroid hormone (PTH). Calcitonin; prolactin.
- Decreased sensitivity to hormones:
 - Altered target response: insulin; growth hormone; PTH; erythropoietin; 1, 25-(OH)₂ vitamin D₃.
 - Decreased circulating hormone levels: 1, 25-(OH)₂ vitamin D₃; erythropoietin. Decreased secretion by other endocrine glands: testosterone, estrogen, progesterone.

- Thyroid hormones: Total and free triiodothyronine (T3) concentrations are Decreased.
- Growth hormone: Plasma GH levels are increased because of increased secretion, impaired clearance, and sensitivity to GH is decreased.
- Prolactin: Significance of increased prolactin levels in CKD leads to amenorrhea, impotence, and gynecomastia.
- Adrenal hormones: clinical abnormalities of glucocorticoid function are not common in CKD and ESRD.
- Insulin and carbohydrate metabolism: Abnormalities of carbohydrate metabolism in CKD include insulin resistance (IR); circulating inhibitors of insulin action; decreased islet cell insulin secretion.
- Sexual dysfunction in males: Clinical features include loss of libido, impotence, testicular atrophy, gynecomastia and infertility. Sexual dysfunction in females: Clinical features loss of libido, irregularity of menstrual, amenorrhea, anovulation and sterility.

Treatment of endocrine disorders is not usually addressed in CKD patients. Considering and planning an endocrine work up/evaluation is mandatory.

5.6 OTHER UREMIC COMPLICATIONS OF CKD

Malnutrition & Uremic Gastropathy

Protein malnutrition is common in patients with advanced chronic kidney disease (CKD) due to lower food intake (principally due to anorexia), decreased intestinal absorption and digestion, and metabolic acidosis. And end-stage renal disease (ESRD). All these derangements can further worsen kidney function, leading to poor patient outcomes.

A low plasma concentration of serum albumin concentration may be indicative of malnutrition.

We suggest measurement of serum albumin every 3 months among patients with eGFRs <20 mL/min/1.73 m²

The management of malnutrition in CKD must be individualized and based on the degree of kidney dysfunction, comorbid condition, baseline nutritional status, and physical functional capacity. Close nutritional counseling and supervision are recommended.

Uremic gastropathy is a term commonly used to describe upper gastrointestinal signs and changes due to uremia. Poor appetite, nausea, vomiting, fatigue, GI bleeding, and uremic fetor are common symptoms.

Treatment can be tried with H₂- H₂-receptor antagonists, such as ranitidine, famotidine..., Protein pump inhibitors, anti-acids & anti-emetics. Uremic gastropathy can persist despite these measures and may require initiation of renal replacement therapy.

We recommend referral for kidney replacement therapy in patients whose malnutrition and/or gastropathy symptoms are refractory to individualized nutritional and medical interventions.

Uremic Pericarditis

Anterior chest pain, shortness of breath, and a pericardial friction rub are the major presentations of uremic pericarditis. It is not associated with ST, or T elevations because it is metabolic without injury to the pericardium.

The development of otherwise unexplained pericarditis in a patient with advanced kidney failure is an indication to institute dialysis (providing there is no circulatory compromise or evidence of impending tamponade). Most patients with uremic pericarditis respond rapidly to dialysis, with a resolution of chest pain as well as a decrease in the size of the pericardial effusion.

We recommend the initiation of dialysis in patients with uremic pericarditis.

Uremic Bleeding

Contributing factors for uremic bleeding include platelet dysfunction, uremic toxins interfering with platelet aggregation, anemia, vessel abnormalities, and von Willebrand factor abnormalities that happen with advanced stages of CKD.

Treatment includes supportive care, anemia correction, desmopressin, cryoprecipitate and even conjugated estrogen,

Uremic bleeding can be an indication of initiation of dialysis.

Metabolic acidosis

Metabolic acidosis in CKD is a common complication of advanced CKD and leads to poor outcomes, such as bone demineralization, muscle mass loss, and worsening of renal function. Metabolic acidosis is assessed by measuring serum bicarbonate concentration and PH where feasible.

We recommend alkali therapy to maintain the serum bicarbonate concentration above 22 mEq/L.

Initial Oral sodium bicarbonate dose of 650mg twice daily should be used. This treatment is also used to slow progression to end-stage renal disease.

Refer for hemodialysis if acidosis persists despite sodium bicarbonate.

NB: caution should be taken with long-term use of sodium bicarbonate due to its fluid retention effects leading to peripheral and pulmonary edema, weight gain, and increased blood pressure

5.7 HYPERKALEMIA

Hyperkalemia is an elevation of the blood potassium level, usually defined as a serum concentration ≥ 5.5 mmol/L. It is a frequently occurring biochemical abnormality, which can result in serious sequelae including arrhythmias and cardiac arrest. Most of the body's potassium is stored intracellular, and the extracellular potassium is maintained at a relatively constant level by a variety of homeostatic mechanisms

Serum potassium may increase via increased intake, reduced urinary excretion, or redistribution from cells/cell damage, e.g. tumor lysis syndrome or rhabdomyolysis/crush syndrome. Under normal circumstances, the body's (particularly the kidneys') ability to excrete potassium can be increased to avoid hyperkalemia. In certain disease states, such as chronic kidney disease (CKD), this compensation may not occur to the same degree.

Pathophysiology

Dietary potassium intake is variable but is usually in the region of 40–200 mmol per day. Ingested potassium undergoes absorption in the gastrointestinal tract, and excretion is predominantly via the urinary route. The principal cells in the distal convoluted tubules are integral to this excretory process. The colon is responsible for approximately 10% of potassium excretion in those with normal kidneys; however, this route of excretion may be more significant in those with impaired renal excretion. It stands to reason that diarrhea and increased gastrointestinal losses as well as sweating could decrease serum potassium levels.

Key factors in promoting potassium excretion include:

- Increased potassium intake or blood concentration
- Delivery of sodium to distal tubule (where it exchanges with potassium)
- Aldosterone secretion

Beta-2-adrenergic receptors and insulin serve to drive potassium into cells via increasing the activity of the Na-K-ATPase pump. This forms part of the rationale for their use in the management of acute hyperkalemia. Conversely, beta-blockade and insulin deficiency will promote potassium efflux.

Etiology

Hyperkalemia can result from increased potassium intake, decreased potassium excretion, or a shift of potassium from the intracellular to the extracellular space. The most common causes involve decreased excretion.

Increased potassium intake

Alone, increased intake of potassium is a rare cause of hyperkalemia, because the mechanisms for renal excretion and intracellular disposition are very efficient. Increased intake may result from the

following:

- High-potassium, low-sodium diets
- Ingestion of potassium supplements
- High concentrations of potassium in IV fluid preparations (eg, total parenteral nutrition formulas)
- Dietary salt substitutes
- Packed red blood cell transfusion (risk peaks at 2-3 weeks of cell storage)

Decreased potassium excretion

Almost all patients with persistent hyperkalemia have impaired renal excretion of potassium. Mild degrees of renal failure generally do not result in resting hyperkalemia, because of compensation by adaptive mechanisms in the kidneys and GI tract. However, once the GFR falls below 15-20 mL/min, significant hyperkalemia can occur, even in the absence of an abnormally large potassium load. The simple lack of nephron mass prevents normal potassium homeostasis.

Medications that can decrease potassium excretion include the following:

- Potassium-sparing diuretics (e.g., spironolactone, triamterene, amiloride)
- NSAIDs
- ACE inhibitors
- Angiotensin-receptor blockers (ARBs)
- Cyclosporine or tacrolimus
- Pentamidine
- Trimethoprim-sulfamethoxazole
- Mineralocorticoid receptor antagonists (MRAs)
- Heparin
- Ketoconazole

Shift of potassium into extracellular space

Like increased intake, this is rarely the sole cause of hyperkalemia, because the mechanisms for renal excretion are very efficient. However, the inability to intracellularly transport potassium exacerbates hyperkalemia in individuals with impaired renal excretion.

Factors that can shift potassium into the extracellular space include the following:

- Metabolic acidosis
- Beta-adrenergic blockade
- Acute tubular necrosis
- Electrical burns
- Thermal burns
- Cell depolarization
- Head trauma

Classification

Mild - 5.5– 5.9 mmol/L

Moderate - 6.0–6.4 mmol/L

Severe - ≥ 6.5 mmol/L and/or ECG changes

Signs and symptoms

Hyperkalemia is often clinically silent with no symptoms and is just apparent in biochemistry results. On occasions where patients do have symptoms, they are often vague and non-specific, including muscular weakness, nausea, muscle pain, lethargy, and paresthesia.

Abnormalities may be detected on electrocardiogram (ECG) which can progress with increasing severity of hyperkalemia. Such changes include “tenting” of T waves, P wave loss/flattening, increased QRS duration, and an eventual “sine wave pattern”.

Approximate Serum K ⁺ Concentration	Illustrative EKG	EKG Changes
4.0		None
6.0		Peaked T-wave
8.0		P-wave flattening Prolonged PR Prolonged QRS
>9.0		Sine wave/asystole

Figure 11: EKG changes in Hyperkalemia

Additional treatments

- Discontinuing drugs which can cause hyperkalemia e.g. enalapril, Losartan, Spironolactone...
- β_2 -Adrenergic agonist - Nebulized albuterol/ salbutamol, 10 – 20 mg helps shift potassium into cells.
- Loop diuretics, particularly in volume-replete or hypervolemic patients who are non-oliguric
- Dietary restriction - low potassium diet is recommended for all patients with persistent serum potassium values greater than 5.5 mmol/L
- Novel oral potassium binders have been incorporated into new guidelines for those with life-threatening hyperkalemia in the form of either sodium zirconium cyclosilicate (SZC) at a dose of 10 grams three times per day for 3 days, or Patiromer at a dose of 8.4 grams once a day.
- Every patient with severe hyperkalemia needs to be referred to a hospital with a setup for hemodialysis. Acute hemodialysis is indicated for potassium removal in AKI or advanced CKD when hyperkalemia is life-threatening and not responding to medical management.
- Mild to Moderate hyperkalemia with no EKG changes can be managed with diet, medication revision, and diuretics as needed.

Mechanism	Therapy	Dose	Onset	Duration
Antagonize membrane depolarization	Calcium	Calcium gluconate, 10% solution, 10ml IV over 10min	1-3 minutes	30-60 minutes
Increase cellular potassium intake	Insulin	Regular insulin, 10IU IV; add dextrose, 50%, 50 ml IV if Plasma glucose < 250 mg/ dl	30 minutes	4-6 hours
	β_2 - Adrenergic agonist	Nebulized albuterol (salbutamol), 10mg	30 minutes	2-4 hours
Remove Potassium	Sodium polystyrene sulphonate	Kayexalate, 20g orally, 8hourly	1 - 2 hours	4-6 hours
	Hemodialysis	Blood flow and dialysate flow as tolerated, avoid excessive K gradient which can cause cardiac ectopy and hypokalemia.	Immediate	Until dialysis completed

Table 16: Summary of severe hyperkalemia management in CKD

5.8 VOLUME OVERLOAD IN CKD



Activity 5.8 - Case Scenarios for group discussion

Instructions - Read the following case scenario and discuss the points mentioned in teams.

Your instructor will create groups.

Ato Getachew is a 64-year-old man who came to the emergency room with bilateral leg swelling with difficulty of breathing over the past week. He has been a known diabetic and hypertensive patient for the past four years. He was admitted with heart failure with preserved ejection fraction secondary to hypertensive heart disease six months back. He has been on follow-up for CKD stage 4 for the past year.

He currently takes Amlodipine 10 mg, Metformin 500 mg Po BID, Furosemide 40 mg Once daily, and Losartan 50 mg Po daily.

He was a smoker (20 pack years) and quit five years ago.

BP – 160/90 mmHg, PR – 94, SaO₂ – 90% with 2-liter intranasal oxygen

The patient is fatigued and short of breath

Chest – Bibasal crackles

Elevated JVP with S3 heart sounds

Positive abdominal shifting dullness, Bilateral pitting edema up to the knees.

Investigations

- Serum Creatinine – 3.8 mg/dl (Baseline – 3.2 mg/dl), K – 5.4, Na – 128, Hgb – 10.2, Urinalysis – Protein +3
- Chest X-ray – Grade III Pulmonary edema with bilateral pleural effusion
- Echocardiography – Concentric LVH, Grade II diastolic dysfunction, EF – 55%

I – Outline problem lists/diagnosis

II – Identify contributing factors that could lead to his current complaints

III – Outline management plan

Time allocated – 20 minutes

Fluid retention and overload may become a problem with worsening CKD severity and usually manifests as edema. Edema is rarely caused by early-stage CKD alone (except in nephrotic syndrome) and is more a feature of advanced-stage CKD.

The potential causes of edema in patients with CKD include

- Reduced water excretion and reduced urine output in advanced CKD,
- Sodium retention and/or excess sodium intake,
- Urine protein loss and low blood albumin in nephrotic syndrome; and
- Medication (amlodipine, nifedipine, steroids).

Clinical Examination

High blood pressure, edema, shortness of breath, signs of pulmonary edema/pleural effusion, and ascites can be signs and symptoms of fluid overload.

Assessment of fluid status includes

- History – Body swelling, shortness of breath, orthopnea, cough
- Physical exam – High blood pressure, raised jugular venous pressure, third heart sounds, rales, peripheral edema, ascites.
- Chest x-ray
- Ultrasound – Inferior vena cava indices, pulmonary indicators, third space fluid collection
- Bioimpedance, if available

Management

Mild ankle edema that is not symptomatic does not usually need to be treated; and may be managed conservatively by raising legs, using stockings, and moderate sodium restriction.

Diuretic therapy with loop and thiazide diuretics should be used for treating ankle edema only after an assessment of volume status has occurred.

Diuretic resistance may occur in stages of CKD, necessitating diuretic doses may need to increase. Refractory edema in advanced CKD is usually an indication to commence dialysis.

Other management considerations

- Reduction of fluid intake – To be titrated based on the patient's urine output
- Restriction of salt intake to less than 2g/day (half teaspoonful/day)
- Monitoring of weight daily- daily weight loss should depend on the degree of fluid overload however aim at weight loss of 0.5L/day
- Diuretics are given concurrently with the above measures.
- Loop diuretics- furosemide- 20- 80 mg/day but can be increased every 2 hours for IV administration and 6 hours for oral administration, maximum of 160-200 mg/dose, do not exceed 600 mg/day

- Torsemide- 10- 20 mg/day increased to a maximum daily dose of 40mg
- Thiazides – This group of diuretics is thought to be less effective in CKD especially when GFR is < 30mL/min. However, several interventional studies suggest that thiazides are effective even among patients with advanced CKD and can be tried. These drugs include Hydrochlorothiazide, Chlorthalidone, and Indapamide.
- Consider referral for dialysis in acute pulmonary edema or fluid overload not responding to the above therapies.

5.9 PREPARATION FOR RRT



Activity 5.9 - Case Scenarios for group discussion

Instructions - Read the following case scenarios and discuss the points mentioned in teams.

Your instructor will create groups of three. Select one individual to present the group's opinion after discussion.

Scenario A

A 28-year-old lady with a diagnosis of advanced CKD is referred to you to be counselled on possible vascular access creation. She is clinically stable, and her recent labs show a serum creatinine value of 6.5 mg/dl. Her GFR estimation was 17.8 ml/min/1.73m². The cause of CKD is hypertensive nephropathy after 10 years of poorly controlled hypertension. She is a patient with renal artery stenosis with poor drug adherence.

Discussion point

- Options for vascular access

Scenario B

A 65-year-old male patient came to your dialysis center. He was sent to you by his nephrologist, and he was counseled to start hemodialysis. He is diabetic for the past 18 years, hypertensive for 10 years, a smoker for 40 years, and has been on follow-up for ischemic heart disease and CKD for the past 03 years. He has had shortness of breath, acidotic breathing, and altered mentation for the past 03 days. His attendants brought up his recent cardiac exam results, and his echocardiography shows ischemic heart disease with an ejection fraction of 40 % and moderate pulmonary hypertension.

Discussion Points

- Options of vascular access for his current condition
- Long term options for vascular access

Scenario C

43 years old lady living with HIV for the past 3 years, not on HAART, and recent CD4 count of 124 presented to your dialysis center for initiation of hemodialysis. She was diagnosed with advanced CKD, secondary to obstructive uropathy, secondary to advanced cervical cancer. The oncologist who was following her has decided on palliative care only. Her younger sister is willing to donate a kidney to her.

Discussion points

- Option of RRT
- Options for vascular access

Time allocated – 20 minutes

All patients with CKD G4 with progressive decline in kidney function should be educated on all modalities of kidney replacement options, including kidney transplantation.

RRT modality selection depends on the goal of care, local resources, capacities of health care facilities, family/social support, patient's own cost of care, level of dependence/independence, preservation of residual kidney function, and underlying medical conditions.

Dialysis Access Preparation

HD vascular access

- Referral for surgical evaluation and creation of new Arteriovenous (AV) access should be done at eGFR 15 – 20 ml/min/1.73 m² or even earlier in patients with unstable and/or eGFR decline of > 10 ml/min/1.73m².
- The type of access depends on the remaining life expectancy and quality of blood vessels. Permanent catheters may be reasonable if life expectancy is less than a year whereas arteriovenous fistula (AVF) or arteriovenous graft (AVG) would be preferred in patients with longer life expectancy.
- After AV access creation, whole arm isometric exercise rather than finger exercise is recommended. Use of heparin, aspirin, clopidogrel or other antiplatelet is not helpful for the sole purpose of AVF maturation or improvement of primary/secondary patency.

If peritoneal dialysis (PD) is planned, access should be placed at least two weeks before RRT. Urgent PD access placement is also an option if medically safe and performed properly.

Initiation of Maintenance Dialysis

When to start?

When patients reach stage 5 CKD (eGFR < 15 ml/min/1.73m²), nephrologists should evaluate the benefits, risks, and disadvantages of beginning RRT.

These are the indications to start the RRT.

- Recurrent Hyperkalemia despite correction of all contributing factors
- Volume overload, leading to pulmonary edema or refractory to diuretics.
- Uremic Pericarditis, unexplained pericardial effusion
- Persistent nausea/vomiting
- Severe metabolic acidosis
- Declining health
- Severe malnutrition attributable to the uremic state

These indications can sometimes appear before stage 5 CKD based on GFR criteria.

There is no survival benefit with early initiation of hemodialysis. In the absence of uremic symptoms or complications with ESRD, later initiation of RRT can be done. The patient's preference for the trial of medical non-dialytic management before dialysis initiation should be considered when the patient is medically stable and asymptomatic.

Criteria for dialysis initiation for those with failed kidney transplants are similar to those with ESKD of native kidneys.

Counseling for Kidney Transplantation

Transplantation is the kidney replacement therapy of choice for suitable patients with advanced CKD. Not all patients are suitable for transplantation. Before proceeding with transplantation, risks and benefits should be compared for each individual patient. KDIGO 2020 clinical practice guideline on the evaluation and management of candidates for kidney transplantation recommends informing and educating all patients with CKD G4 – G5 (GFR < 30 ml/min/1.73m²) about kidney transplantation regardless of their socioeconomic status, sex, gender identity, race, or ethnicity.

Potential kidney transplant candidates should be referred for kidney transplant evaluation at least 6 to 12 months before anticipated dialysis initiation to facilitate/work up to living donors and plan for possible preemptive transplantation. Likewise, all CKD patients already on dialysis should be sent for transplant evaluation once they are medically stable, and kidney failure is deemed irreversible.

Kidney transplantation should not be considered in patients with:

- Multiple Myeloma, Light chain deposition diseases.... (Unless they have received curative treatments)
- Decompensated cirrhosis
- Severe irreversible obstructive/restrictive lung diseases
- Severe uncorrectable and symptomatic cardiac diseases
- Progressive central neurodegenerative diseases
- Underlying etiology with high risk of recurrence and graft loss

Transplant evaluation should be delayed for patients with:

- Unstable psychiatric disorders
- Morbid obesity
- Ongoing substance use
- Poor adherence
- Active infection (Except Hepatitis C)
- Active malignancy
- Active symptomatic cardiac disease
- Active symptomatic peripheral arterial diseases
- Recent stroke or transient ischemic attack
- Acute hepatitis
- Severe hyperparathyroidism
- Active symptomatic Gastrointestinal or hepatobiliary disease
- Unaddressed psychosocial issue

Estimated survival should be acceptable to send/consider a patient on dialysis for kidney transplantation. Age alone is not a contraindication for transplantation, but consideration of age should be in the context of the presence of comorbidities.

Transplant evaluation and counseling will be performed by a multidisciplinary team that includes a minimum of transplant physicians, a surgeon, a nurse, and a psychiatrist/psychologist. Detailed counseling should be done before transplantation, usually by the transplant physician or nurse. This counseling should include details of pre-transplant workup, details of the surgical procedure and possible outcomes, post-operative treatment and follow-up plans, medications, adherence, and psychosocial and legal issues.

5.10 CHAPTER SUMMARY

- Anemia and CKD-MBD are very common and early complications of CKD which necessitate treatment.
- As CKD reaches stage 5 (eGFR <15 ml/min/1.73m²), patients start manifesting signs and symptoms related to uremia.
- Uremia symptoms include nausea, vomiting, fatigue, sexual dysfunction, malnutrition, pericarditis, platelet dysfunction, and neuropathy.
- All patients with CKD G4 with progressive decline in kidney function should be educated and prepared for all modalities of kidney replacement options including kidney transplantation.

CHAPTER 6: NUTRITION IN CKD



Time Allocated: 1 hour and 15 minutes



Chapter description: This chapter describes nutritional screening and management of patients with chronic kidney disease (CKD).



Primary Objective

By the end of this chapter, participants will be able to discuss the nutritional management of CKD patients.

Enabling objectives

By the end of this session, participants should be able to:

- Describe the epidemiology of nutritional disorders in CKD.
- Describe causes of under nutrition and Protein energy wasting (PEW) among CKD patients.
- Discuss nutritional management in CKD
- Describe nutrition in CKD in the Ethiopian Context

Chapter outline

- 6.1. Causes of under nutrition and Protein energy wasting (PEW) among CKD patients.
- 6.2. Nutritional Management in CKD
- 6.3. Nutrition in CKD in the Ethiopian context
- 6.4. Chapter summary

6.1 CAUSES OF UNDER NUTRITION AND PROTEIN ENERGY WASTING (PEW) AMONG CKD PATIENTS.



Activity -6.1

Activity type: Case scenario

A 74-year-old woman with history of CKD secondary to diabetes is visiting her nephrologist with concerns of anorexia, nausea, and occasional vomiting. During her evaluation, it is noted that her serum creatinine has risen compared with her most recent (6 months prior) levels, from 2.5 mg/dL to 3.6 mg/dL (eGFR 19 and 12 mL/min/1.73 m respectively). She has been mostly at home with limited activity since her husband died 6 months ago. Which problem(s) may best explain her recent nutrition and functional decline?

Advanced CKD-uremia Depression

Decreased physical activity

All of the above

Time Allocated : 5 minutes

As chronic kidney disease (CKD) progresses, the requirements and utilization of different nutrients change substantially. These changes are accompanied by multiple nutritional and metabolic abnormalities that are observed in the continuum of kidney disease. To provide optimal care to patients with CKD, an understanding of the applicable nutritional principles and the methods for assessing nutritional status, establishing patient-specific dietary needs, and preventing or treating potential or ongoing nutritional deficiencies and derangements is essential.

6.1.1 Epidemiology

Individuals with CKD are at risk for a spectrum of nutritional disorders that encompass nutrition, protein-energy wasting (PEW), and electrolyte disturbances. They also face other challenges such as obesity, cardiovascular disease, and maintenance of a high-quality diet within the constraints of reduced glomerular filtration.

Correctly diagnosing PEW is challenging because the suggested criteria are extensive and not always easily assessed in clinical settings. Studies suggest the worldwide prevalence ranges from 11% to 54% in persons with CKD stages 3-5 and is between 28% and 54% in patients requiring dialysis. The global prevalence of protein-energy malnutrition in persons with CKD is difficult to estimate because

it varies by region and country and lacks a single diagnostic test that is highly accurate, reproducible, and easy to perform in the clinical setting. Hence, the diagnosis is usually based on a combination of history and clinical examination.

6.1.2 Etiology and Implications of Nutritional and Metabolic Derangements in Kidney Disease

Multiple factors affect nutritional and metabolic status in patients with moderate to advanced kidney disease, and this can lead to adverse consequences. Accordingly, prevention and treatment strategies should involve an integrated approach to reduce nutrient depletion along with interventions that would avoid further losses and replenish already wasted store. Table 17 describes possible causes of PEW in patients with CKD.

Cause	Mechanism	Complications Amenable to Nutrition Interventions
Reduced intake	Poor appetite (uremic toxicity, inflammation); spontaneous decline in protein intake	Anorexia, nausea, dry mouth, dyspnea
Restrictive dietary intake	Unmonitored restriction of protein and other nutrients	Hypokalemia, unintentional weight loss
Inflammation	Protein catabolism, suppressed albumin synthesis, increased REE	Unintentional weight loss, muscle wasting
GI function	Delayed gastric emptying, impaired motility	Gastroparesis, constipation
Metabolic acidosis	Proteolysis, catabolism of branched-chain amino acids, suppression of albumin synthesis	Hyperkalemia, taste changes
Abnormal hormone response	Altered CHO and lipid metabolism; insulin resistance	Hyperglycemia, hyperlipidemia
Abbreviations: CHO , carbohydrate; GI , gastrointestinal; REE , resting energy expenditure.		

Table 17: Dietary and Metabolic Changes with Uremic Wasting and Symptoms That Can Respond to Nutritional Therapy, adapted from Nutrition in Kidney Disease: Core Curriculum 2022.

6.2 NUTRITIONAL MANAGEMENT IN CKD

Components of nutrition management in CKD

The 2020 updated Kidney Disease Outcomes Quality Initiative (KDOQI) Nutrition guideline in CKD statements focus on 6 primary areas:

1. Nutritional assessment

2. Medical nutrition therapy (MNT)
3. Dietary protein and energy intake
4. Micronutrient and Electrolytes
5. Nutritional supplementation

6.2.1 Nutritional Assessment

The measurement of nutritional status does not lend itself to one simple test, and a panel of measures is required. Table 18 summarizes some of the methods used for assessment of nutritional status.

Area	Assessments
Physical Examination Assessment of dietary intake	Diet history, food diaries, appetite assessment questionnaires
Anthropometric measurements	Body weight, height, body mass index Percentage weight change Skinfold thickness Midarm muscle circumference
Body composition	Bioelectrical impedance Dual-energy x-ray absorptiometry (DEXA) Near-infrared reactance Neutron activation Total body potassium
Biochemical determinations	Serum electrolytes Serum proteins PNA, PCR Serum cholesterol Creatinine index*
Nutritional scoring systems	Subjective global assessment, malnutrition inflammation score (MIS) Geriatric nutritional risk index (GNRI)
Immunologic assays	Blood lymphocytes Delayed cutaneous hypersensitivity tests
Functional tests	Grip strength
*The creatinine index is measured as the sum of creatinine removed from the body (measured from the creatinine removed in dialysate, ultrafiltrate, and/or urine), any increase in the body creatinine pool, and the creatinine degradation rate. See also www.kidney.org/professionals/kdoqi/pdf/KDOQI2000NutritionGL.pdf. PCR, Protein catabolic rate (mathematically identical to PNA); PNA, protein equivalent of total nitrogen appearance.	

Table 18: Assessment of nutritional status, adapted from Comprehensive clinical nephrology, 6th edition.

Routine Nutrition Screening: KDOQI recommendation

1. In adults with CKD 3-5D or post transplantation, it is reasonable to consider routine nutrition screening at least biannually with the intent of identifying those at risk of protein-energy wasting.

Frequency of Body Weight/BMI and Body Composition Assessment

In adults with CKD 1-5D or post transplantation who are clinically stable, it is reasonable to measure body weight and BMI and to monitor for changes in body weight/ BMI and body composition as needed.

- At least every 6 months in patients with CKD 1-3

6.2.2: Medical Nutrition Therapy (MNT)

Individualized management of nutritional intake is a crucial aspect of care for individuals diagnosed with any stage of CKD. Nutritional needs change throughout the disease course, from the earlier stages of CKD to the post-transplant period. The metabolic abnormalities and comorbid diseases that often accompany CKD further emphasize the need for specialized nutrition health care. Therefore, it is essential that such individuals receive tailored nutrition assessment and counseling in the form of MNT.

MNT requires nutrition screening and assessment on nutritional status to provide individualized treatment for specific disease states. Patients with CKD are on a dynamic nutrition trajectory according to their disease stage and MNT is needed at each stage of CKD. Metabolic abnormalities and acid-base and fluid, and electrolyte balance often change as CKD progresses.

Nutritional recommendation for CKD patients on specific diets according to KDOQI 2020 updated guideline on nutrition in CKD

Grade	Definition
High (A)	We are very confident that the true effect lies close to that of the estimate of the effect.
Moderate (B)	We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.
Low (C)	Our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.
Very low (D)	We have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.
Table ©2013 GRADE Working Group; reproduced from the GRADE handbook ³ ; with permission of the copyright holder.	

Table 19: Quality of Evidence Grades, according to KDOQI 2020 updated guideline on nutrition in CKD.

In the context of the following KDOQI recommendations, “metabolically stable” indicates

- The absence of any active inflammatory or infectious diseases,
- No hospitalization within 2 weeks,
- Absence of poorly controlled diabetes and consumptive diseases such as cancer
- Absence of antibiotic or immunosuppressive medications, and
- Absence of significant short-term loss of body weight.

6.2.3. Dietary Protein and Energy Intake

Activity 6.2.3: Group work



Instruction:

Form 5 groups and each group is expected to calculate protein intake for a 60-kg adult CKD patient (for non-dialysis patients) based on specific amount of protein foods per day.

Time allocated; 10 minutes

A. CKD Patients Not on Dialysis and Without Diabetes

In adults with CKD 3-5 who are metabolically stable, we recommend, under close clinical supervision, protein restriction with or without ketoacid analogs, to reduce risk for end-stage kidney disease (ESKD)/death (1A) and improve quality of life (QoL) (2C):

- a low-protein diet providing 0.55 – 0.60 g dietary protein/kg body weight/day, or
- a very low-protein diet providing 0.28 – 0.43 g dietary protein/kg body weight/day with additional ketoacid/amino acid analogs to meet protein requirements (0.55 – 0.60 g /kg body weight/day)

B. CKD Patients Not on Dialysis and With Diabetes

In the adult with CKD 3-5 and who has diabetes, it is reasonable to prescribe, under close clinical supervision, a dietary protein intake of 0.6 – 0.8 g/kg body weight per day to maintain a stable nutritional status and optimize glycemic control.

C. Statement on Protein Type

In adults with CKD 1-5 D (1B) or post transplantation (OPINION), there is insufficient evidence to recommend a particular protein type (plant vs animal) in terms of the effect on nutritional status, calcium or phosphorus levels, and lipid profile.

D. Statements on Dietary Patterns

Mediterranean Diet

In adults with CKD 1-5 not on dialysis or post transplantation, with or without dyslipidemia, we suggest that prescribing a Mediterranean Diet may improve lipid profiles (2C).

Fruits and Vegetables

In adults with CKD 1-4, we suggest that prescribing increased fruit and vegetable intake may decrease body weight, blood pressure, and net acid production (NEAP) (2C).

E. Energy intake KDOQI recommendation on energy intake in CKD patients:

In adults with CKD 1-5D (1C) or post-transplantation (OPINION) who are metabolically stable, we recommend prescribing an energy intake of 25-35 kcal/kg body weight per day based on age, sex, level of physical activity, body composition, weight status goals, CKD stage, and concurrent illness or presence of inflammation to maintain normal nutritional status.

6.2.4. Micronutrient and Electrolytes

Micronutrients include vitamins, trace elements, and electrolytes and are essential for optimal biological function. The main concern in patients with CKD is that as a result of reduced dietary intake or dietary restrictions, comorbid conditions, and/or losses through the dialysis procedure, deficiencies in certain vitamins and trace elements may develop. However, excess intake of certain micronutrients may also occur, leading to vitamin toxicity (e.g., vitamin C, vitamin D) or clinical complications (e.g., sodium, potassium, phosphorus). Because the delicate balance between under consumption and overconsumption is also influenced by the stage of CKD and each person's unique needs and risk factors, developing an optimal dietary strategy in persons with CKD can be challenging.

A. Vitamins

Due to concern beginning decades ago over possible micronutrient deficiencies, particularly with water-soluble B vitamins, vitamin supplements tailored to patients with CKD were developed. A reasonable strategy would be to supplement only those individuals who by history or examination exhibit impaired dietary intake and/or nutrient loss for sustained periods of time.

B. Sodium intake

Sodium consumption should be limited to less than 100mmol/d (2.3g) to help control blood pressure and limit extracellular volume expansion. Of particular interest in CKD is the role of sodium reduction in improving the pharmacologic effect of antihypertensive medication, thereby controlling hypertension. Replacing bread, processed meats, and cheese with fresh meats, fish, brown rice, legumes, and whole wheat pasta can reduce sodium intake.

C. Potassium intake

KDOQI recommendation on potassium intake

Dietary Potassium Amount

In adults with CKD 3-5D or post transplantation, it is reasonable to adjust dietary potassium intake to maintain serum potassium within the normal range.

Dietary and Supplemental Potassium Intake for Hyperkalemia or Hypokalemia

In adults with CKD 3-5D (2D) or post-transplantation (OPINION) with either hyperkalemia or hypokalemia, we suggest that dietary or supplemental potassium intake be based on a patient's individual needs and clinician judgment.

D. Phosphorus intake

Dietary phosphorus intake should be adjusted to maintain serum phosphorus levels in the normal range. Phosphorus should ideally be obtained from plant-based foods, such as whole grains, legumes, and pulses, because the phosphorus is typically less well absorbed and whole grain foods have a higher nutritional value compared with processed foods that contain phosphate additives.

E. Calcium intake

The intake of dietary calcium in patients with CKD stages 3-4 should be to achieve 800 to 1,000mg daily to maintain a neutral calcium balance. The intake goals for more advanced disease may be complicated by concurrent use of vitamin D analogs and calcimimetics.

KDOQI recommendation on calcium intake

Total Calcium Intake

- In adults with CKD 3-4 not taking active vitamin D analogs, we suggest that a total elemental calcium intake of 800-1,000 mg/d (including dietary calcium, calcium supplementation and calcium-based phosphate binders) be prescribed to maintain a neutral calcium balance (2B).

6.2.5. Nutritional supplementation in CKD

When intake from food alone is inadequate, supplementation with nutrition formulations can be used to meet nutritional requirements to prevent or treat malnutrition. In CKD, nutrition is associated with poorer outcomes, and using nutritional interventions is warranted in these patients. Oral nutritional supplements (ONS), enteral tube feeding, and parenteral nutrition may be used when clinically indicated.

KDOQI recommendation on Oral Protein-Energy Supplementation

In adults with CKD 3-5D (2D) or post-transplantation (OPINION) at risk of or with protein-energy wasting , we suggest a minimum of a 3-month trial of oral nutritional supplements to improve nutritional status if dietary counseling alone does not achieve sufficient energy and protein intake to meet nutritional requirements.

6.3. NUTRITION IN CKD IN THE ETHIOPIAN CONTEXT

Nutritional Consideration	Rationale	Examples of Interventions
Maintaining health and minimizing risks from comorbid conditions (eg, diabetes, CVD, hypertension, obesity)	Dietary pattern impacts individual on disease risk more than individual nutrients or foods. Lower salt intake reduces risk factors.	• Cook from fresh ingredients when possible. • Include vegetables and whole grains with meals every day. • Choose legumes or plant-based meat alternatives, reduce meat portions, and limit processed meats. • Personalize meal plans to meet energy needs.
Addressing barriers to changing eating behaviors	Food choices are multifactorial.	• Individualize strategies for addressing identified barriers. • Use behavioral therapy techniques such as self-monitoring and self-directed goal setting.
Hyperkalemia	Consider all possible causes before reducing high nutritional value fruits and vegetables. Use fruit and vegetables to reduce net acid load and provide dietary fiber.	• Stage 1: Address possible nondietary causes such as hyperglycemia, acidosis, constipation, recent medication changes, or use of potassium-sparing diuretics. • Stage 2 (if required): Reduce lower-nutritional value foods such as potato chips, fruit juices, and chocolate. • Stage 3: Peel, chop, and boil vegetables, access pictorial or color-coded resources for lower-potassium-containing fruits and vegetables, and maintain recommended number of servings per day.
Hyperphosphatemia	Consider food additives, protein requirements, and dialysis adequacy.	• Limit processed meats, processed cheese, and processed cheese products. • Choose fresh meat, poultry, or fish without added phosphates (read food labels). • Legumes, soy products, nuts, and whole grains have lower phosphate availability as they contain nondigestible phytates. • Ensure phosphate binder doses are matched to mealtimes and protein/phosphorus intake.
Optimizing nutritional status	Reduce mortality risk, and improve quality of life.	• Personalize meal plans to meet energy needs. • May need supplemental nutrition when fatigue is limiting factor or intake is poor.
Abbreviation: CKD, chronic kidney disease; CVD, cardiovascular disease.		

Table 20: Nutrition Interventions in CKD When Individualized Medical Nutrition Therapy Is Not Available.
Adapted from *Nutrition in Kidney Disease: Core Curriculum 2022*.

6.4. CHAPTER SUMMARY

- Nutrition plays very important role in CKD management and under nutrition is associated with increased mortality among CKD patients.
- Routine nutritional screening should be done on specific time intervals for all CKD patients, including those on dialysis.
- When applicable, nutritional management among CKD patients should be offered in the form of MNT.
- Specific nutritional recommendations can be applied from internationally accepted nutrition guidelines, mainly KDOQI nutrition guideline in CKD.

CHAPTER 7: MANAGEMENT OF MAJOR INFECTIONS AND ROUTINE VACCINATION IN CKD



TIME ALLOCATED: 1 hour and 30 minutes



Chapter description: This chapter provides overviews of the management of major infections and vaccinations in chronic kidney disease



Primary objective:

By the end of this chapter participants will be able to:

- Describe the management of major infections and vaccinations in chronic kidney disease

Enabling objectives: by the end of this chapter participants will be able to:

- Identify the major infections encountered in chronic kidney disease
- Describe the diagnosis and therapeutic approach of the major infections in chronic kidney disease
- Describe Routine vaccinations in chronic kidney disease

Chapter outline

- 7.1 Major infections, diagnosis and therapeutic approach of the major infections in chronic kidney disease
- 7.2 Routine Vaccinations in chronic kidney disease
- 7.3 Chapter Summary

7.1 MAJOR INFECTIONS, DIAGNOSIS AND THERAPEUTIC APPROACH OF THE MAJOR INFECTIONS IN CHRONIC KIDNEY DISEASE



Activity 7.1 Case Scenario

Instruction: Read the following case scenario and discuss the points mentioned in teams. Your instructor will create groups.

Mr. Tesfay B is a 44-year-old man who is known hypertensive and chronic kidney disease patient on follow up for the past three years. He presented to a medical outpatient clinic with a three-month history of intermittent fever, dry cough, anorexia, night sweats, nausea and generalized weakness and progressive abdominal swelling and discomfort of one month duration. He had also experienced unintentional weight loss. He currently takes Amlodipine 10 mg, Furosemide 40 mg once daily, and Losartan 50 mg Po daily.

Vital sign: BP – 140/80 mmHg, PR – 94/min, SPaO₂ – 90% with atmospheric oxygen, Temperature 37.8 degree centigrade, RR=28 breaths/min

Chest examination – bilateral lower one third lung zone absent entry

Abdominal examination: Positive abdominal shifting dullness and fluid thrill

MSS examination: No peripheral edema

CNS examination: Conscious and oriented to TPP

Investigations:

- WBC- 7000/microliter, Neutrophil of 42%, Lymphocyte 40%, Hgb – 6.2g/dl, MCV 94 fL and PLT- 374,000/microliter
- ESR- 44mm/Hr.
- Urinalysis – Protein +3, Many WBC per HPF, blood + 3
- Serum Creatinine – 8.8 mg/dl (Baseline – 5.2 mg/dl), K – 4.4, Na – 128,
- AST-7IU/L, ALT 18 IU/L, Serum Albumin 2.6g/dl
- Chest X-ray – blunted left costophrenic angle with meniscus sign (left pleural effusion)

I – Outline problem lists of the patient

II – Identify possible differentials for his current presentation

III – Describe the diagnostic approach necessary to reach at a diagnosis

IV- Outline the management plans of the patient

Time allocated – 20 minutes

Infections are a major cause of kidney injury either directly by invasion, indirectly by immune mediated mechanisms, or through their treatment. Common sub-acute and chronic viral infections (such as hepatitis B virus, hepatitis C virus and HIV) induce glomerular disease which can progress to end-stage renal disease (ESRD). In individuals with chronic kidney disease (CKD), infections and their treatments can cause worsening of kidney function and pose a significant source of morbidity and mortality. Infections are a major cause of death in ESRD patients, second only to cardiovascular disease, and contribute to significant morbidity in patients with earlier stages of CKD. Infectious complications are 3 to 5 times more common in CKD than in non-CKD patients and up to 8 times more common in dialysis patients. From this perspective, major infections such as pneumonia, sepsis, and urinary tract infection can be seen as major sources of morbidity, which is consistent with the reduced immunological function reported for patients with chronic kidney disease. After a major infectious episode, cardiovascular disease event rates are markedly increased, suggesting that infection may play a synergistic role in the inflammatory hypothesis of vascular disease, which is prominent in CKD patients. Prophylactic vaccines are a strategy to attempt to reduce morbidity related to infections in CKD.

7.1.1 Tuberculosis and chronic kidney disease

A. Risk factors and diagnostic approach of tuberculosis in chronic kidney disease

Patients with CKD are more susceptible to tuberculosis (TB) infection or reactivation of latent TB infection (LTBI) due to underlying immunosuppression. The relative risk of developing active TB in CKD or hemodialysis patients is 10 to 25 times higher than in the general population.

Decline in cell-mediated immunity begins as early as CKD stage 3 and worsens in later stages as kidney function declines and uremic toxins accumulate, increasing the risk of latent TB reactivation and active TB disease.

High mortality is observed in CKD patients with TB, often due to delayed diagnosis and the side effects of anti-TB medications. Typical TB symptoms are less common in CKD patients than in the general population. Instead, they often present with non-specific systemic symptoms such as fever, anorexia, and weight loss, which can mimic symptoms of uremia and delay diagnosis. Extrapulmonary TB occurs in 60% to 80% of CKD patients with TB, often with disseminated disease, making diagnosis even more challenging. Therefore, a high index of suspicion is essential, particularly for extrapulmonary TB presentations.

Diagnosis is further complicated by the immunosuppressive state in CKD. Health workers should maintain a high level of suspicion for TB in any CKD patient presenting with unexplained fever, appetite loss, weight loss, night sweats, and elevated ESR.

Once TB is confirmed in a CKD patient, renal dose adjustments of anti-TB drugs are necessary to enhance efficacy and reduce toxicity. Treatment duration may be extended based on the clinical condition, especially in cases of severe or extensive disease.

Generally, standard drug dosages are used, and if dose modification is needed, reducing dosing frequency is preferred over lowering the dose, to avoid subtherapeutic levels.

The CKD epidemic has a growing impact on both regional and global TB epidemiology, particularly in resource-limited settings where CKD is rising, and TB remains endemic. Therefore, it is recommended that all CKD and dialysis patients undergo TB risk assessment, clinical evaluation, chest X-ray, and microbiological testing if risk factors are identified.

B. Anti tuberculosis medications and chronic kidney disease

Drug	eGFR≥60ml/min	eGFR30-59 ml/min	eGFR <30ml/min	Intermittent Hemodialysis	Peritoneal dialysis
Rifampicin	10 mg/kg q24h		No dose adjustment required		
Isoniazid	5 mg/kg q24h		No dose adjustment required		
Pyrazinamide	25-35 mg/kg q24h	25-35 mg/kg q24h	25-35 mg/kg 3 times a week	25-35 mg/kg 3 times a week after Hemodialysis	No dose adjustment required
Ethambutol	15-25 mg/kg q24h	15-25 mg/kg q24h	15-25 mg/kg 3 times a week	15-25 mg/kg 3 times a week after Hemodialysis	15 mg/kg q48h
Moxifloxacin	400 mg q24h	No dose adjustment required			
Protionamide	15-20mg/kg/d in divided doses	No dose adjustment required	eGFR≥10 to 30: No dose adjustment	250 mg q12h	250 mg q12h
			eGFR<10: 250 mg q12h		

Cycloserine	10-15mg/kg/d in Divided doses	250mgq12h	250 mg q24h	250mg q24h given After dialysis	250 mg q24h
Bedaquiline	400 mg q24h for 2 weeks then 200mg thrice weekly	No dose adjustment required	Use with caution and consider therapeutic drug monitoring		
Linezolid	600 mg q24h	No dose adjustment required			
Para-amino salicylic acid	8-12g/d in divided doses	No dose adjustment required	4g q12h	4g q12h	4g q12h
Streptomycin, Amikacin, Capreomycin	15mg/kg/d	15mg/kg individual doses with dosing interval extended to achieve an undetectable plasma trough level prior to each dose			
Clofazimine	100 mg q24h	No dose adjustment required			

Table 21: Anti-tuberculosis medication dose adjustments in chronic kidney disease

Anti-TB medications, some of which are nephrotoxic, also lead to worsening of the stage of kidney disease. Of the first line agents (EHRZ), ethambutol and pyrazinamide are excreted via the kidneys. Ethambutol accumulation in CKD can lead to irreversible ocular toxicity while Pyrazinamide can lead to accumulation of uric acid due to impaired urate excretion. For patients having MDR TB with CKD, they should be managed in consultation with a Nephrologist.

The treatment of TB in patients with mild renal impairment, and with glomerular filtration rate (GFR) between 30 to 60mL/min, should be individualized using standard drugs in dose ranges in the lower range of usual recommendations for patients with normal renal function, with careful monitoring of side effects. In patients with GFR <30, first line drugs RHZE should be given every other day (see Table 18). Careful monitoring of patients is essential as side effects (mainly neuropsychiatric problems, hepatitis and optic neuropathy) are noted to occur at higher levels in patients with renal failure and especially those on dialysis. Hemodialysis often leads to the elimination of most Anti TB drugs and medications are usually given after dialysis. For the second line drugs including aminoglycosides and quinolones, doses should be adjusted according to the GFR.

7.1.2 HBV and HCV infection and chronic kidney disease

A. Clinical profile and diagnosis of HBV and HCV infection in chronic kidney disease

Kidney disease is an important extra-hepatic manifestation of HBV and HCV infection. HCV associated renal disease can either be an immune-complex glomerulonephritis such as membranous glomerulonephritis, membranoproliferative glomerulonephritis, cryoglobulinemic glomerulonephritis and IgA nephropathy, or immune-mediated vasculitis (Polyarteritis nodosa).

The disease can present as either nephritic syndrome (hypertension, hematuria, proteinuria, edema), or nephrotic syndrome (high level proteinuria, edema, hyperlipidemia, hypoalbuminemia and lipiduria). The special characteristics of this patient group (immunosuppressive effect of chronic kidney disease, polypharmacy, susceptibility for nosocomial transmission, immunosuppressive therapy after renal transplantation) and the related changes in clinical course of liver disease often have long-term implications on morbidity and mortality and affect the patients' management.

Although HBV-associated renal disease usually resolves uneventfully in children, it rarely reaches spontaneous remission in adults, with most of them (30–50%) progressing to end-stage renal disease. Screening of all patients with CKD for hepatitis B (HepBsAg) and hepatitis C (Anti-HCV) infection is recommended to determine if these are the causes of kidney disease and to plan further management. It's also recommended referral of patients with CKD who are found with hepatitis B or hepatitis C infection to hepatologist to co-manage it with Nephrologist.

B. Management of HBV and HCV infection in chronic kidney disease

Treatment for of HBV associated kidney disease involves the use of interferon's (conventional IFN- α , or pegylated IFN- α -2a) or oral antiviral agents that consist of either one of nucleotides (adefovir dipivoxil, tenofovir disoproxil fumarate, tenofovir alafenamide) or nucleoside (lamivudine, entecavir, and telbivudine) reverse transcription inhibitors for treatment. Dose adjustments of these drugs according to creatinine clearance, and monitoring for nephrotoxicity in patients on Adefovir and Tenofovir are warranted. The use of entecavir in patients with advanced kidney disease may be associated with lactic acidosis and may require dose adjustments according to the GFR. Treatment for HCV infection in CKD: relies on the use of directly acting antiviral (DDAs) e.g. Sofosbuvir (renal dose at GFR < 30). Interferon therapy should be avoided due to prolonged therapy, poor tolerance, an unsatisfactory sustained viral response.

Promoting access to DAAs with referral to hepatologist as HCV is a curable infection even in advanced CKD and reducing risk of HCV transmission in dialysis centers, along with universal immunization for HBV, can make it possible to eliminate viral hepatitis within the next decade as a global health problem.

7.1.3 HIV Infection and chronic kidney disease

A. Risk factors and etiologies of CKD in patients with HIV infection

HIV associated kidney disease is a common complication of HIV infection and results from direct viral invasion of the glomerular visceral and parietal epithelial cells causing HIV-associated nephropathy, non-collapsing focal segmental glomerulosclerosis. Other causes of kidney dysfunction in HIV infection are due to immunocomplex formation, comorbid kidney disease, as well as kidney injury resulting from prolonged exposure to antiretroviral therapy or from opportunistic infections. The largest subset of HIV infected individuals at higher risk for developing kidney disease include those using injection drugs, especially cocaine, strong family history of kidney disease, dyslipidemia, smoking and those who receiving certain nephrotoxic medication.

HIV causes accelerated kidney dysfunction and there is need for close evaluation of HIV patients to avoid progression to ESRD. Urinalysis should be performed in all HIV-positive individuals to detect worsening or new onset of proteinuria or hematuria. Where feasible, quantification of proteinuria (spot urine albumin-to-creatinine or protein-to-creatinine ratio) should also be performed. Annual monitoring of eGFR if low risk of CKD, bi-annual if high risk of CKD with more frequent monitoring is recommended in persons who are clinically unstable, severely immunocompromised, or viremic. Adequate control of traditional risk factors including diabetes and hypertension is of paramount importance to halt the risk of progression to ESRD in HIV patients.

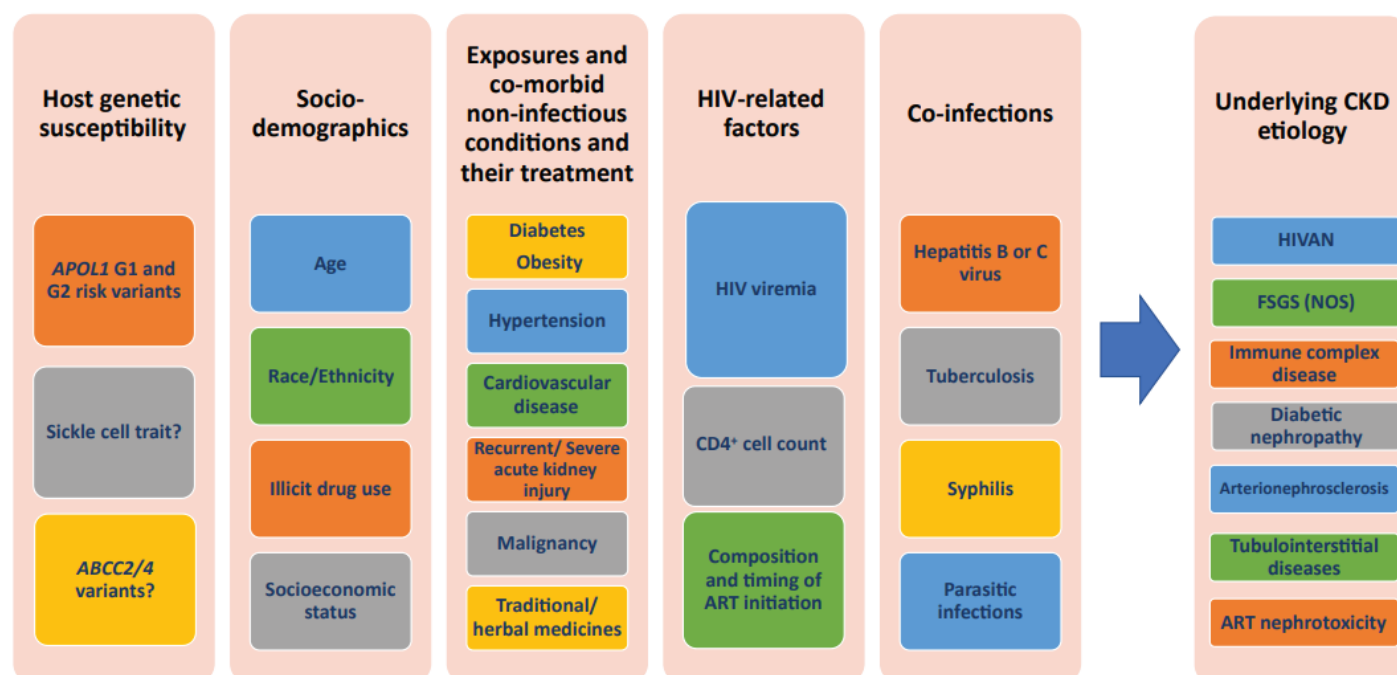


Figure 12: Risk factors and underlying etiologies of CKD in HIV-positive individuals

APOL1, apolipoprotein L1; ABCC, ATP-binding cassette transporter proteins; ART, antiretroviral therapy; CKD, chronic kidney disease; FSGS (NOS), focal segmental glomerulosclerosis, not otherwise specified; GN, glomerulonephritis; HIVAN, HIV-associated nephropathy

(Adopted from Kidney disease in the setting of HIV infection: conclusions from a kidney disease: Improving Global Outcomes (KDIGO) Controversies Conference)

B. ART medication in chronic kidney disease

In treatment of patients with HIV and CKD, there is a need for appropriate adjustment of antiretroviral therapy doses according to eGFR. Tenofovir, the backbone of HAART, requires dose adjustment when eGFR is <50 ml/min. Avoidance of nephrotoxic ART (tenofovir, indinavir, atazanavir, lopinavir) in patients at high risk of CKD progression (TDF in eGFR <50ml/min, diabetes mellitus, uncontrolled hypertension, history of cardiovascular disease)

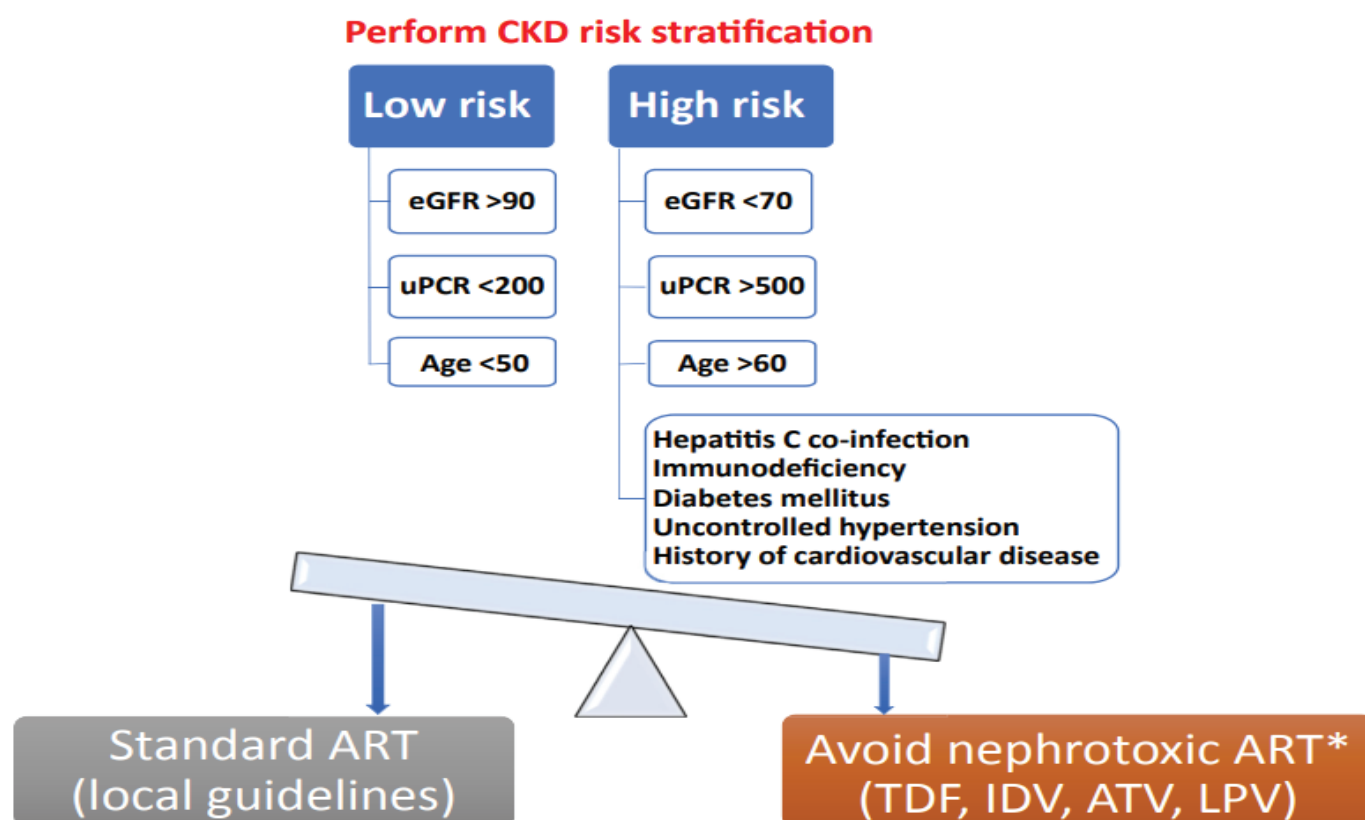


Figure 13: Recommendations at starting ART in CKD patients

*If suitable alternatives are available. ART, antiretroviral therapy; ATV, atazanavir; CKD, chronic kidney disease; eGFR, estimated glomerular filtration rate (CKD-EPI, expressed in ml/min per 1.73 m²); IDV, indinavir; LPV, lopinavir; TDF, tenofovir disoproxil fumarate; uPCR, urine protein-to-creatinine ratio (values above are in mg/g; multiply by 0.10 to obtain values in mg/mmol).

C. Kidney disease screening and monitoring in patients with HIV infection

Early detection and diagnosis of kidney disease are essential for preventing or slowing further decline in kidney function and improving outcomes in HIV infected persons. It's advocated that evaluation for both proteinuria kidney dysfunction is warranted as CKD screening in HIV positive patients.

Individuals with underlying impaired kidney function and those initiated on drugs with potential renal toxicities likely benefit from more vigilant renal follow up. Persons receiving tenofovir, in particular, may benefit from more frequent assessments of their kidney function, serum phosphate levels, and urinalysis to monitor for early signs of nephrotoxicity.

CKD screening CKD monitoring Nephrology referral

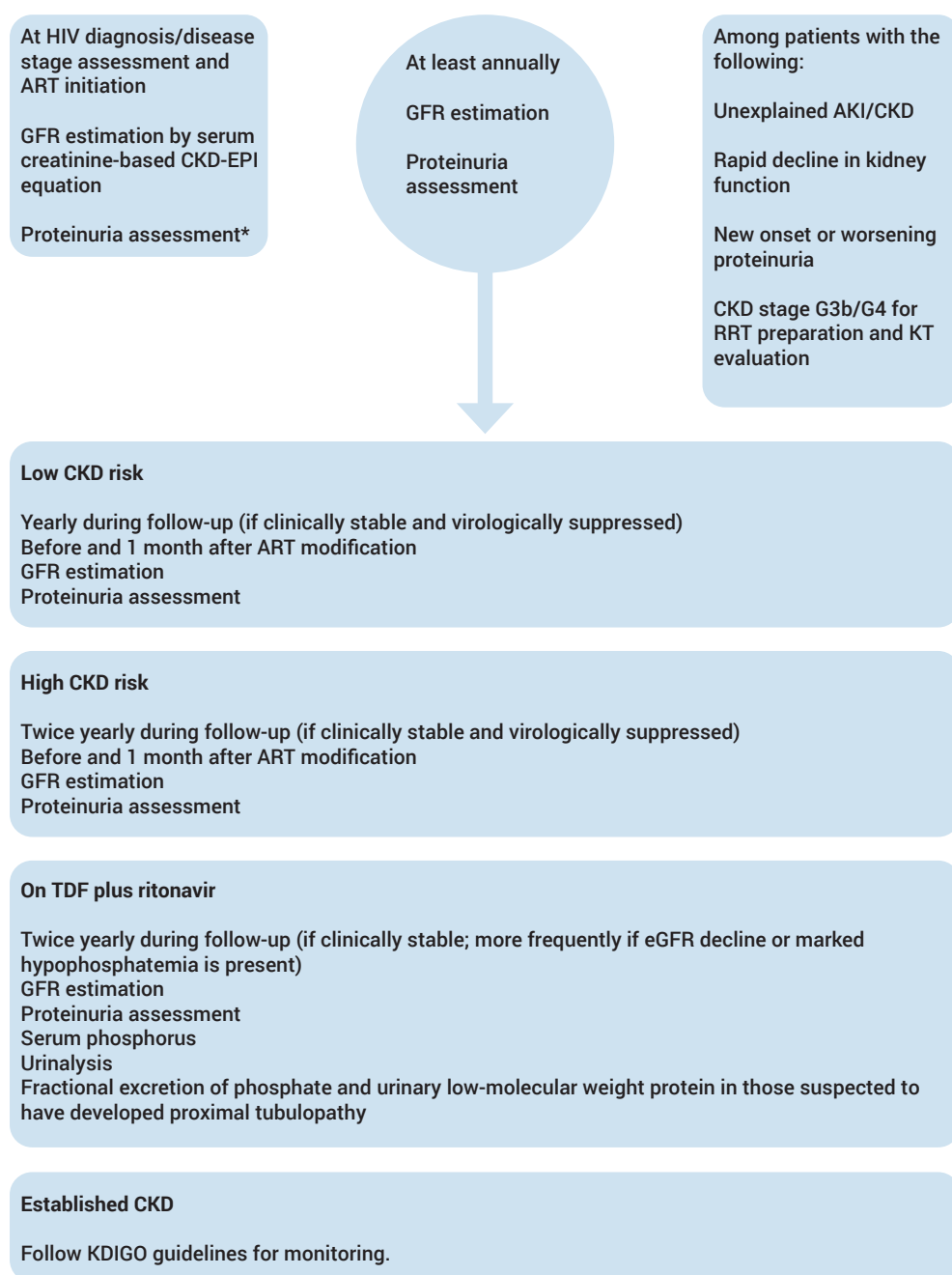


Figure 14: Recommendations for kidney disease screening and monitoring in HIV-positive adults

AKI, acute kidney injury; ART, antiretroviral therapy; CKD, chronic kidney disease; CKD-EPI, CKD Epidemiology Collaboration; eGFR, estimated glomerular filtration rate; GFR, glomerular filtration rate; KDIGO, Kidney Disease: Improving Global Outcomes; TDF, tenofovir disoproxil fumarate

7.2 VACCINATION IN CHRONIC KIDNEY DISEASE



Activity 7.2 Case Scenario

Instruction: Read the following case scenarios and discuss the points mentioned in teams. Your instructor will create groups of two each and select one individual to present the group's opinion after discussion.

A 40 years old female with hypertension and chronic kidney disease is adherent on her medication with anti hypertensive and routine follow up. She is currently presented with cough, fever, shortness of breath of one week duration. Her basic work up showed WBC count of 17, 000 with granulocyte count of 90%. Her chest x- ray showed pneumonic consolidation in the left lower lung zone. She recently was having worsening of her kidney profile from the baseline due to recurrent infections arising from the chest making her hit the stage four of chronic kidney disease with estimated GFR 19/min/1.72m². She has multiple similar episodes of infections and hospitalizations in the last one year.

- I. As she advances to advanced stage chronic kidney discuss on the need for screening and prophylactic vaccinations to prevent vaccine preventable infections related morbidity
- II. Outline the list of vaccinations needed in this patient?
- III. Discuss the dose, schedule and need of booster doses for each vaccinations needed

Time allocated – 10 minutes

Infection remains a major cause of death and hospitalization among patients with chronic kidney disease (CKD). Patients with CKD are susceptible to infectious organisms due to the many facets of uremia-associated immune deficiency. These include increased production of reactive oxygen radicals, increased polymorphonuclear neutrophil apoptosis, reduced antibody-producing capacity of B lymphocytes and depletion of memory CD4⁺ and CD8⁺ T lymphocytes. The loss of this function is important as vaccines often use proteins for their antigens and the resulting antibody response is T-cell dependent. Another key feature of immunity in CKD includes impaired antigen presentation by monocytes (15).

To reduce morbidity and mortality due to infections, encouraging prophylactic vaccinations among patients with CKD is paramount preventative health maintenance. Vaccination plays a crucial role in curbing vaccine preventable infection in patients with CKD and kidney transplant recipients and

should be done early in the course of CKD or prior to RRT when possible as response rates for many vaccines are inversely related to CKD stage. The risk of bacterial infection and sepsis particularly, pulmonary and genitourinary sources increase with the decline in kidney function. Patients with CKD require the routine adult immunizations including annual influenza and decennial tetanus, diphtheria, and acellular Pertussis (Tdap). All age-appropriate vaccines are also recommended, such as the HPV vaccine and the recombinant varicella zoster vaccine (RZV).

7.2.1 Hepatitis B vaccination in CKD

Patients with CKD who are not infected and not yet immune would benefit from being protected against these risks by vaccination against hepatitis B. Hepatitis B vaccination is recommended for all CKD patients as it provides effective protection against the infection and remains 'standard-of-care' in the hemodialysis population despite immunological challenges. For primary care providers, obtaining hepatitis B surface antibody titer values of previously immunized patients and/or initiating hepatitis B vaccination/booster would be prudent concurrent with nephrology referral, particularly in patients with advanced CKD stages.

Patients with uremia who were vaccinated before they required dialysis have higher seroprotection rates and antibody titers. As seroconversion rates after vaccination in individuals with ESRD are lower than in the general population, various strategies have been devised to try to enhance the immune response, such as using a double antigen dose (40 µg) of one of the recombinant hepatitis B vaccines and providing a booster dose if the antigen titer falls ≤ 10 international units/L after an initial response (19). Double dose (40 mcg) and four doses given have better seroprotection rate (close to 90%) compared to 45–67% with conventional 3 dose schedule.

Hepatitis B vaccination in CKD: recommendations on dosage and schedule

- Patients should receive four doses of hepatitis B vaccine as early in the course of disease as possible
- Recombinant hepatitis B vaccine is recommended
- Use special formulations of vaccine (40 mcg/ml) or two 1 ml 20 mcg doses given at one site.
- Dose schedule should be 0, 1, 2, and 6 months
- Vaccine should be given intramuscular in deltoid regions
- Assess antibody titer to hepB surface antigen (anti-HBs). First titer should be done 1–2 months after the primary course is completed and annually thereafter
- Booster dose should be given if anti-HBs titer falls below 10 mU/ml
- Revaccination with full doses is recommended for persons who do not develop protective antibody titer after primary course.

7.2.3 Pneumococcal vaccination

In patients with CKD, pneumonia is a significant cause of infection-related hospitalization and mortality, and streptococcus pneumonia has been recognized as the most common bacterial etiology of community-acquired pneumonia in the general population and in patients with CKD. Thus, pneumococcal vaccination should be administered to all patients with CKD as early in the disease as possible. CKD patients over the age of 19 should receive a dose of PCV13 first, followed by a dose of PPSV23 at least 8 weeks later. The second PPSV23 dose is recommended 5 years after the first PPSV23 dose for persons aged 19–64 years. In PPV-23-vaccinated persons, PCV-13 should be administered at least 1 year after the PPV-23 dose.

7.2.4 Summary of routine prophylactic vaccinations in CKD

Vaccination plays a crucial role in curbing vaccine preventable infection in patients with CKD and kidney transplant recipients. Vaccination should be done early in the course of CKD or prior to kidney transplantation when possible. It is incumbent upon all healthcare providers to not only stay abreast of the rapidly evolving evidence and recommendations regarding this area but to also continue to update clinical practice regarding vaccines for long-recognized infectious threats, such as influenza, pneumococcal disease and chronic hepatitis B infection, to mitigate the burden of infectious diseases on this particularly vulnerable patient population.

Vaccine	Type	Age	Dose	Vaccination schedule/routes of administration	Booster doses
Hepatitis B, EngerixB®	Inactivated	≥20 years	40 mcg	0, 1, 2, and 6 months/IM	Yes, when antiHBs<10 IU/l
		<20 years	10 mcg	0, 1, and 6 months/IM	Yes, when antiHBs<10 IU/l
Influenza	Inactivated	>3 years	15 µg	Each year/IM before onset of influenza season	No
Varicella	Inactivated	112 years	0.5 ml	One single dose/SC	No
		Adolescents and adults	0.5 ml	Two doses of 0.5 ml/SC Given 4-8 weeks apart	Second dose 4 weeks after
Hepatitis A, Havrix®	Inactivated	>17 years	1440 U	Two doses 0, 6–12 months/IM	No

Measles, mumps, and rubella	Live attenuated	>18years	0.5ml	One single dose/SC Administer if no evidence of immunity: 2 doses of MMR at least 28 days apart	No
Poliovirus	Inactivated	<18years	0.5ml	Three doses with an Interval of 12 months	No (revaccination 1 year after the third dose)
Diphtheria and tetanus toxoids	Inactivated	7years	0.5ml	Three doses/IM (0, 1, 6-12 months)	Booster dose with (Td) every 10 years
Pneumococcal	Inactivated	>19years	1 dose	1 dose PCV 13 followed by PPSV23 with minimum interval of 8 weeks between PCV 13 and PPSV23 can be considered)	Administer 1 dose of PPSV23 ≥5 years later

Table 22: Summary of routine prophylactic vaccinations recommendations for chronic kidney disease

7.3 CHAPTER SUMMARY

- Infection remains a major cause of death and hospitalization among patients with chronic kidney disease.
- Patients with CKD are susceptible to infectious organisms due to the many facets of uremia-associated immune deficiency.
- Infections and their treatments including tuberculosis, HIV, HBV and HCV cause worsening of kidney function and pose a significant source of morbidity and mortality.
- To reduce morbidity and mortality due to infections, encouraging prophylactic vaccinations among patients with CKD is paramount preventative health maintenance.
- Vaccination plays a crucial role in curbing vaccine preventable infections

CHAPTER 8: CKD IN SPECIAL POPULATION



Allocated time - 2 hours



Chapter description – This chapter discusses CKD in children and women. It highlights the key differences in the diagnosis, classification, and management of CKD in children compared to adults. Additionally, it explores the risk factors for CKD in women and outlines appropriate management approaches for women living with the condition



Chapter Objective

By the end of this chapter participants will be able to:

- Discuss the peculiarities of CKD diagnosis and management in children and women

Enabling objectives

By the end of this session, you will be able to:

- Describe the variation in the diagnosis of CKD between children and adults
- Discuss the causes of CKD in children
- List risk factors and indications for screening in CKD
- List the difference regarding categorization of CKD in children versus adult
- Describe measures to prevent CKD progressions
- Describe measures to Manage complications of CKD
- Identify unique risk factors of CKD In women
- Describe contraceptive option for women with CKD

Chapter outline

- 8.1. Diagnosis of CKD in children
- 8.2. Epidemiology and causes of CKD
- 8.3. Risk factors and indication for screening in CKD
- 8.4. Difference regarding categorization of CKD in children versus adult
- 8.5. Measures to prevent CKD progression
- 8.6. Complication of CKD management
- 8.7 Identify unique risk factors of CKD In women
- 8.8 Describe contraceptive option for women with CKD
- 8.9 Chapter Summary

8.1 CHRONIC KIDNEY DISEASE IN CHILDREN

It is kidney damage that persists for more than 3 months with health implications. The diagnosis will be based on structural or functional kidney abnormalities or decreased GFR.

Peculiarities in children regarding diagnosis of CKD

1. Duration of 3 months cannot be applied in neonates and infants below 3 months of age
2. GFR below 60 cannot be applied for those below 2 years of age because of developmental immaturity of the renal system
3. To estimate the GFR, a modified bedside Schwartz formula will be used, which uses serum creatinine, the patient's height, and a constant (k) for all pediatric populations.

Estimated $GFR = k \times \text{height in centimeter} / \text{creatinine in mg/dl}$, where K is 0.413

A new equation (named CKiD U25) which uses serum creatinine and Cystatin C, gender, and age of the child are the future equations of possible CKD screening tools.

4. Serum creatinine is an unreliable marker in the first 72 hours of life.

8.2 EPIDEMIOLOGY AND CAUSES OF CKD IN CHILDREN

Worldwide, CKD is underreported in children as well as adults due to the asymptomatic nature of the early stages of the disease. Data from different regions on CKD stage 3 or more shows the incidence is around 11.9 in Europe and 2.8 to 15.8 in Latin America per million of age-related populations (pmarp). There is a lack of data in low-resource settings, which makes the problem underestimated, but a single-center study in sub-Saharan Africa ranged from 1 to 3 pmarp. Figures on the prevalence of childhood CKD In our county are limited.

The common causes of CKD in children below the age of 5 years are congenital anomalies of the kidney and urinary tract (CAKUT) and genetic or metabolic diseases, while after the age of 5 years, glomerular and tubular causes will take a lead, followed by congenital malformation of the kidney.

Age below 5 years	Age above 5 years
CAKUT	Glomerulopathy
Dysplasia/hypoplasia of the kidney	Hemolytic uremic syndrome
Obstructive uropathy	Focal segmental glomerulosclerosis
Reflux nephropathy	Alport syndrome
Posterior urethral valve	Systemic disease associated (HUS,SLE,vasculitis)

Metabolic or genetic disorders	Tubulointerstitial disease
Polycystic disease	Chronic tubulointerstitial disease
Congenital nephrotic syndrome	Cystinosis/CAKUT
Oxalosis	Nephrotoxic drugs /nephronophthisis
Glomerulopathy	

Table 23: Causes of CKD based on the age

Activity8.2-Case scenario-1



Instruction

Read the case scenario in your manual, form a small group, discuss and answer the question. Present it to the large group

4-year-old child presented with fatigability and limping during walking of 3 weeks duration. Physical examination, he has stunting and pallor, otherwise no finding.

- What detailed Hx you would like to ask?
- How do you work him up
- What possible cause can you list

Time Allocated: 10 minutes

8.3. RISK FACTORS AND SCREENING OF CKD IN CHILDREN

Since the early stage of CKD is difficult to diagnose, we need to have a high index of suspicion of CKD and screen children in the following condition.

Red flags for CKD	Risk factor for CKD
Unexplained Failure to thrive	Abnormal renal ultrasound imaging(CAKUT)
Unexplained anemia	Syndromic child, exposure to nephrotoxic drug, previous AKI
Bony deformity or treatment-resistant rickets	Spinal cord defect(spina bifida -or spina bifida occult)
polyuria and recurrent UTI	preterm or low birth weight
	Systemic disease known to involve the kid

Table 24: screening indication for CKD in children

Additionally, a history of oligohydramnios or polyhydramnios in utero must be explained since it may be a sign of kidney disease in utero.

8.4. CATEGORIZATION OF CHRONIC KIDNEY DISEASE

Chronic kidney disease is categorized according to cause, estimated GFR and albuminuria. Adult categorization works for children too. The exceptions are:

1. Staging of CKD based on GFR only works for those above 2 years of age. The staging of CKD below the age of 2 years is by age-specific standards of GFR, and it is classified as CKD with:
 - A. Normal GFR
 - B. Moderate reduction in GFR if the value lies between 1 and 2 standard deviations below the mean for age
 - C. Severe reduction in GFR if the value is ≥ 2 Standard Deviation below the mean for age
2. Urine albuminuria category: albuminuria test is not highly sensitive with children, since tubulointerstitial disease (tubular proteinuria) produces low molecular weight protein and will not be picked by the albuminuria test. So we may need to utilize proteinuria in place of albuminuria test or do a 24 hour urine protein in selected conditions. The interpretation of result should account age of the child.

Age (months)	Mean GFR + - standard deviation(ml/min/1.73m ²)
<1.2	52 + OR - 9
1.2 -3.6	61.7 + 14.3
3.6-7.9	71.7+ 13.9
7.9-12	82.6 +OR - 17.3
12-18	91.5 + 17.8
18-24	94.5 + 18.1
> 24	104.4+ 19.9

Table 25: NORMAL INFANT AND CHILDREN GFR, BASED ON CR-EDTA

Principles of Treatment of CKD

The goal of treatment relies on

1. Treating underlining cause of CKD if possible
2. Retard progression of the disease
3. Treat complications of CKD
4. Prepare for renal replacement therapy

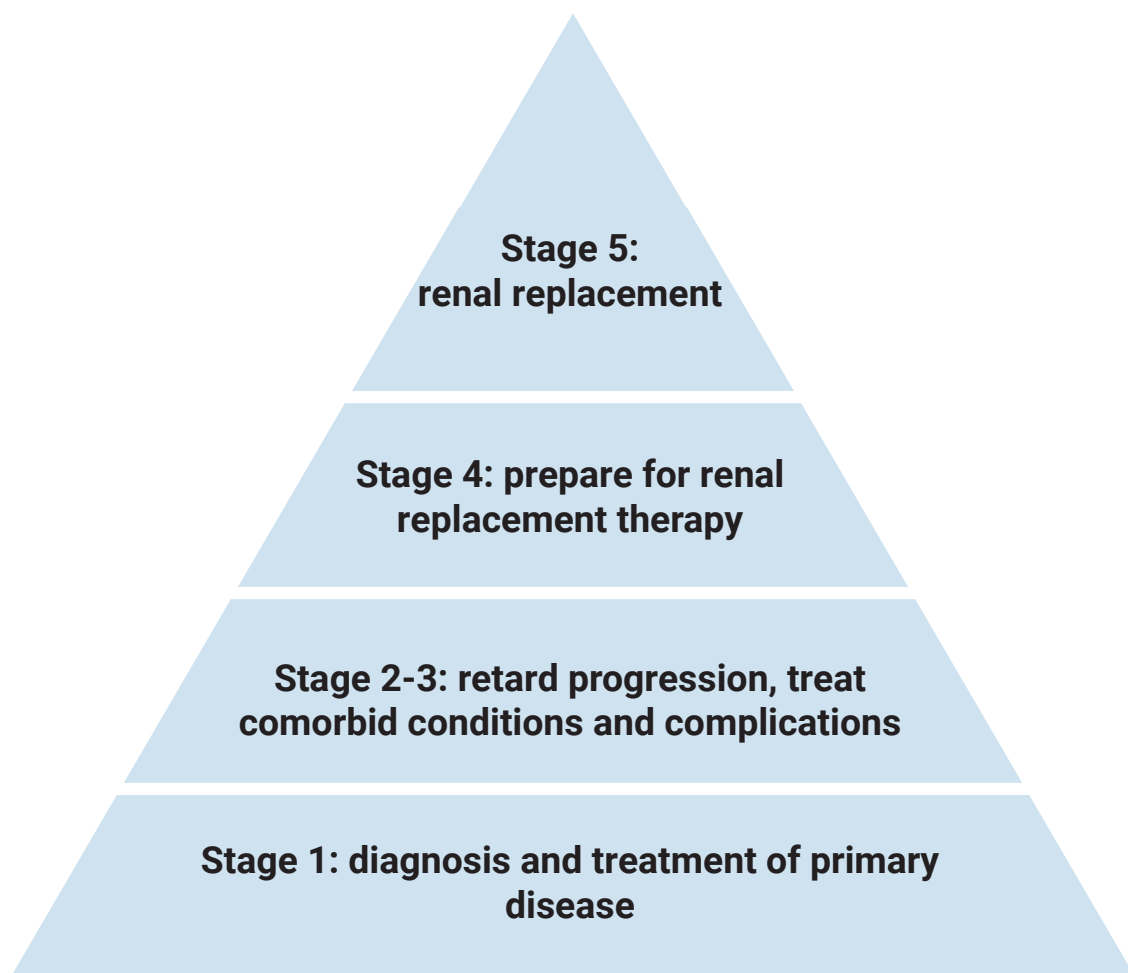


Figure 15: pyramidal representation of CKD progression and management, adapted from Pediatric Nephrology, 2014.

8.5. RETARD PROGRESSION OF CKD

8.5.1. Hypertension

In children, BP should be measured using age and size-appropriate BP apparatus. Hypertension is diagnosed when the BP lies above or equal to the 95th centile of normative value for age, gender, and height percentile till age 13 years, after that hypertension will be diagnosed like adults.

For children aged 1 to 13 years
Normal BP : < 90 th percentile
Elevated BP: > or =90 th percentile to < 95 th percentile
Stage 1 HTN :>= 95 th centile to < 95 th percentile + 12 mmHg
Stage 2 HTN >= 95 th percentile + 12 mmHg

Table 26: . Diagnosis of hypertension in children

-
- All CKD children require annual checkups with 24 hr ABPM for diagnosis, and once on management to assess the adequacy of treatment.
 - Target blood pressure control is below 50th centile as long as no symptoms of hypotension.⁹⁴

ACE inhibitors (enalapril, lisinopril) and angiotensin II blockers (losartan) are the drug of choice irrespective of the degree of proteinuria. Creatinine and potassium should be checked within 2 to 4 weeks of initiation and should be reduced or discontinued if there is > 30 % rise of creatinine. If patient tolerates increase the dose. These drugs should be avoided if the GFR is below 30 and if you are in a setup where you cannot strictly follow renal function test or start renal replacement therapy.

In patients with evidence of fluid overload, diuretics are the initial choice. In advanced kidney disease furosemide works better than hydrochlorothiazide, escalate the dose targeting adequate diuresis.

In case of uncontrolled hypertension after monotherapy combination treatment with (ACE inhibitor/ARB and calcium channel blocker, ACE inhibitor/ARB and diuretics) are options.

In refractory hypertension (defined by uncontrolled BP with three drugs on optimal dose and one of the drugs being diuretics, or controlled BP with 4 drugs) aldosterone blocking agent should be started with potassium monitoring. But adherence, thyroid abnormalities and obstructive sleep apnea needs to be assessed first.

The other antihypertensive drugs include beta-blockers such as propranolol and atenolol. Newer beta-blockers like carvedilol also have antiproteinuric effects and a longer half-life. Vasodilators such as hydralazine, and centrally acting agents like clonidine and minoxidil, can be administered as adjunctive agents in cases of uncontrolled hypertension.

Supportive measures with Salt and water restriction and Exercise should be applied.

8.5.2. Proteinuria

All children with CKD should have their protein-to-creatinine ratio or albumin-to-creatinine ratio measured annually. If these tests are not available, a 24-hour urine protein test should be performed.

Antiproteinuric agent like ACE inhibitors or ARB should be started, with low dose initially and escalate till desired outcome or as long as no hypotensive symptoms. A raise of creatinine by 30 % from the baseline or Hyperkalemia resistant to medical management warrant drug discontinuation. The goal is to reduce the proteinuria to less than 300 mg/24 hours.

8.6. TREATMENT OF COMPLICATIONS OF CKD

8.6.1. Anemia

Anemia definition in children with CKD depends on the age of a child.

- Age below 5 years-hgb of <11 gm/dl
- Age 5 to 12 years-hgb<11.5 gm/dl
- Age 12-15 years –hgb<12 gm/dl, after 15 years definition is same as adult

Anemia workup, diagnosis, and monitoring in children is the same as in adults.

For iron depleted state Start iron treatment with 2 to 3 mg per kg to a maximum of 6 mg/kg, it should be given in an empty stomach. If the anemia is not responding for oral therapy there is a place of IV iron.

IV iron 1–2 mg/kg (maximum of 5 mg/kg/ week or 100 mg/dose) of elemental iron per week for 8–10 weeks. The type of iron, preparation of iv infusion is same like adults.

ESA treatment should be considered when the hemoglobin is below 10 gm/dl and corrected iron store.

- Erythropoietine -Starting dose of 150 iu/kg divided in to two doses per week.
- Darbepoetin: starting dose of 0.5µg/kg/week
- Escalate or deescalate dose by 25 % if the hgb response is not according to the target.

8.6.2. BONE MINERAL DISEASE (BMD)

The clinical manifestation depends on the stage of CKD and the age of the child. Infants generally present with rickets and motor developmental delay while in older children they will present with bone pain, limping, knocked knee, valgus or varus deformity, and fracture with variable degrees of vascular calcifications.

The evaluation and monitoring of CKD-MBD is similar to adults, but the assessment should begin starting from stage 2 CKD unlike adults. Additionally, a child should be evaluated for height and height velocity incrimination during every visit and give a special emphasis during growth spurt (infancy and adolescent).

Principle of management of BMD relies on:

- Control of hyperphosphatemia
- Control of serum calcium
 - Calcium supplementation

- Vitamin D therapy
- Correction of 25 OH vitamin D stores
- Control of PTH

1. Phosphate management

The goal is to keep the serum phosphate level to age-appropriate normal range for early CKD and to the lower limit of normal range for advanced CKD.

Nutritional restriction, especially canned and processed food. For infants on formula milk, choose low phosphate containing formula. Inorganic phosphate containing substances needs to be avoided since it is completely absorbable like antacid and some antihypertensive drugs. Intensive phosphate restriction from diet may significantly compromise the protein intake and affect the growth of children so the restriction should not be in the expense of protein, in this condition, it's better to add phosphate binders.

Calcium based phosphate binders are the first choice as long as there is no hypercalcemia in the child

Drug	Calcium Content	Phosphate Bound per mg Ca++ Absorbed	Available formulation	Dosing
Calcium carbonate	40 %	1 mg/8 mg	500 mg	Do not exceed 2000 mg/day, all source of calcium
Calcium acetate	25 %	1 mg/3 mg	475 mg or 950 mg	
Sevelamer	-	400 mg sevelamer binds to 32 mg phosphate	800 mg/2400 mg sachet	120-160 mg/kg in 3 divided doses

Table 27: Phosphate management

2. Calcium management

The goal is to maintain serum calcium to an age-appropriate normal level. Severe deficiency (serum calcium below 7.5 mg/dl or 0.8 mmol/lit) or symptomatic patient IV correction is required. Otherwise oral calcium 1 to 2 gm of elemental calcium per day in divided doses given.

3. Vitamin D management

To maintain the best gut calcium absorption, the serum vitamin D level should be kept in the range of 30-48 ng/ ml. Vitamin D can be administered on a daily, weekly, or monthly basis.

Vitamin D level	Dose in IU orally
Between 30-48 ng/ml	1000 daily
30 to 20 ng/ml	2000 daily
20 to 5 ng/ml	4000 dailies
Less than 5 ng/ml	8000 daily for 4 wks then 4000 daily for 2 months

Table 28: Vitamin D management

4. Active vitamin D treatment

In children who continue to have elevated PTH levels despite normalization of serum phosphorus and vitamin D levels, active vitamin D should be initiated. The recommended dose of calcitriol is 0.05 mcg/day for children weighing less than 10 kg, 0.1 to 0.5 mcg/day for those weighing 10 to 20 kg, and 0.25 mcg/day for children weighing more than 20 kg.

Active vitamin D should be avoided in children with high normal or elevated serum calcium levels. Despite optimized conventional management, persistent and severe hyperparathyroidism in the presence of high or high-normal Calcium levels, cinacalcet can be tried in dialyzed children older than 3 years.

Parathyroidectomy is the final treatment option, if all the above medical care failed to control the PTH level.

8.6.3. Metabolic acidosis and Fluid, electrolyte management

Metabolic acidosis is not only a complication of CKD, but it also contributes to disease progression. This is common once the GFR is below 50 ml/min/1.73m². If the underlying cause is tubulointerstitial or obstructive uropathy, acidosis may happen even in the early stages of CKD.

In children, it significantly affects growth, so it needs to be managed optimally. The goal is to keep the bicarbonate level at 22meq/l or above but avoid alkalosis.

- Sodium bicarbonate tablets (650 mg = 8 mEq of base) or
- Bicitra (1 mEq sodium citrate/mL), 1 meq/kg start dose

Unlike adults, children may have CKD due to tubulointerstitial disease which may lead them to have polyuria, hyponatremia, and hypokalemia. Universal salt, potassium, and fluid restriction shouldn't be applied to all kids with CKD.

8.6.4. Cardiovascular complications and Dyslipidemia

Left ventricular hypertrophy and cardiovascular mortalities are also common in children especially those started on dialysis. Annual echocardiography screening is recommended.

Lipid profiles should be evaluated annually in CKD children and for those on lipid-lowering treatment every 2 to 3 months.

Hyperlipidemia in children is defined as lipid levels >95th percentile for age and gender

Children with dyslipidemia should practice therapeutic lifestyle changes, which include restriction in saturated fat intake, an increase in fiber diet, and moderate physical activity. If LDL cholesterol is 160 mg/dl and non-HDL cholesterol 190 mg/dl, statin therapy (atorvastatin 5–10 mg/day, simvastatin 20 mg/day) is recommended in children.

8.6.5. Growth and Nutritional evaluation with supportive care

Children with CKD are prone to malnutrition, growth failure, and puberty delay. Decreased intake, high metabolic demand, acidosis, renal loss of fluid and electrolytes, growth hormone production failure as well as resistance and high level of PTH are some of the factors attributing to this.

Assess the adequacy of nutritional intake (use dietary recall or 3 days dietary records).

Take anthropometry measurements (estimated dry weight, weight for age standard deviation, height or length for age standard deviation, BMI for height -age-, head circumference) and height velocity should be performed.

Blood tests like serum albumin level has its own limitations.

Anthropometric evaluation should be performed every 3 to 6 months using the WHO growth curve and plotted on the chart of the child.

For children under 3 years of age, and those with severe acute malnutrition or excessive urinary and electrolyte losses, assessment may be needed every 1 to 3 months.

The daily recommended dietary allowance (RDA) of healthy children of the same age is recommended. 100 % of RDA of protein should be offered and no protein restriction. Children on hemodialysis or peritoneal dialysis should take an additional 0.4 to 0.8 gm/kg/day of protein.

For those with affected anthropometry, an additional calorie is recommended to have catch-up growth. If they cannot tolerate oral feeding or fail to gain weight, tube or gastrostomy tube feeding is an option.

Multivitamin supplementation, especially for those on hemodialysis is required. However excess vitamin A (leads to hypercalcemia and hyperlipidemia) and vitamin C (may cause oxalate deposition) intake should be avoided.

Despite adequate nutritional supplementation and optimal management of CKD complications such as acidosis, anemia, hyperparathyroidism, and hyperphosphatemia, if the child's growth does not improve, recombinant human growth hormone (rhGH) therapy may be considered—provided there is potential for growth, a height SDS of less than -2 SD, and a height velocity below -1 SD.

Active rickets or a slipped capital femoral epiphysis should be resolved and wrist and hip X-rays, bone age estimation should be done before starting growth hormone therapy

- The initial dose of GH is 0.05 mg/kg/day SC or 30 IU/m² / week (0.35 mg/kg/week).

Adequate response is defined if the growth velocity is > 2cm/year over the baseline before rhGH.

Indications to stop growth hormone therapy:

- CKD stages 2–4: PTH level >400 pg/ml; restart GH when PTH level is <200 pg/ml.
- CKD stage 5: PTH level >900 pg/ml; restart GH when PTH level is <450 pg/ml.
- In all stages of CKD, if the patient develops a slipped capital femoral epiphysis or symptomatic high turnover renal osteodystrophy.
- Epiphyseal closure.
- Other complications of growth hormone, including pseudotumor cerebri.

Neurodevelopment

In CKD children's brain development will be exposed to toxins which will prone them to have variable degrees of encephalopathy, motor developmental delay, seizure and hypotonia.

Surveillance for risk factor of neurodevelopment abnormality, meticulous medical management with multidisciplinary approach is needed to limit the sequelae.

Preventing acute insult

Avoid dehydration or nephrotoxic drugs, if it is a must then Drug dose adjustment should be applied.

Immunization

Immunization should be up-to-date based on the EPI schedule. All children should receive live attenuated vaccine before transplantation. Avoid live attenuated vaccines on those children on immunosuppressant drugs.

-
- All children with CKD should receive a yearly influenza vaccine
 - **Pneumococcal** vaccine should be given to children with nephrotic syndrome and with CKD
 - **Hepatitis B vaccination** should be provided to all children with CKD or undergoing dialysis if not immune.

Transition to adult care

Adolescent age is a risky period due to adherence issues. The risk of advanced disease loss of graft for transplant children is high at this period. Frequent counseling with a multi-disciplinary team approach is needed.

Counseling and preparation

Counseling of caregivers on the option of RRT should start when the child is in stage 4 CKD. Chronic hemodialysis, chronic peritoneal dialysis, and renal transplantations are the choices of management, but Preemptive renal transplantation is an ideal choice for children.

- the indication for renal replacements therapy depends on;
1. Estimated GFR, value below 15 can be considered but other factors like-Clinical well-being of the child and Laboratory derangements resistant to medical management

8.7 SPECIAL CONSIDERATIONS IN WOMEN

It is increasingly recognized that sex (biological attributes) and gender (sociocultural factors) factors across individuals contribute to differences in kidney health and disease.

There is substantial literature demonstrating that both sex and gender-related factors (e.g., puberty, menstrual patterns, hormonal contraception, pregnancy and pregnancy-related complications and menopause) play important roles in the risk, progression, complications, and treatment of kidney disease.

Women are at an increased risk of CKD. Some risk factors for CKD such as recurrent urinary tract infections, autoimmune conditions such as systemic lupus erythematosus are more common in women. They may also suffer hypertensive disorders (pregnancy-induced hypertension and pre-eclampsia) and AKI in pregnancy.

UNIQUE RISK FACTORS OF CKD IN WOMEN

1. Pregnancy –Hypertensive disorders in pregnancy
2. Pregnancy conditions that cause AKI –Puerperal sepsis, hyperemesis gravidarum, post-partum hemorrhage

-
-
3. Autoimmune disorders
 4. Recurrent Urinary tract infections

Pregnancy

Pregnancy is a unique challenge and is a major cause of acute kidney injury (AKI) in women of childbearing age; AKI and pre-eclampsia (PE) may lead to subsequent CKD.

People with CKD are at risk for adverse pregnancy- associated outcomes, including:

- Progression of their underlying CKD,
- A flare of their kidney disease
- Pre-eclampsia,
- Preterm delivery, and small for gestational age infants.

The risks increase with CKD progression thus posing potentially challenging ethical issues around conception and maintaining of pregnancies.

The severity of CKD is associated with risk of adverse pregnancy outcomes. A multidisciplinary approach to preconception counseling and management of pregnancy is necessary to achieve optimal outcomes for both the person with CKD and the infant.

Pre pregnancy Counseling

All women with CKD should receive prepregnancy counseling, especially those with CKD stages 3 to 5, but women with CKD 1 or 2 may also have worse pregnancy outcomes than the general population. Issues that should be covered in counseling:

A. Maternal Risks

- Accelerated decline in glomerular filtration rate, sometimes precipitating dialysis during pregnancy or soon after in women with advanced chronic kidney disease
- Severe maternal hypertension with risk for stroke
- Superimposed preeclampsia with kidney, hepatic, thrombotic, or bleeding and neurologic risks
- Exacerbation of proteinuria, including nephrotic syndrome with risks for thrombosis

B. Fetal Risks

- Fetal growth restriction or intrauterine fetal death from placental insufficiency
- Preterm delivery, with both short-term and long-term consequences
- Teratogenicity and/or fetotoxicity of some drug therapies for kidney disease

Activity 8.7



Role play

Martha is a 35-year-old lady. She has been told to have CKD few months back. Her creatinine is 2.5mg/dl and no proteinuria on dipstick. She does not have children. She has history of pregnancy loss at gestational age of 24 weeks 5 years back. Her tears fall down her cheeks when she talks about her pregnancy. She tells you that it was a twin pregnancy, and she had severe preeclampsia at time. The pregnancy was terminated. She regrets not having children and she has strong desire to be pregnant. How would you counsel her? (Clue stage her CKD)

Time Allocated: 15 minutes

Pregnancy in women with CKD

Pregnancy presents unique challenges among women with CKD.

The hemodynamic changes observed in normal pregnant women make it difficult to identify CKD during pregnancy. Increased blood volume, decreased systemic vascular resistance, and increased cardiac output determine glomerular hyperfiltration, expressed by a 50% increase in GFR (serum creatinine in the normal range: 0.4–0.6 mg/dL) and a slight increase in proteinuria (the threshold for elevated proteinuria in pregnancy has been set at a higher level of 300 mg/day).

In early stages of CKD, episodes of AKI arising out of peripartum volume changes, pre-eclampsia, puerperal sepsis, hyperemesis gravidarum, peripartum hemorrhage and severe malaria in pregnancy may lead to worsening of the CKD and progression to ESKD.

More advanced stages of CKD are associated with adverse maternal and fetal outcomes including preterm delivery, low birth weight and worsening of kidney disease.

Estimating GFR in pregnancy

Considering that the estimated GFR (eGFR) is not validated in pregnancy, the current recommendation is to assess GFR through serum creatinine. Health workers should monitor kidney function using serial creatinine measurements and UACR to detect reduction in kidney function in pregnancy. Patients with rising creatinine and protein should be referred to nephrologists.

Diagnosis of pre-eclampsia in CKD

The diagnosis of pre-eclampsia in the absence of CKD is based on the presence of proteinuria and hypertension—features that are also seen in CKD. This overlap makes it difficult to distinguish

between the two conditions, especially when follow-up is inadequate. Health workers should suspect pre-eclampsia if there is worsening proteinuria and hypertension at 20 weeks of gestation or later. Uterine artery Doppler may be used to assess placental blood flow and help diagnose pre-eclampsia superimposed on CKD.

8.8 CONTRACEPTION OPTION FOR WOMEN WITH CKD

Health-care providers should offer contraceptive and pre-pregnancy counseling to patients with CKD. Although fertility is often reduced—mainly due to a higher prevalence of anovulatory cycles—pregnancy is still possible at all stages of CKD, including during dialysis and after kidney transplantation.

Clinicians should counsel women of reproductive age with end-stage renal disease (ESRD) on contraceptive options if pregnancy is not desired. Progestogen-only methods are considered safer due to their lower risk of thromboembolism and hypertension. Both medicated and non-medicated intrauterine devices (IUDs) are also effective and suitable contraceptive options for women with CKD.

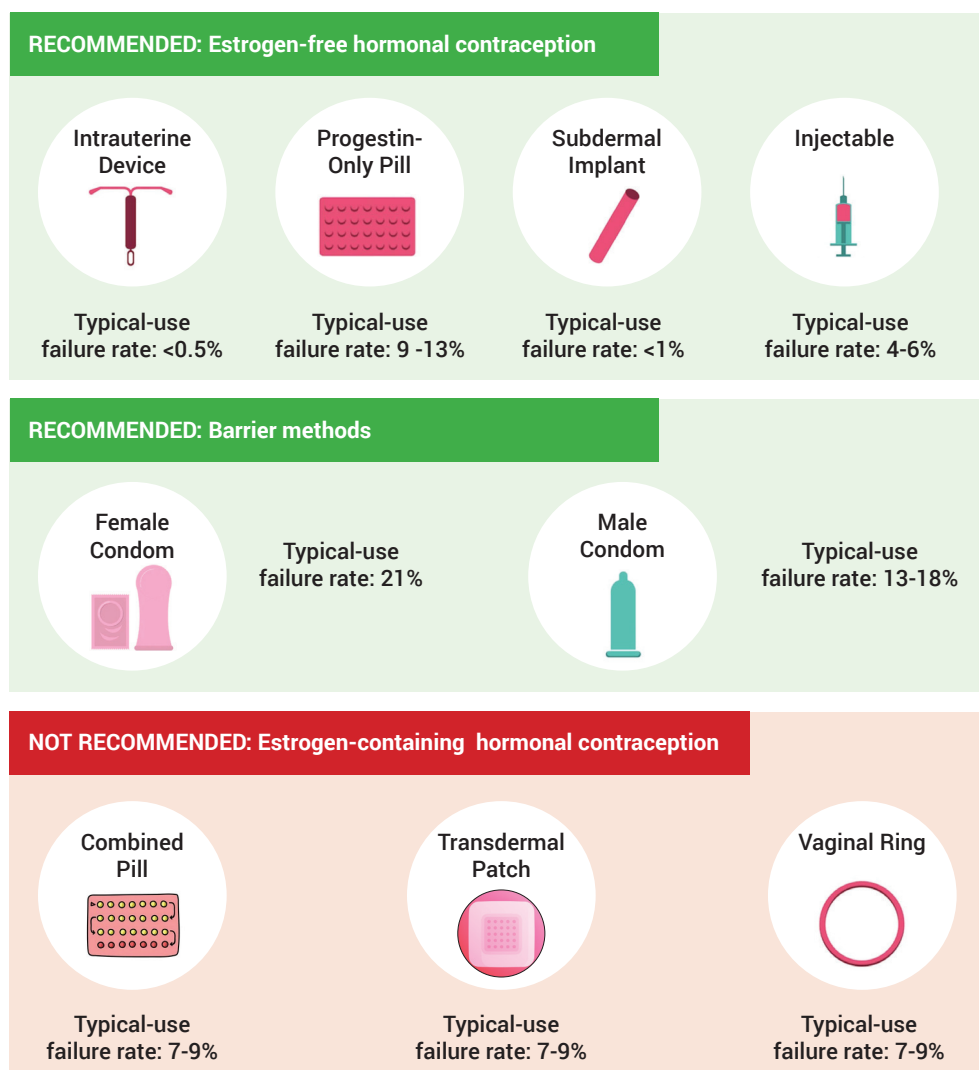


Figure 16: Contraception options for women with CKD (Adapted from Integrated Chronic Kidney Disease Management Guidelines for Uganda)

Activity 8.8



Case discussion

Elsabeth has stage 4 CKD. Her underlying renal disease was systemic lupus erythematosus. She has been counselled about pregnancy in CKD and she opted to try pregnancy after renal transplant. She came to you for contraception advice. What will you recommend?

Time Allocated: 10 minutes

8.9 CHAPTER SUMMARY

- Understanding the cause of CKD in children is critical for early recognition of the disease and to prevent further progression
- Definition and staging of CKD in children needs modification on certain criteria
- Early identification and specific therapy of CKD complication may retard progression of the disease.
- Control of hypertension and proteinuria are cornerstones of therapy in CKD
- Frequent counselling and multi-disciplinary team work is vital when we treat children who needs transition of care to adult unit.
- People with CKD are at risk for adverse pregnancy- associated outcomes
- Health-care providers should offer contraceptive and pre-pregnancy counselling to these patients

CHAPTER 9: CKD PROGRAM MANAGEMENT



Time Allocated: 2 hours and 15 minutes



Chapter Description: This chapter describes screening and management of CKD at different levels of healthcare system in Ethiopia and monitoring and evaluation of CKD program.



Chapter Objective:

By the end of this chapter, participants will be able to describe screening and management of CKD at different levels and monitoring and evaluation of CKD program.

Enabling Objectives:

By the end of this chapter participants will be able to:

- Describe the screening tests to be made available at different levels of the health care delivery system in Ethiopia
- Explain CKD management strategies at different levels of the health care delivery system
- Define key terms related to monitoring and evaluation
- Demonstrate the monitoring and evaluation activities of CKD program
- Identify essential indicators and data elements used in CKD program monitoring
- Describe appropriate data collection and reporting tools for CKD services
- Discuss the CKD program data to inform decision-making and improve service delivery

Chapter Outline

- 9.1 Screening tests required at different levels of the healthcare delivery system in Ethiopia
- 9.2 CKD management strategies at different levels of the health care delivery system in Ethiopia
- 9.3 Understanding monitoring and evaluation
- 9.4 Data collection and reporting tools for CKD services
- 9.5 Key indicators and data elements to be used in CKD program monitoring
- 9.6 Chapter summary

9.1 SCREENING TESTS TO BE MADE AVAILABLE AT DIFFERENT LEVELS OF HEALTHCARE IN ETHIOPIA

Activity 9.1 Group Discussion



Instruction:

- Instruct them to divide you into 4 small groups and to select group presenters.
- Ask them to Discuss and write down the tests required at different levels of the health care delivery system in Ethiopia
- Tell them to write their answers on a flip chart and select 2 groups to present their work to the larger group.

Time Allocated: 10 minutes

Effective diagnosis and management of diseases require the availability of appropriate diagnostic tests at various levels of health service delivery. In Ethiopia, the healthcare system is structured into three main tiers: primary, secondary, and tertiary levels. Each level is equipped to provide specific types of laboratory and diagnostic services based on capacity, resources, and scope of care.

This table outlines the key diagnostic tests recommended at different levels of health facilities in Ethiopia. It serves as a guide for healthcare providers, program planners, and policymakers to ensure that essential diagnostic services are accessible, appropriate, and aligned with the level of care provided.

Health facility	Screening Tests required	Action
Health posts	BP screening	• Health education about CKD and its risk factors • Identify risk and refer for CKD screening
Rural health center	BP screening Urine dipstick Point of care UACR (if available) Glucometer	• Screening of patients at risk of CKD should be done every year • Refer to Primary hospital for diagnosis of CKD

Primary Hospitals and Urban health centers	Serum creatinine Urine dipstick UACR Ultrasound Glucometer HepBSAg, Anti-HCV, HIV screen, TPHA	• Perform UACR and creatinine at the HF or through referral • Perform urine dip stick • Confirm diagnosis (from eGFR and proteinuria) • Treat to slow progression of CKD
General Hospital	Serum creatinine Urine dipstick UACR Ultrasound Glucometer Serum electrolyte HepBSAg, Anti-HCV, HIV screen, TPHA	• Confirm CKD, determine the cause and treat • Treat complications such as anemia, hyperkalemia • comprehensive conservative care • eGFR<30: refer to nephrologist
Specialized hospital	Serum creatinine Urine dipstick UACR Ultrasound Glucometer Serum electrolyte HGBA1C Cystatin C	• Confirm CKD, determine the cause and treat • Treat complications such as anemia, hyperkalemia • comprehensive conservative care • eGFR< 30: refer to nephrologist • eGFR<15: refer for kidney transplant

Table 29: Screening tests at all tiers of the health care delivery

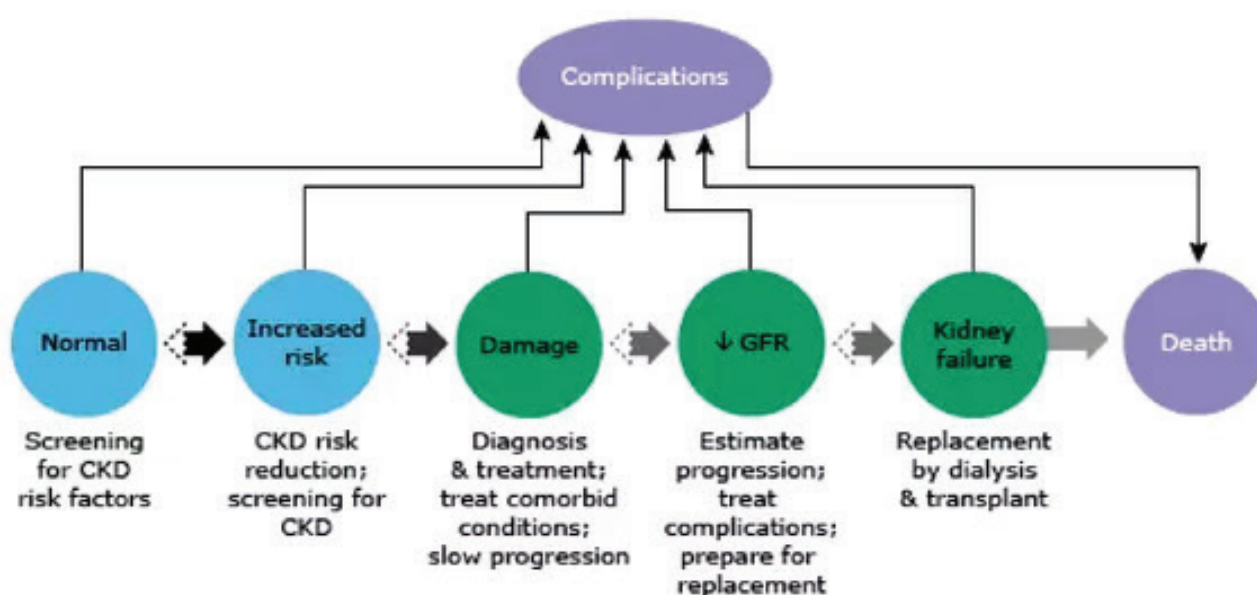
9.2 CKD MANAGEMENT AT DIFFERENT LEVELS OF THE HEALTH CARE SYSTEM

Grading of CKD:

CKD is graded as low risk, moderately increased risk, high risk and very high risk (Figure 1) according to the level of albuminuria and estimated GFR. It is very important to grade all patients diagnosed with CKD because these grades have management and prognostic implications.

1. Low risk – eGFR of >60 with normal to mildly increased albuminuria (UACR- $<30\text{mg/g}$ OR $<3\text{mg}/\text{mmol}$)
2. Moderate risk – eGFR of 45- 59 with normal to mild albuminuria (UACR – $<30\text{ mg/g}$) OR eGFR of $>60\text{ ml/min/ }1.73\text{m}^2$ with moderately increased albuminuria (UACR – $30\text{-}300\text{mg/g}$)
3. High risk – eGFR of $30\text{-}44\text{ ml/min/ }1.73\text{m}^2$ with mild albuminuria OR eGFR of $45\text{ –}59\text{ ml/min/ }1.73\text{m}^2$ with moderate albuminuria OR eGFR of $>90\text{ ml/min/ }1.73\text{m}^2$ with severely increased albuminuria.
4. Very high risk – eGFR of $<30\text{ ml/min/ }1.73\text{m}^2$ with normal to mild albuminuria OR eGFR of $30\text{ ml/min/ }1.73\text{m}^2$ to $44\text{ ml/min/ }1.73\text{m}^2$ with moderate albuminuria OR eGFR of $45\text{- }<60$ with severe albuminuria.

Conceptual model for chronic kidney diseases



This diagram presents the continuum of development, progression and complications of chronic kidney disease (CKD) and strategies to improve outcomes. Green circles represent stages of CKD; aqua circles represent potential antecedents of CKD, lavender circles represent consequences of CKD; and thick arrows between circles represent the development, progression and remission of CKD. “CKD” is defined as the presence of either kidney damage or decreased kidney function for three or more months, irrespective of cause (underlying illness and pathology). “Complications” refers to all complications of CKD, including complications of decreased GFR, albuminuria and cardiovascular disease. Complications may also arise from adverse effects of interventions to prevent or treat the disease. The horizontal arrows pointing from left to right emphasize the progressive nature of CKD. Dashed arrowheads pointing from right to left signify that remission is less frequent than progression.

Figure 17: Conceptual Model for CKD

9.2.1 Low Risk Patients

- eGFR of $>60\text{ml/min/1.73m}^2$ with normal to mildly increased albuminuria (UACR $<30\text{ mg/g}$ or $<3\text{ mg/mmol}$) or a negative urine dipstick test

Health Posts and Family health team at HC

- Health education about CKD and its risk factors
- Identify risk and refer for CKD screening

Rural Health centers

- Screen and manage risk factors for CKD
- Screen for CKD with dip stick
 - If proteinuria of 1+, repeat after 3 months, if still positive refer to PH for possible diagnosis of CKD
 - If proteinuria $> 1+$, refer to PH for diagnosis of CKD
- Screening of patients at risk of CKD should be done every year
- Educate on:
 - Adequate hydration
 - Balanced and healthy diet (low salt diet, adequate fruits and vegetables etc.)
 - Regular exercise and weight reduction
 - Avoidance or reduction of alcohol intake
 - Avoidance or cessation of tobacco use including smoking cigarettes, shisha, chewing tobacco etc.
 - Avoidance of unlicensed herbal medication
 - Indiscriminate or prolonged use of analgesics especially NSAIDS and illicit drug use eg. heroin, cocaine, etc

Counsel patients on adherence to medication once diagnosis is confirmed at PH and referred back to HC for follow-up.

9.2.2 Moderate Risk Patients

- eGFR of 45- 59 with normal to mild albuminuria (UACR – $<30\text{ mg/g}$) or a negative urine dipstick test, or
- eGFR of $>60\text{ ml/min/ }1.73\text{m}^2$ with moderately increased albuminuria (UACR – $30\text{-}300\text{mg/g}$) or trace/+ urine dipstick test ≥ 89

Manage At URBAN Health Center & Above

Aims of care:

- Investigate to determine underlying cause
- Reduce progression of kidney disease (manage albuminuria)

- Restrict use of nephrotoxic drugs
- Health education and promotion of health lifestyle
- Screen and Manage risk factors for CKD like DM/HTN, infections(immunization)
- Screen and treat complications particularly anemia.
- Review Creatinine & Albuminuria 6 – 12 months

9.2.3 High Risk Patients

- eGFR of 30-44 ml/min/ 1.73m² with mild albuminuria or a negative urine dipstick test or
- eGFR of 45 –59 ml/min/ 1.73m² with moderate albuminuria or trace/+ urine dipstick test
- eGFR of >60 ml/min/ 1.73m² with severely increased albuminuria or ++ or greater urine dipstick test

Manage At General Hospital and Above

Goals of care:

- Medication review / renal dose appropriate for eGFR
- Education & promotion of healthy lifestyle
- Screen and Manage risk factors for CKD
- Screen and treat complications such as Hyperkalemia, BMD, anemia,
- Restrict use of nephrotoxic drugs
- Appropriate referral to a nephrologist when indicated
- Review creatinine & albuminuria every 1- 3 month

9.2.4 Very High Risk

- eGFR of <30 ml/min/ 1.73m² with normal to mild albuminuria or any urine dipstick test result, or
- eGFR of 30 ml/min/ 1.73m² to 44 ml/min/ 1.73m² with moderate albuminuria or trace/+ urine dipstick test or
- eGFR of 45-59 ml/min/ 1.73m with severe albuminuria or ++ or greater urine dipstick test

Manage at Regional Referral Hospital & Above

Goals of care

- Review medication / ensure renal dose appropriately for eGFR
- Provide health education about CKD, its risk factors, and progression
- Promote a healthy lifestyle
- Manage risk factors and causes for CKD
- Screen and treat complications
- Restrict the use of nephrotoxic drugs
- Prepare for kidney replacement. If comprehensive conservative care has been instituted patients may than be referred back to HC.
- Review creatinine and albuminuria every 1 month

9.3 UNDERSTANDING MONITORING AND EVALUATION

Activity 3



Reflective Thinking

Instructions

Consider the following questions and write down your thoughts.

1. Why is it important to monitor CKD prevention and management programs?
2. What are the key differences between monitoring and evaluation?
3. What are the essential components of an effective M&E system?

Time: 20 minutes

It is important to monitor CKD Prevention and Management Programs:

1. Early Detection and Timely Intervention

CKD often progresses silently, with symptoms appearing in advanced stages. Regular monitoring ensures that screening and early detection programs are effectively reaching at-risk populations (like people with hypertension, diabetes, and cardiovascular diseases), allowing timely intervention to slow disease progression.

2. Improving Patient Outcomes

By tracking program performance, it's possible to identify gaps in care — such as missed screenings, poor follow-up, or inadequate treatment — and address them promptly. This helps reduce complications like kidney failure, cardiovascular events, and premature death.

3. Resource Allocation and Program Efficiency

CKD care, especially at later stages, is resource-intensive and costly. Monitoring helps determine whether prevention efforts are working, potentially reducing the long-term burden on the healthcare system by shifting focus towards cost-effective, early-stage interventions.

4. Data-Driven Decision Making

Monitoring provides critical data for assessing trends, measuring impact, and making evidence-based decisions. It supports advocacy for policy changes, resource mobilization, and scaling up successful strategies.

5. Integration with Other NCD Programs

Since CKD shares common risk factors with other non-communicable diseases (like hypertension and diabetes), monitoring allows for better coordination and integration of services, leading to more comprehensive and patient-centered care.

6. Raising Community Awareness and Engagement

Regular reporting and feedback from monitoring activities help raise public awareness about CKD risks, symptoms, and the importance of early check-ups. It also encourages patient engagement and adherence to prevention and management plans.

7. Ensuring Quality of Care

Monitoring helps maintain standards by assessing whether healthcare providers follow national guidelines and protocols for CKD prevention, diagnosis, and management. This ensures consistent, high-quality care across different service delivery points.

Definition of terms

Health Management Information System (HMIS): is the routine collection, aggregation, analysis, presentation, and utilization of health and health related data.

Purposes of HMIS:

- Availing accurate, timely and complete data
- Strengthening the use of locally generated data for evidence-based decision making

Monitoring: is a routine tracking of progress towards the performance standards that were set.

- Is a continuous function
- Relies on the systematic collection of data on specified indicators
- Informs management and stakeholders of an ongoing development of intervention

Monitoring addresses the following questions:

- To what extent are planned activities actually realized?
- Are we making progress toward achieving our objectives?
- What services are provided, to whom, when, how often, for how long, and in what context?
- What is the quality of the services provided?
- What is the cost per unit service?

To effectively monitor and evaluate, a robust HMIS is essential.

- Though monitoring and evaluation are interdependent, they differ in purpose, approach, and implementation.
- Particularly Monitoring involves:
 - Keeping good records at health facility,
 - Reviewing health facility's records regularly,
 - Compiling data, and
 - Analyzing key indicators

Evaluation: Is the periodic assessment of program impact and value.

- It involves judging, appraising, or determining the worth, value, and quality of a program or project.
- It involves comparing the present situation with the past in order to find out to what extent the original purpose has been achieved (effect).

Evaluation addresses the following questions:

1. What outcomes are observed?
2. What do the outcomes mean?
3. Does the program make a difference?

Note: Though monitoring and evaluation are interdependent, they differ in purpose, approach and implementation.

Data flow involves the following:

1. Data Capturing & Collection – patient-level forms (patient chart, intake, follow-up forms, CKD screening tally sheet or HTN/DM treatment cohort register)
2. Data Analysis & Presentation – facility-level compilation of indicators
3. Data Reporting – aggregate reporting to woreda, zone, region, national levels
4. Data Use – evidence-based planning, resource allocation, and program improvement .

Facility Based HMIS

Recording: The practice of capturing data on patients' management over time and across clinical sites and writing information either directly on paper forms and/or entering into a computer.

Reporting: The routine tracking (monitoring) of priority program management information and its intended aggregated patient outcome data (evaluation) at the facility, woreda, Zone, Region and national level.

Indicator: An indicator is a measurable number, proportion, percentage or rate that suggests the extent of achievement of a program or the level of some condition in the population.

Program information reported to the next level should be complete, accurate, timely, and consistent.

It is essential for program management and specifically used to:

- Systematically evaluate patient progress and treatment outcome,
- Identify problem and measure its magnitude,
- Monitor and evaluate the program performance,
- Perform evidence-based planning (e.g. resource allocation for drug and laboratory supplies, designing interventions, etc.)

Therefore, health care workers should always ensure that records are always up-to-date, complete, accurate and reliable.

9.4 DATA RECORDING, AND REPORTING TOOLS OF CKD SERVICES

The information system CKD is currently under development. Recording and reporting formats are to be used for ensuring accurate documentation of patient-level data, which are then transferred into registers serving as sources for monthly reports on CKD indicators generated by health facilities. These tools are designed based on the CKD service flow within health facilities, as outlined in the figure below.

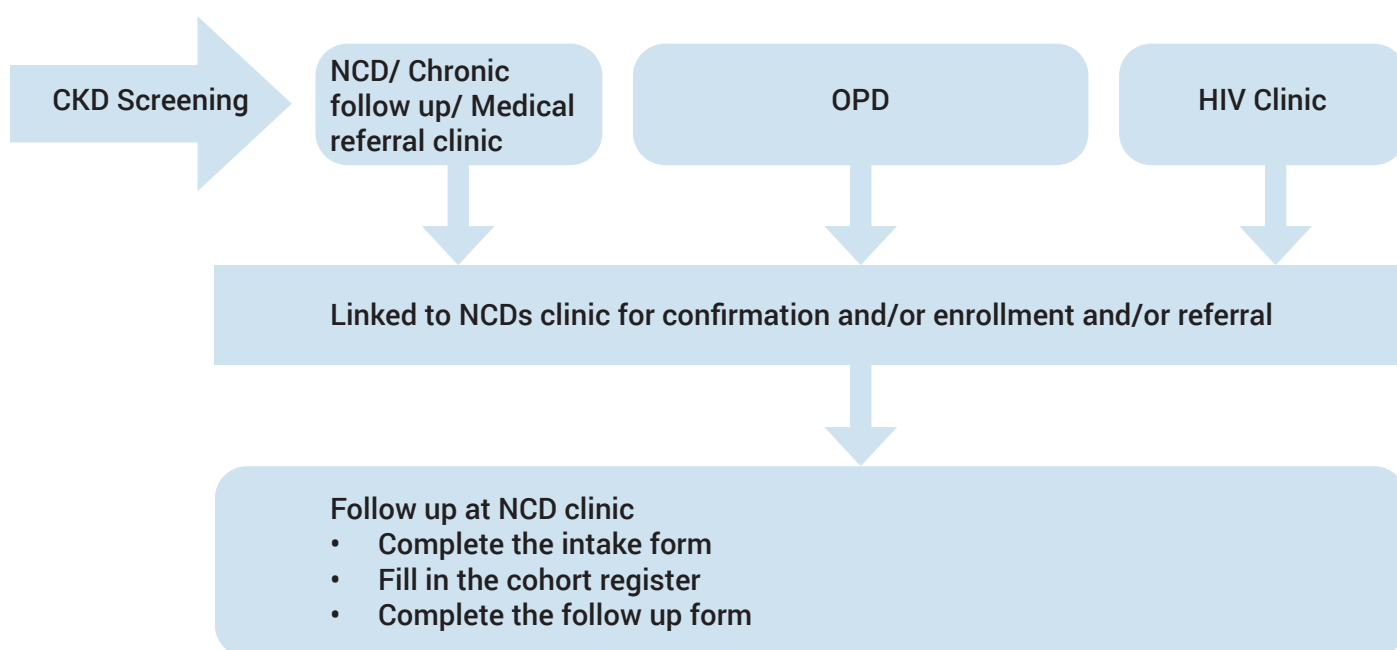


Figure 18: CKD service flow at health facilities

The tools for recording CKD services are summarized below:

1. CKD screening documentation tools:

DM/HTN/CKD Screening Tally Sheet: Simple log of urine protein and/or creatinine $\pm$ BP $\pm$ FBG/RBG) for all at-risk clients. It is kept at triage, OPD, NCD clinic, HIV clinic, etc. The reportable data elements are directly compiled from it.

DM, Hypertension and CKD Treatment Cohort Register: It is a longitudinal register and prepared integrated with the mentioned NCDs. It is used to register and follow up clients who are confirmed to have hypertension (HTN) and/or diabetes mellitus (DM) and/or CKD and enrolled in care. It is kept at NCD/ chronic follow up clinic of HCs or hospitals. The reportable data elements are directly compiled from the register by the end of each reporting period. The service provider completes the information required by the register as soon as the service is provided. Cohort registers are kept at NCD/Chronic follow-up/medical referral clinic. (See annex 3 for Cohort register – CKD integrated.)

- **The Enrollment cohort:** This includes patients registered with diabetes, hypertension and CKD during a specific Ethiopian month and year. The purpose is mainly to assess the number of patients enrolled to care in the reporting period. The cohort for these patients will be the month and the year they were registered, not the date they were initiated on drug treatment. New patients enrolled will be registered as they come in their respective cohorts.

Individual medical records: The hypertension/diabetes/CKD intake and follow up forms are used to document patients' past and present medical records, risk factors, action taken and treatment plans including CKD. These formats are used for newly enrolled patients and patients on follow-up care and treatment. Both are used for patient clinical decision making and help maintain quality of services. Both intake and follow up forms are prepared by integrating hypertension, diabetes and CKD programs.

Hypertension/Diabetes/CKD intake form: The purpose of this form is to record the clinical notes and services provided to newly enrolled hypertension/diabetes/CKD patients. This form includes CKD test results. It is used at HCs, hospitals, and private clinics authorized to provide this service. The form is pre-printed on A4-size paper and stored in an integrated individual folder. The HTN/DM intake form is also integrated into the EMR at facilities using electronic systems. The service provider completes the form during patient enrollment into hypertension/diabetes/CKD care. (See Annex 1).

Hypertension/Diabetes/CKD follow up form: This form is used to document the periodic health status of patients during follow-up for chronic illnesses such as HTN, DM, and CKD. It is used to record the ongoing clinical notes and services provided to enrolled patients. This form is used at HCs, hospitals, and private clinics authorized to provide such services. It is printed on A3-size paper and kept in an integrated individual folder. The follow-up form is also integrated into the EMR at applicable facilities. The service provider completes the form during each follow-up visit at the NCD/chronic follow-up clinic. (See Annex 2).

Patient flow and monitoring for CKD

A. Screening at NCD unit/OPD/ HIV clinic

- Complete screening tally sheet
- Clients with proteinuria or elevated creatinine will be referred to NCD/chronic follow up/medical referral clinic for diagnosis confirmation

B. Enrollment of clients diagnosed to have CKD at NCD clinic

- Complete intake form
- Complete HTN and DM/CKD register for month zero.

C. Follow up of clients enrolled for HTN/DM/CKD

- Follow up form completed
- Update cohort register accordingly.

D. Complete follow-up forms for subsequent visits and update the cohort register accordingly.

9.5 KEY INDICATORS FOR CKD SERVICE

Activity 9.5: Demonstration



Instruction: Review the patient details below and complete treatment cohort register provided.

- Name: XX
- Age/sex: 56/Female
- Health facility: Bole HC
- Patient's history: She is known case of hypertension for the past 6 months and is on treatment, and today came with a newly diagnosis of DM.
- On reviewing her prescription reports, she is on amlodipine 10 mg daily. However, her BP was not under control.
- BP: 131/99 mmHg FBS: 198 mg/dl
- The medical officer decides to modify her management as per the protocol.
- Date of registration: 8.1.15
- Unique NCD treatment number: 00002
- Address: Bahir Dar, Kebele 03, House no. 154
- Phone number: Not available
- Alternative number: +251911***** (spouse number)
- Relevant history: She smokes tobacco, does not consume alcohol, and her elder brother died due to heart attack a year ago.
- On examination, height: 157 cm and weight: 76 kg. The medical officer prescribes the following medicines:

- The patient is advised to return to the clinic after one month.

The lab results are as follows as on 8.2.13:

- Total cholesterol: 226 mg/dl
- Fasting blood sugar: 225mg/dl
- Urine protein +:
- Serum creatinine: 1.7 mg

The patient's follow-up visits are as follows:

The patient did not return for treatment in Yekatit. On 6 Megabit, the nurse called the patient. The patient reported that she was feeling fine, so there was no need to continue treatment.

- 8.2.15 – BP 129/81 mmHg, FBS 230mg/dl, serum creatinine 1.8 advised to continue Amlodipine.

Metformin plus Dapaglifozin was also added in the regimen.

- 08.3.15 – BP 117/73 mmHg, RBS 122 advised to continue the same drug.
- 08.4.15– BP 120/72 mmHg, HbA1C 6.8 advised to continue the same drug.
- 08.5.15– BP 124/76 mmHg, RBS 115 advised to continue the same drug. 233

Time: 40 minutes

To improve the prevention, early detection, and management of CKD among people with hypertension and diabetes, it's important to routinely track a few key indicators. These indicators help us understand how well we're doing, where we're making progress, and where more effort is needed.

Below are the essential CKD indicators health workers should record and report regularly.

The table below summarizes reportable data elements for CKD services along with frequency of reporting and level of report.

A. For Patients with Hypertension

What to Measure	What to Record/ disaggregation	How Often	Where to Report From
1. Number of hypertension patients screened for CKD using urine dip stick	Age, sex, and test result	Monthly	Health Posts, Health Centers, Clinics, Hospitals
2. Proportion of those screened with $\geq 1+$ proteinuria.	Age, sex	Monthly	Health Centers, Clinics, Hospitals

3. Number of hypertensive patients that received creatinine test for CKD screening.	Age, sex, and test result	Monthly	Health Centers, Clinics, Hospitals
4. Proportion of those screened with eGFR <60ml/min/1.73m ²	Age, sex	Monthly	Health Centers, Clinics, Hospitals
5. Number of patients with confirmed CKD that started treatment.	Age, sex, and whether newly initiated or previously in care and, Type of treatment or referral.	Monthly	Health Centers, Clinics, Hospitals

Table 30: CKD Indicators among hypertension patients

B. For Patients with Diabetes

What to Measure	What to Record/ disaggregation	How Often	Where to Report From
6. Number of diabetes patients screened for CKD using urine dip stick	Age, sex, and test result	Monthly	Health Posts, Health Centers, Clinics, Hospitals
7. Proportion of those screened with >_1+ proteinuria.	Age, sex	Monthly	Health Centers, Clinics, Hospitals
8. Number of diabetes patients that received creatinine test for CKD screening.	Age, sex, and test result	Monthly	Health Centers, Clinics, Hospitals
9. Proportion of those screened with eGFR <60ml/min/1.73m ²	Age, sex	Monthly	Health Centers, Clinics, Hospitals
10. Number of patients with confirmed CKD that started treatment.	Age, sex, and whether newly initiated or previously in care and, Type of treatment or referral.	Monthly	Health Centers, Clinics, Hospitals

Table 31: CKD Indicators among DM patients

9.6 CHAPTER SUMMARY

- It is very important to grade all patients diagnosed with CKD because these grades have management and prognostic implications.
- It is important to monitor CKD programs to ensure early detection, timely intervention, and improved patient outcomes by identifying care gaps and preventing complications.
- Monitoring supports efficient resource use and informed decision-making by evaluating program performance, guiding policy, and prioritizing cost-effective, early interventions.
- It also promotes integration, public awareness, and quality care by aligning CKD efforts with other NCD programs, educating communities, and ensuring adherence to national guidelines
- HMIS is the routine collection, aggregation, analysis, presentation, and utilization of health and health related data.
- Evaluation is the periodic assessment of program impact and value.
- Data flow involves data capturing, collection, analysis, presentation, reporting, and use.

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ANNEXES

ANNEX I

Federal Ministry of Health

FMOH DIABETES MELLITUS & HYPERTENSION INTAKE FORM

Health Facility Name: _____ Date of Enrollment (DD/MM): ____/____/20____

Medical Registration Number: _____ Unique NCD Treatment number: _____

I. PATIENT IDENTIFICATION

Name of Patient: _____ Sex: _____ Age: _____ Telephone Number: [Company Phone]

Patient consented to be contacted: Yes ☐ No ☐

II. TREATMENT SUPPORTER CONTACT DETAILS

Full Name: _____ Address: _____
Phone Number: _____

III. PATIENT REFERRAL INFORMATION (circle appropriate one)

- | | | |
|-----------------------|------------------|--|
| 1. Previously in care | 5. In patient | 9. Community (workplace, HP, Camping etc.) |
| 2. Self-referral | 6. Emergency OPD | 10. Other Health Facilities |
| 3. ART Clinic | 7. ANC | |
| 4. TB clinic | 8. General OPD | |

11. PAST MEDICAL HISTORY AND USE OF MEDICATIONS

Disease/Condition	Yes	No	Disease/Condition	Yes	No
Diabetes Mellitus (if yes duration of diagnosis) Month(s) _____ or Year(s) _____			Hypertension (if yes duration of diagnosis) Month(s) _____ or Year(s) _____		
Any symptoms and signs of long term complications of DM identified on eyes, kidneys, nerves at time of diagnosis			Hypertension Related Heart Disease: underline the appropriate one (Ischemic Heart Disease, Hypertensive heart disease, Heart Failure, left ventricular hypertrophy, hypertension related cardiomyopathy)		
Presence of DM-related co-morbidities (cardiovascular diseases such as Stroke, ischemic heart disease, PAD, hypertension and dyslipidemia) at time of diagnosis			Vascular diseases related to hypertension: underline the appropriate one /Stroke, transient ischemic attack, Peripheral Artery Disease, Coronary Artery Disease (Acute coronary syndrome, Chronic coronary syndrome)/		
Foot ulcer or infections or Amputation related to diabetes			Kidney Disease related to HTN: Underline the appropriate one - hypertensive nephrosclerosis, proteinuria, acute kidney injury, chronic kidney disease		
Kidney disease related to DM: Underline the appropriate one- Diabetic Kidney Disease, acute kidney injury, chronic kidney disease					
Any admissions for diabetes complications (if already diagnosed)			Cholesterol abnormality		
For Female Patients					
-for female patients; Past history of delivering big baby(>4kg) or history of Diagnosed GDM			Diagnosed Gestational Hypertension (Pregnancy induced hypertension)		
Use of medications					
			Drugs like steroids, contraceptives, chemotherapy		

PREVIOUS TREATMENT HISTORY

DIABETES TREATMENT	Dose	Duration	Remark
HYPERTENSION TREATMENT			
DYSLIPIDEMIA TREATMENT			
Treatment for other complications/diseases			

V. FAMILY HISTORY OF CARDIOVASCULAR DISEASE AND DIABETES (father, mother, siblings/ one or more)

S. No	Disease /Situation	Yes	No	Unknown	Cause
1.	HTN				
2.	DM				
3.	Stroke				
4.	Heart attack				
5	CKD				
6.	Death				

VI. Presence of NCD risk factors

S. N	Risk Factors	yes	NO	Remark/Add detail or amount/
1.	Tobacco use (Smoking, chewing, Shisha)			
2.	Physical inactivity			
3.	Harmful Alcohol use			
4.	Khat Use			
5.	Unhealthy Diet (high Salt, Sugar, Fat) (Low Fruit and Vegetable)			

VII. CURRENT SYMPTOMS

Symptom	Yes	No	Symptom	Yes	No	Symptom	Yes	No
Hyperglycemia (Polyuria, Polydipsia, Polyphagia)			Peripheral Neuropathy (numbness, tingling, burning, pain, weakness)			Ischemic Heart Disease (Chest pain, shortness of breath, fatigue, dizziness, sweating, nausea, palpitations, discomfort in the arms, neck, jaw, or back.		
Hypoglycemia (<i>Hunger, Sweating, Confusion, tremor, fainting</i>)			Retinopathy			Other Specify:		
Decay			Diabetic Foot Ulcer					
Nephropathy (Fatigue, swelling (legs or face), foamy urine, decreased or changed urine output, loss of appetite, nausea, persistent itching, muscle cramps, anemia)								
For Women (15-49 years)								
Current: LMP ____/____/____				EDD if Pregnant: /____/____				

VIII. ANTHROPOMETRY & VITALS SIGNS

Weight: _____ Kg Height: _____ cm BMI: _____ (Kg/m²) Waist Circumference _____ cm
 BP: _____ / _____ mmHg(_____) PR: _____ /min RR: _____ /min Temperature _____ °C

IX. PHYSICAL EXAMINATION

Organ/system		Normal	Abnormal	Specify Abnormal Finding
HEENT				
Glands (Thyroid)				
Respiratory				
CVS	Heart			
	Peripheral arteries			
Abdomen and GUS				
MSS- Foot exam				
Integumentary System & Injection site				
Nervous System	Mental status			
	Motor			
	Sensory			

X. BASELINE LABORATORY FINDINGS

Tests	FBS/ RBS	Hg A1c	Urinalysis is- Ketone	Urinalysis - Protein	24 hours urine protein or U/A (microscopy)	Creatinine	Serum K+	Total Cholesterol/ LDL & HDL	Fundos copy	ECG
Result										

XI. FINAL DIAGNOSIS

o o Diabetes		o o Hypertension	
o Type of DM (Circle one)	o Complications (circle that apply)	o Stage of Hypertension (circle that apply)	o Cardiovascular Risk o Score (circle that apply)
1. Type 1 2. Type 2 3. Gestational DM 4. Other Specific Types	1. Diabetic Retinopathy 2. Diabetic Neuropathy 3. Diabetic nephropathy If CKD-G1/G2/G3a/G3b/G4/G5 4. Macro-vascular (CAD, PAD, Stroke or TIA), 5. Diabetic Foot Ulcer, 6. DKA/HHS 7. Other	o Grade 1 o (≥140-90 to 159/99 mmHg) Grade 2 (≥160/100-179/109 mmHg) Grade 3 (≥180/110 mmHg) • HTN emergency • HTN Urgency	Type of CV risk chart used: 1. Lab based 2. Non-lab based CV Risk Score: 1. <5% 2. 5% to <10% 3. 10% to <20% 4. 20 to <30% 5. ≥30%
Other chronic/ mental illness		Hypertension complications (circle that apply) 1. CKD - G1/G2/G3a/G3b/G4/G5 2. TIA/Stroke 3. Hypertensive heart disease 4. Peripheral arterial disease	

XII. TREATMENT PLAN:

NON-PHARMACOLOGICAL MANAGEMENT

Key Areas	HLC on Regular Physical Activity	HLC to Stop Tobacco use	HLC to Eat a Healthy Diet	HLC to Avoid Harmful use of alcohol	Advice on Weight reduction	Advice on Foot care,	Advice on dental care	Education on Injection technique	Education on Side effects of drugs	Treatment Adherence Counselling
Mark if done										

PHARMACOLOGIC TREATMENT

Diabetes Treatment	Formulation and Dose	Frequency	Start date
Oral Hypoglycemic Agents			
o Metformin			
o Glibenclamide or Glimepiride or gliclazide			
O Dapagliflozin or Canagliflozin or Empagliflozin			
O Vildagliptin			
o Other			
Insulin			
o NPH (isophane) insulin			
o Regular Insulin			
o Premixed insulin			
Hypertension Treatment			
Amlodipine			
Hydrochlorothiazide			
Lisinopril			
Candesartan			
o Others (specify)			
Dyslipidemia Treatment			
o Atorvastatin, Simvastatin			
Antiplatelet			
o Aspirin			
o Clopidogrel			
Other Treatment			

ANNEX II



ጤና ሚኒስቴር - ኢትዮጵያ
MINISTRY OF HEALTH-ETHIOPIA
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HEALTHY PEOPLE FOR PROSPEROUS ETHIOPIA

Federal Ministry of Health, Ethiopia DM and Hypertension Follow Up Form

Name of Patient: _____ Age: _____ Sex: _____ Medical Registration Number: _____

Address: Zone/Sub-city _____ Woreda: _____ Kebele: _____ H. No. _____ Tel: _____

Date of Enrollment ____/____/____ Full Initial Diagnosis _____

Visit Type Scheduled Unscheduled	Date of Visit (__/__/__)	History and Risk factors		Anthropometry and Vital signs					Physical Examination					Laboratory Investigations					Outcome	Treatment Plan and Adherence				Referral/Consultations	Clinician's Name	Next Appointment ____/____/____
		Symptoms	Risk factors	Waist circumference (cm)	Height (cm)	Weight (kg)	BMI (kg/m ²)	BP (mmHg)	HEENT (Oral/Dental)	CVS	Skin/Inj. site	MSS/Foot Exam	Nerv. Syst.	Nerv. Syst. FBS / HBA1C	U/A (Protein, Ketone)	Cr	Total Cholesterol /LDL	Life-style Changes		Regimen /Dose	# months dispensed	Adherence / Why				
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)	(s)	(t)	(u)	(v)	(w)	(x)	(y)	(z)	



ጤና ሚኒስቴር - ኢትዮጵያ
MINISTRY OF HEALTH-ETHIOPIA
የኢትዮጵያ ጤና ሚኒስቴር

Federal Ministry of Health, Ethiopia DM and Hypertension Follow Up Form

How to fill the DM and Hypertension Follow up format

Symptoms(DM) (c)	Symptoms(HTN) (c)	Risk factors status (d)	B/P (mmHg) (i)	Cardiovascular (k)	Musculoskeletal Exam/ Foot exam (m)		
HyperG: Polyuria, Nocturia, Polydipsia, Fatigue, DKA: Nausea, Vomiting, fast and deep breathing, weakness, worsening Poly symptoms, abdominal pain. HypoG: Hunger, palpitations, lightheadedness, dizziness, sweating, extreme fatigue, confusion, bad dreams Infections: F: fever, C: cough, SL: skin lesions, UTI: dysuria, frequency DFU: Diabetic Foot Ulcer PN: tingling, numbness, burning sensation of Feet and hands. AutoN: Postural dizziness, Sexual dysfunction VisB: Visual Blurring	Dysp: Dyspnea Ortp: Orthopnea PND: Paroxysmal Nocturnal Dyspnea LSw: Leg swelling Ang: Angina IntCl: Intermittent claudication HemP: Hemiparesis /Hemiplegia SpA: Speech Abnormality Seiz: Seizure HA: Headache PhP: Photophobia NkP: Neck pain and Stiffness Other UOP: Decreased Urine Output	Tbco: Tobacco Use including shisha PhIn: Physical inactivity Alc: Alcohol use (>2 standard drinks for men and >1 standard drinks/day for women for >5days per week). Salt: High Salt consumption Sug: High Sugar Consumption Fat: High Fat Consumption Veg: Low Vegetable and fruit consumption Khat: Khat Consumption Ht(m), Wt(kg) and BMI (kg/m²) (f, g and h) - Measure height of an adult at enrollment and children every visit - Measure weight at initial assessment and at each visit - Calculate BMI as: Weight in Kg divided by the square of height in meters(kg/m ²) BMI Interpretations: • <18.5—Malnourished • 18.5-24.9—Normal • 25-29.9—Overweight • 30-39.9—Obese • >40—Morbid Obesity	- Measured at every visit with Goal of <140/90 mmHg (check for orthostatic changes when indicated) HEENT(Oral/Dental) (j) GOH: Adequately hygienic POH: Poor Oral Hygiene DNTC: Dental Caries Ging: Gingivitis DNTL: Dental Loss Cand: Oral Candidiasis	1. Normal Cardiac Exam 2. S3 and/or S4 gallop 3. Murmurs 4. Neck Veins engorged or JVP raised 5. Signs of arrhythmia Skin and Injection site (l) 1. Normal findings 2. Atrophic injection site 3. Hypertrophic injection site 4. Signs of infection 5. Wrong injection site	1. Dry skin /Callus/ deformity/ 2. Ulcer 3. Absent/asymmetric pulses, poor capillary filling 4. Reduced or absent light touch, Vibration or position senses 5. Decreased/Absent DTR 6. Infected wound 7. Pedal edema 8. Pretibial Edema 9. Normal findings Neurologic and Dilated eye exam (n) • Neurologic: o Motor, Sensory and Eye Exam • Dilated eye exam should be done for o all type 2 patients annually starting from Dx and o all type 1 annually 5yrs from Dx o Hypertensive patients with suspected hypertensive urgency or emergency		
Lab tests (o,p,q and r)	Outcome (s)	Life Style Changes and education (t)	Treatment (u,v)	Treatment (u,v)	Treatment (u,v)	Adherence (w)	Referrals/ Consultations (x)
1. FBS (to be done at every visit) or HBA1C every 3 months for DM patients. 2. U/A • Protien (P-,P+ to P++++)—check annually • Ketone (K-,K+ to K++++)—Type 1 DM every visit and for Type 2 when FBS/RBS>300mg/dl • Microscopy to see Casts 3. RFT done annually for Type 2 DM;for Type 1 DM 5 years after diagnosis	• Controlled • Uncontrolled • Dead • Lost to follow up • Transferred out	1. Cessation of smoking 2. Exercise(150min per week) 3. Healthy Diet 4. Reduce alcohol use 5. Medication adherence 6. Medication side effects 7. Daily Foot inspection 8. Family screening 9. Injection Techniques	1. Insulin 1a. NPH insulin 1b. Regular insulin 1c. Premixed insulin 2. Oral hypoglycemic Agent 2a. Glimeclamide 2b. Metformin 2c. Glimepiride 2d. Other	3. Antihypertensives o Am—amlodipine o HCT—Hydrochlorothiazide o Lis—Lisinopril o Others:specify	4. Antiplatelets ASA —Aspirin Clop —Clopidogrel 5. Statins Smv —Simasatin Atv —Atorvastatin Others -specify	• good (missed <20% of the doses last month) • Poor (missed ≥20% of the doses last month)W	Refer eligible Diabetic patients to Ophthalmologist for dilated eye exam and document the feed back

Health Facility: _____ Woreda: _____ Region _____ Month----- Cohort Year _____

Entry Point: Col (8)

1. Previously in care
2. Self referral
3. Referred from ART/.
4. Referred from TB
5. Referred from OPD
6. Referred from Emerg Dep.
7. Referred from ANC
8. Referred from another facility
9. Referred by HEW

BMI: Col (10)

- 1.Underweight(<18.5)
- 2.Normal(18.5-24.9)
- 3.Over weight(25-29.9)
- 4.Obese(30-39.9)
- 5.Morbid Obesity(>40)

Diagnosis: Col (15)
Types of Diabetes:
1.Type I
2.Type II
3.GDM
4 others

Grading of hypertension:

1. Grade 1
2. Grade 2
3. Grade 3 (3.1.hypertensive urgency;
3.2. Hypertensive emergency)

Risk factors: Col (12)

- 1.Tobacco Use
- 2.Physical inactivity
- 3.Harmful use of Alcohol
- 4.Khat Use
- 5.Unhealthy Diet
- 6.Others

10 years CVD risk Col (14)
 -Lab based risk category 1.<10% (low)
 2. 10%- 20% (moderate)
 3. >=20% (High)
Non-Lab based category
 4. <10% (low to moderate)
 5.>=10% (High)

Antidiabetic Drugs

1. NPH insulin
2. Regular insulin
3. Premixed insulin
4. Glibneclamide
5. Metformin
6. Glimepiride
7. Gliclazide
8. Dapagliflozin
9. Canagliflozin
10. Vildagliptin
11. Empagliflozin
12. Others - Specify name

Antihypertensives

1. Amlodipine
2. Hydrochlorothiazide
3. Lisinopril
4. Enalapril
5. Nifedipine
6. Atenolol
7. Candesartan
8. Others - Specify name and dose

Statins

1. Simvastatin
2. Atorvastatin
3. Rosuvastatin
4. Lovastatin
5. Pravastatin
6. Others - Specify name and dose

Outcome Status
HTN Rx Outcome
1.HTN Controlled
2.HTN Not Controlled
3.Dead
4.Lost to follow up (LTF)
5.Transferred out (TO)

DM Rx Outcomes
1.DM Controlled
2.DM Not Controlled
3.Dead
4.Lost to follow up (LTF)
5.Transferred out (TO)

Result of TB diagnosis (2010)
1.TB
2. No TB
3. Not decided

Result of CKD diagnosis (17)
 1. Grade 1 3. Grade 3a 5. Grade 4
 2. Grade 2 4. Grade 3b 6. Grade 5

